

Compressing Global Dynamics of Deep Nonlinear Feature Learning

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Abstract

We give a modular proof of three trajectory-compression constructions for Gaussian-initialized deep nonlinear networks with learned hidden features. Legendre compression retains $n^{5/4+o(1)}$ moving coordinates and fixed dense mixers. Harmonic compression retains $O((\log n)^{3d+2})$ real coordinates for the whole input sphere. Taylor compression retains $O((\log n)^3)$ real coordinates on a fixed panel declared at initialization. All three have error negligible relative to actual independent dense-run variability, uniformly over their query domain and the complete training trajectory, including the fitted limit. The two selected constructions also attain absolute error Y/n . The common ingredients are finite residual activity, analytic response approximation, and geometry-preserving reduction; the selected dynamics use an additional readout correction to cancel unstable feedback. This copy isolates the theorem and its proof: it makes no empirical, finite-precision, or efficient-preprocessing claim.

1 Setup and the headline theorem

There are $m \geq 2$ training pairs (x_a, y_a) with $x_a \in \mathbb{R}^d$, $\|x_a\|_2 = \sqrt{d}$, width n , and a fixed number $L \geq 2$ of hidden layers. Write $v = x/\sqrt{d}$. The first matrix is $W^{(1)} \in \mathbb{R}^{n \times d}$, the hidden matrices are $W^{(\ell)} \in \mathbb{R}^{n \times n}$, and the readout is $w \in \mathbb{R}^n$. The activation may depend on the layer; the superscript is suppressed when the same equation applies to every layer. When a response is indexed by the normalized query v , its network input is $x = \sqrt{d}v$. The canonical network is

$$\begin{aligned} z^{(1)}(x) &= W^{(1)}v, & z^{(\ell)}(x) &= W^{(\ell)}h^{(\ell-1)}(x), & h^{(\ell)}(x) &= \phi_\ell(z^{(\ell)}(x)), \\ f_n(x) &= \frac{w^\top h^{(L)}(x)}{n}, & r_a &= f_n(x_a) - y_a, & \mathcal{L} &= \frac{1}{m} \sum_{a=1}^m r_a^2. \end{aligned} \quad (1)$$

The residual-free backward responses are

$$\delta_a^{(L)} = w \odot \phi'_L(z_a^{(L)}), \quad \delta_a^{(\ell)} = \phi'_\ell(z_a^{(\ell)}) \odot W^{(\ell+1)\top} \delta_a^{(\ell+1)}. \quad (2)$$

At a passive input use the same forward and backward recursions, without supplying a label or adding a training term. Dots denote physical training time. Gradient flow with block mobilities $(n, 1, \dots, 1, n)$ is exactly

$$\begin{aligned} \dot{W}^{(1)} &= -\frac{2}{m} \sum_a r_a \delta_a^{(1)} v_a^\top, \\ \dot{W}^{(\ell)} &= -\frac{2}{nm} \sum_a r_a \delta_a^{(\ell)} h_a^{(\ell-1)\top} \quad (2 \leq \ell \leq L), & \dot{w} &= -\frac{2}{m} \sum_a r_a h_a^{(L)}. \end{aligned} \quad (3)$$

First-layer entries are independent $\mathcal{N}(0, 1)$; hidden entries are independent $\mathcal{N}(0, 1/n)$; all blocks are independent; and $w(0) = 0$. Nothing freezes the hidden features.

Activations, data, and constants. Each ϕ_ℓ is real on the real axis, holomorphic on $|\operatorname{Im} z| < a$ for a common $a > 0$, and has bounded first derivative on that strip. Its values need not be bounded. For explicit estimates put

$$\beta = \max \left\{ 10, 1 + \max_{\ell} |\phi_\ell(0)|, 16/a, \max_{\ell, j \in \{1,2\}} \sup_{|\operatorname{Im} z| \leq a/2} |\phi_\ell^{(j)}(z)| \right\}.$$

The second derivative is bounded on the smaller strip by Cauchy’s formula. Define the uncentered population feature Gram and the label scale by

$$\begin{aligned} Q_{ab}^{(0)} &= v_a^\top v_b, & Q_{ab}^{(\ell)} &= \mathbb{E}[\phi_\ell(Z_a)\phi_\ell(Z_b)], & Z &\sim \mathcal{N}(0, Q^{(\ell-1)}), \\ \gamma &= \lambda_{\min}(Q^{(L)}) > 0, & Y &= \|y\|_2 / \sqrt{m} > 0. \end{aligned} \tag{4}$$

Assume $m/\gamma \geq 1$ and the same sufficient small-label condition

$$0 < Y \leq \frac{\gamma}{m} \beta^{-30L}. \tag{5}$$

No input-rank, centering, orthogonality, or label-sign hypothesis is imposed. All data, m, d, L, γ, Y , and activations are fixed separately as n grows. Throughout the stated bounds, $C > 0$ depends only on activations and depth, never on n, m, d, γ, Y or a passive-panel size. Its value may change between occurrences. All other constants are defined in the statement or proof using them.

Query domains and probability. For a query set $\mathcal{X}$, use

$$\|f - g\|_{\mathcal{X}} = \sup_{t \in [0, \infty]} \sup_{x \in \mathcal{X}} |f(t, x) - g(t, x)|. \tag{6}$$

The endpoint $t = \infty$ means the fitted limit. For Legendre and Harmonic, $\mathcal{X} = \{x : \|x\|_2 = \sqrt{d}\}$. For Taylor, fix $p \geq 0$ additional inputs on this same sphere before initialization and take $\mathcal{X} = \{x_1, \dots, x_{m+p}\}$. Their labels are never supplied. Let $\tilde{f}_n$ be an independently initialized dense run on the same training problem.

An eventual assertion at confidence $1 - \delta$ means: for every fixed admissible problem and $0 < \delta < 1$, there is a finite width threshold such that the assertion holds at each larger individual width. That threshold may depend on every fixed problem parameter, including the positive Y , and is not proved polynomial or practically small. No simultaneous event over infinitely many independently initialized widths, uniform growing-data limit, or uniform limit as $Y \downarrow 0$ is asserted. For a ratio below, missing fitted limits or a zero denominator count as failure.

All neural-network statements below inherit this setup and these probability conventions. Every storage count charges a nonprimitive activation evaluator’s description and scratch separately.

What must be remembered. The following is the complete persistent notation. A row with several entries counts each entry separately; layer and sample indices do not create new objects. The two uses of δ are counted separately.

Symbols	Meaning	Count
n, m, d, L, p	width, training count, dimension, depth, passive count	5
$x, v = x/\sqrt{d}, y, t$	input, normalized input, labels, physical time	4
$W, w, z, h, r, \delta_a^{(\ell)}$	weights, readout, preactivations, features, residuals, backward responses	6
$\phi, a, \beta, Q, \gamma, Y$	activations, strip, envelope, population Gram, gap, label RMS	6
$\mathcal{L}, \mathcal{X}, \ \cdot\ _{\mathcal{X}}$	loss, query domain, trajectory norm	3
δ, C	failure probability, activation/depth constant	2
$f_n, \tilde{f}_n, f_{\text{model}}, f_{\text{NTK}, n}$	dense, independent dense, compressed and frozen-kernel predictors	4
		30

Subscripts naming a compression method specialize f_{model} . Ordinary mathematical operations and bound indices have their usual meanings. Every other symbol is bound to the statement and proof that introduce it: it can be forgotten afterward. In particular, approximation orders, clocks, source spaces, selected metrics, and numerical ledgers are not additional global notation. Later arguments use the stated conclusions, not private coefficients from their proofs.

Theorem 1.1 (Compression of the complete learning trajectory). *Under the preceding assumptions, each construction can be chosen so that*

$$\frac{\|f_{\text{model}} - f_n\|_{\mathcal{X}}}{\|f_n - \tilde{f}_n\|_{\mathcal{X}}} \xrightarrow{\mathbb{P}} 0. \quad (7)$$

Here f_n is the construction's coupled ordinary dense reference, and both norms use the same promised query domain and all physical times, including the fitted endpoint. Sufficient real-coordinate counts are

$$\text{Legendre moving state: } Cm \left(\frac{m}{\gamma}\right)^3 n^{5/4} [\log(en)]^2 e^{\sqrt{\log(en)}/2}, \quad (8)$$

$$\text{Harmonic total state: } C(m+d)^2 + \left(\frac{C}{d}\right)^{d+1} \left(\frac{Ym}{\gamma}\right)^4 [\log(en)]^{3d+2}, \quad (9)$$

and the Taylor total state is at most

$$C(m+p)^2 \left[\left(\frac{Ym}{\gamma}\right)^4 [\log(en)]^3 + \left(\frac{m}{\gamma}\right)^2 [\log(en)]^2 [\log(e + \log(en))]^2 \right] + C(m+p)d. \quad (10)$$

Legendre additionally retains $(L-1)n^2$ fixed mixer coordinates. Harmonic and Taylor include the data, selected mixers, metrics, initial copies, and working arrays. Their predictions additionally satisfy

$$\|f_{\text{model}} - f_n\|_{\mathcal{X}} \leq Y/n \quad (11)$$

at every fixed confidence for sufficiently large width, with the same storage orders. The theoretical initializers use the dense initialization, training data, and finitely many initialization derivatives; no later trained weight or dense training trajectory is an input. Subsequent evolution is an autonomous nonlinear system with the retained state.

These are sufficient sizes, not optimality claims. Dense itself has $(L-1)n^2 + n(d+1)$ weight coordinates. Data storage is additional to the Legendre moving-state count. Coordinate counts do not bound precision or preprocessing time. The finite-panel theorem makes no accuracy promise on inputs first supplied after initialization. Zero labels give the stationary zero predictor instead.

1.1 The compressed dynamics are nonlinear and feature-learning

A moving kernel alone does not distinguish nonlinear learning from a deep linear network. We therefore separate two comparisons. In this subsection, $f_{\text{NTK},n}$ is the predictor obtained by freezing the coupled dense reference’s initial tangent kernel. Since $w(0) = 0$, this is exactly its readout-only flow with all hidden features fixed at initialization.

Theorem 1.2 (Compression preserves nonlinear feature learning). *Under the compression theorem’s assumptions, suppose additionally that every activation is nonaffine and $|x_a^\top x_b| < d$ for $a \neq b$. For Taylor, augment $\mathcal{X}$ by at most four deterministic passive witness inputs, declared before initialization without their labels. There are $c_*, t_* > 0$, depending on the fixed problem but not on width, such that for every fixed $0 < t \leq t_*$, each compressed predictor satisfies, with probability tending to one as $n \rightarrow \infty$,*

$$\inf_{a \in \mathbb{R}^d, b \in \mathbb{R}} \sup_{x \in \mathcal{X}} |f_{\text{model}}(t, x) - a^\top x - b| \geq c_* t, \quad (12)$$

$$\sup_{x \in \mathcal{X}} |f_{\text{model}}(t, x) - f_{\text{NTK},n}(t, x)| \geq c_* t^3. \quad (13)$$

The storage orders and approximation conclusions are unchanged; Taylor’s bound uses the augmented panel count, at most $p + 4$ instead of p .

In the dense reference, every hidden layer satisfies

$$\left(\frac{1}{n} \sum_{a=1}^m \|h^{(\ell)}(t, x_a) - h^{(\ell)}(0, x_a)\|_2^2 \right)^{1/2} \geq c_* t^2 \quad (1 \leq \ell \leq L) \quad (14)$$

with the same probability convention. In the first layer, the nonlinear part of the feature increment itself has positive mean-square size of order t^4 , in the precise sense of (116).

The first inequality excludes every deep linear predictor, even with biases, an arbitrary optimizer and time-dependent weights. The second excludes the reference network’s frozen-feature training. The internal statement further rules out a frozen nonlinear feature component accompanied only by an affine feature change. It does not identify the compressed coordinates with dense neurons. The additional hypotheses belong only to this activity theorem; the broader compression theorem does not require them.

The mechanism has three time scales. The initially zero readout first grows at order t ; hidden features move at order t^2 ; the output departs from frozen-feature training at order t^3 . The leading departure in the label direction is a sum of squared hidden-layer accelerations, so contributions from different layers cannot cancel. Separately, the initial feature kernel has nonlinear input components that no nonzero label vector can cancel completely on the sphere. Both signals persist while compression error vanishes. The proof in Section 7 uses only initial Gaussian conditioning and real short-time estimates, not a width-independent complex-time radius or a trained population-flow theorem. These are early-time certificates, not endpoint or test-risk comparisons.

Proof architecture. First establish fitting and finite residual activity without a source approximation. A common probabilistic source lemma then controls forward and backward responses on the two analytic domains. Legendre approximates their history integrals. The other two methods share one geometry-preserving selection and one corrected nonlinear optimizer; only their finite source bases differ. A last-layer innovation and a deterministic analytic estimate give a dense lower bound at a training input. The final section combines these ingredients. Every dependency of that argument is supplied below; the older decoder, numerical setup-cost alternatives and empirical material are outside this theorem-focused copy.

2 Initialization, fitting, and a signed perturbation estimate

Lemma 2.1 (Initialization and real fitting). *For this lemma put*

$$H = \max_{0 \leq j \leq L} \sqrt{Q_{11}^{(j)}}, \quad \mathbf{H}(t) = [h^{(L)}(t, v_1), \dots, h^{(L)}(t, v_m)].$$

For the parameter tuple $\theta = (W^{(1)}, \dots, W^{(L)}, w)$ use the norm

$$\|\theta\|_{\text{par}}^2 = \|W^{(1)}\|_F^2/n + \sum_{j=2}^L \|W^{(j)}\|_F^2 + \|w\|^2/n.$$

The initialized training Grams satisfy

$$\left(\frac{h_0^{(j)}(v_a)^\top h_0^{(j)}(v_b)}{n} \right)_{a,b=1}^m \xrightarrow{\mathbb{P}} Q^{(j)} \quad (1 \leq j \leq L). \quad (15)$$

With probability tending to one,

$$\begin{aligned} \frac{\|W_0^{(1)}\|_{\text{op}}}{\sqrt{n}} &\leq 8, & \max_{2 \leq j \leq L} \|W_0^{(j)}\|_{\text{op}} &\leq 8, \\ \sup_{\|v\|=1, 1 \leq j \leq L} \frac{\|h_0^{(j)}(v)\|}{\sqrt{n}} &\leq \frac{3H}{2}, & \frac{H_0^\top H_0}{mn} &\succeq \frac{\gamma}{2m} I_m. \end{aligned} \quad (16)$$

On this event the flow is global, converges in parameters, fits the labels, and satisfies, for all $t \geq 0$,

$$\begin{aligned} \frac{\|W^{(1)}(t)\|_{\text{op}}}{\sqrt{n}} &< 9, & \max_{2 \leq j \leq L} \|W^{(j)}(t)\|_{\text{op}} &< 9, \\ \sup_{\|v\|=1, 1 \leq j \leq L} \frac{\|h^{(j)}(t, v)\|}{\sqrt{n}} &< 2H, & \frac{\mathbf{H}(t)^\top \mathbf{H}(t)}{mn} &\succeq \frac{\gamma}{4m} I_m \end{aligned} \quad (17)$$

$$\begin{aligned} \frac{\|r(t)\|_2}{\sqrt{m}} &\leq Y e^{-\gamma t/(2m)}, & \frac{\|w(t)\|}{\sqrt{n}} &\leq 2Y \sqrt{\frac{m}{\gamma}}, \\ \int_t^\infty \|\dot{\theta}(u)\|_{\text{par}} du &\leq \frac{2\|r(t)\|_2}{\sqrt{\gamma}}. \end{aligned} \quad (18)$$

The output converges uniformly on the sphere, with

$$\sup_{\|v\|=1} |f_n(\infty, v) - f_n(t, v)| \leq C \frac{m}{\gamma} \frac{\|r(t)\|_2}{\sqrt{m}}. \quad (19)$$

These assertions hold jointly for all rectangular cavities deleting any fixed bounded number of coordinates per layer, with the original width- n variances, flow factors, and denominators unchanged.

Proof. All auxiliary coefficients in this proof are local. Put $\lambda = \gamma/m$, $\rho = \|r\|_2/\sqrt{m}$,

$$s = \max \left\{ 1, \max_j \sup_{|\text{Im } z| \leq a/2} |\phi_j'(z)| \right\}, \quad b = \max_j |\phi_j(0)|,$$

and retain H as defined in this lemma's statement. Thus $|\phi_j(z)| \leq b + s|z|$ on the half strip. Define

$$\begin{aligned} D_j &= s(9s)^{L-j}, & U_1 &= D_1, & U_j &= 2HD_j \quad (j \geq 2), \\ F_1 &= sU_1, & F_j &= s(2HU_j + 9F_{j-1}), & F &= F_L. \end{aligned} \quad (20)$$

Equal input norms give $Q_{aa}^{(j)} = Q_{11}^{(j)}$, hence $\lambda \leq H^2$. The coefficients satisfy

$$H \leq (2\beta)^L, \quad F = s^2 \left[(9s)^{2L-2} + 4H^2 \sum_{j=0}^{L-2} (9s)^{2j} \right], \quad 8H\sqrt{F} \leq \beta^{5L}. \quad (21)$$

The first inequality follows from Minkowski's inequality and $\sqrt{Q_{11}^{(j)}} \leq b + s\sqrt{Q_{11}^{(j-1)}}$. For the last, summing the geometric series gives $F \leq (21/20)H^2s^2(9s)^{2L-2}$, and then $8H\sqrt{F} \leq H^2(9s)^L \leq 36^L\beta^{3L} \leq \beta^{5L}$. Consequently the assumed cap $Y \leq \lambda\beta^{-30L}$ implies $Y \leq \lambda/(8H\sqrt{F})$, which will close the real-flow estimates.

A maximal $1/4$ -separated subset of the unit sphere in $\mathbb{R}^k$ has at most 9^k points: disjoint radius- $1/8$ balls fit inside the radius- $9/8$ ball. Approximating each side of a maximizing bilinear pair gives $\|M\|_{\text{op}} \leq 2 \max |u^\top Mv|$ over these nets. Each pairing for $W_0^{(\ell)}$, $\ell \geq 2$, or $W_0^{(1)}/\sqrt{n}$ is $N(0, 1/n)$. The exponential Gaussian tail and a union bound prove the operator assertions, with failure bounded by $2(L-1)e^{-(8-2\log 9)n} + 2e^{-(8-\log 9)n+d\log 9}$.

On any fixed finite query list, condition on the preceding initialized layers. Each next empirical Gram is an average of independent Gaussian-row feature products, with conditional mean $\Psi_\ell(C)_{ab} = \mathbb{E}\phi_\ell(Z_a)\phi_\ell(Z_b)$, $Z \sim N(0, C)$. On bounded covariance sets the conditional variances are bounded, by linear growth and Gaussian fourth moments. Moreover Ψ_ℓ is continuous on the entire positive semidefinite cone: write $Z = C^{1/2}G$ with a standard Gaussian vector G of the matching dimension, use continuity of the positive square root, and dominate by a fixed Gaussian polynomial. Conditional Chebyshev and induction therefore prove convergence of each empirical Gram to $Q^{(\ell)}$, including when intermediate covariances are singular. Apply this to the training inputs and a fixed sphere net of mesh $h \leq H/[4(8s)^L]$. On the operator event the initialized layer- ℓ feature map, in neuron RMS, is $(8s)^\ell$ -Lipschitz. Its norm at net points is eventually at most $\sqrt{5/4}H$, with probability tending to one; the off-net increment is at most $H/4$. This proves the sphere bound and the training Gram assertion in (16).

Here is also a uniform justification for the cavity assertion, without a union bound over deletion sets. On a sphere net of mesh n^{-2} , the preceding RMS bounds and conditional Gaussian tails give $\max_{i,\ell,v} |h_{0,i}^{(\ell)}(v)| = O(\sqrt{\log(en)})$ with probability tending to one. The net has only a fixed power of n points; the coordinate Lipschitz bound is at most $\sqrt{n}(8s)^\ell$, so the same bound holds off the net. Zero-extend each cavity's features into $\mathbb{R}^n$. At a deleted layer, at most a fixed number of original coordinates are removed. At an undeleted coordinate the activation difference is at most s times the preactivation difference. Thus successive RMS differences are bounded by $8s$ times the preceding difference plus $O(\sqrt{\log(en)/n})$. At fixed depth their supremum, simultaneously over all such cavities and sphere queries, tends to zero. Submatrix operator norms do not increase. Taking strictly smaller initial feature and Gram margins before this comparison gives (16) for every cavity. This also shows directly that each cavity's training Gram has the same limit $Q^{(L)}$.

Stop any one of these flows at the first failure of (17). Backward propagation gives $\|\delta_a^{(\ell)}\|/\sqrt{n} \leq D_\ell\|w\|/\sqrt{n}$. In Euclidean mobility coordinates $(W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$, the vector field is minus the gradient of ρ^2 . Consequently

$$-\partial_t \rho^2 = \|\dot{\theta}\|_{\text{par}}^2 \geq \lambda \rho^2.$$

The last inequality uses only the readout contribution and the stopped Gram gap. For any stopped interval $[t, T]$, Cauchy-Schwarz gives

$$\left(\int_t^T \|\dot{\theta}\|_{\text{par}} \right)^2 \leq \left(\int_t^T \frac{\|\dot{\theta}\|_{\text{par}}^2}{\rho} \right) \left(\int_t^T \rho \right) \leq 2\rho(t) \frac{2\rho(t)}{\lambda}.$$

If a residual vanishes, all velocities vanish, so the estimate extends without division by zero. In particular $\|w\|/\sqrt{n} \leq 2Y/\sqrt{\lambda}$. Sample Cauchy–Schwarz bounds each normalized hidden-block velocity by $2\rho U_\ell \|w\|/\sqrt{n}$. Integration and forward subtraction give

$$\begin{aligned} \frac{\|W^{(1)} - W_0^{(1)}\|_F}{\sqrt{n}} &\leq \frac{8U_1 Y^2}{\lambda^{3/2}}, & \|W^{(\ell)} - W_0^{(\ell)}\|_F &\leq \frac{8U_\ell Y^2}{\lambda^{3/2}}, \\ \sup_v \frac{\|h^{(\ell)} - h_0^{(\ell)}\|}{\sqrt{n}} &\leq \frac{8F_\ell Y^2}{\lambda^{3/2}}. \end{aligned} \quad (22)$$

Indeed each later feature difference is bounded by s times $2H\|W^{(\ell)} - W_0^{(\ell)}\|_F$ plus nine times the preceding feature difference, which is exactly the recursion defining F_ℓ . The label restriction implies $8FY^2/\lambda^{3/2} \leq \sqrt{\lambda}/(8H^2) \leq 1/(8H)$. Thus the operators remain below $8 + 1/8$, the features below $3H/2 + H/8$, and the smallest singular value of $H/\sqrt{mn}$ above $(1/\sqrt{2} - 1/8)\sqrt{\lambda} > \sqrt{\lambda}/2$. Every stop has a strict margin. At fixed width the bounded parameters and locally Lipschitz vector field give global continuation. The finite parameter length proves convergence, and residual decay proves fitting. The argument uses only the retained dimensions and the original denominators n , and hence also proves cavity fitting.

Finally the same forward recursion gives

$$\|\dot{w}\|/\sqrt{n} \leq 4H\rho, \quad \sup_v \|\dot{h}^{(L)}(v)\|/\sqrt{n} \leq 2F\rho\|w\|/\sqrt{n}.$$

Therefore

$$\sup_{\|v\|=1} |f_n(\infty, v) - f_n(t, v)| \leq \frac{16}{\lambda} \left(H^2 + \frac{FY^2}{\lambda} \right) \rho(t).$$

The coefficients H, F depend only on activations and depth, and the headline cap with $\lambda \leq 1$ gives $Y^2/\lambda \leq 1$. This proves (19), including its uniform endpoint. $\square$

Lemma 2.2 (Signed perturbation estimate). *Let a differentiable prediction map F take Euclidean parameter space to sample space with inner product $\langle u, v \rangle_m = m^{-1}u^\top v$. Write $r = F(\theta) - y$, $r' = F(\theta') - y$, $\rho = \|r\|_m$, $\rho' = \|r'\|_m$, and $J = DF(\theta)$, $J' = DF(\theta')$. Suppose $\dot{\theta} = -2J^*r + \mathcal{E}$ and $\dot{\theta}' = -2J'^*r'$. Here J^* and J'^* are adjoints for the stated parameter and sample inner products. If a constant $K \geq 0$ satisfies, for $e = \|\theta - \theta'\|$,*

$$\|r - r' - J'(\theta - \theta')\|_m \leq Ke^2/2, \quad \|(J - J')(\theta - \theta')\|_m \leq Ke^2,$$

then, on every finite interval,

$$e(t) \leq \left(e(0) + \int_0^t \|\mathcal{E}(u)\| du \right) \exp \left\{ K \int_0^t (\rho' + 3\rho) du \right\}. \quad (23)$$

Proof. Put $d = \theta - \theta'$ and $R = r - r' - J'd$. Exact subtraction gives

$$\frac{1}{2}\partial_t e^2 = -2\|r - r'\|_m^2 + 2\langle r - r', R \rangle_m - 2\langle r, (J - J')d \rangle_m + \langle d, \mathcal{E} \rangle \leq K(\rho' + 3\rho)e^2 + e\|\mathcal{E}\|.$$

Apply the integrating factor to $\sqrt{e^2 + \epsilon^2}$ and let $\epsilon \downarrow 0$. This proves the assertion also through zeros of the parameter discrepancy. $\square$

3 One source argument on two analytic domains

The following interface supplies the analytic approximation domains and real-time bounds used by the compression constructions. Its proof contains the stopped-cavity argument and all its auxiliary coefficient estimates.

Proposition 3.1 (Analytic sources and real-time response bounds). *Fix the finite query lists below before initialization. Eventually at each fixed confidence, the following hold on one event. In this proposition put*

$$T = 32 \frac{m}{\gamma} \log(en), \quad r_t = \frac{\gamma}{\beta^{30L} Y^2 m \sqrt{(d+3) \log(en)}}, \quad r_q = \frac{1}{\beta^{3L} \sqrt{(d+3) \log(en)}}.$$

Write $k^{(L)} = w$, $k^{(j)} = W^{(j+1)\top} \delta^{(j+1)}$ ($j < L$) for the backward responses before the activation derivative. The four source families are

$$h^{(j)}, \quad W^{(j)}(0)h^{(j-1)}, \quad \delta^{(j)}, \quad W^{(j+1)}(0)^\top \delta^{(j+1)}, \quad (24)$$

omitting the forward image for $j = 1$ and reverse image for $j = L$.

(i) The forward and backward fields are holomorphic on a neighborhood of

$$([-r_t, T + r_t] + i[-r_t, r_t]) \times \{v \in \mathbb{C}^d : v^\top v = 1, \|\operatorname{Im} v\|_2 \leq \sinh r_q\}.$$

Query holomorphy is relative to the quadric; for $d = 1$ use $\{-1, 1\}$. Here and for $t \in [0, \infty]$ over the real sphere, the first-layer operator divided by $\sqrt{n}$ and all hidden operators have norm at most ten, and

$$\frac{\|h^{(j)}\|_2}{\sqrt{n}} \leq \beta^{3L}, \quad \frac{\max\{\|\delta^{(j)}\|_2, \|k^{(j)}\|_2\}}{\sqrt{n}} \leq 16\beta^{4L} \frac{Ym}{\gamma}. \quad (25)$$

Each source coordinate has modulus at most $\beta^{6L} \sqrt{n}$ on the complex domain.

(ii) On each fixed finite query list, the predictions are holomorphic near $[-r, T + r] + i[-r, r]$, where

$$r = \frac{m}{\gamma \sqrt{\log(en)}}, \quad |f_n(t, \sqrt{d}v)| \leq 16\beta^{6L} \frac{Ym}{\gamma}. \quad (26)$$

On the entire real training trajectory,

$$\sup_{t \in [0, \infty]} \max_{a, j, i} |k_{a,i}^{(j)}(t)| \leq 32\beta^{21L} \frac{Ym}{\gamma} \sqrt{\log(en)}. \quad (27)$$

(iii) For a declared finite panel containing the training inputs, there are $0 = t_0 < \dots < t_J = T$ and $r_b > 0$, $0 \leq b < J$, with $t_{b+1} - t_b \leq r_b/2$. The sources are holomorphic near $|t - t_b| \leq r_b$, with the operator, RMS and coordinate bounds in (i): forward queries range over the panel, backward queries over training inputs. The partition and radii depend deterministically only on n, m, γ, Y , activations and depth, and

$$J \leq 1 + \beta^{30L} \left(\frac{Ym}{\gamma} \right)^2 \sqrt{\log(en)} + \beta^{16L} \frac{m}{\gamma} \log \left(1 + \frac{4 \log(en)}{\log 2} \right). \quad (28)$$

Proof. We prove stronger intermediate domains with explicitly calculated bounds, then compare them with the displayed interface. All symbols introduced from here through the end of this proof are private to this argument; the subsequent constructions use only the proposition's stated bounds. All probability statements concern one sufficiently large width, with the data, depth, activations and confidence fixed. Write $\lambda = \gamma/m$, $Y = \|y\|_2/\sqrt{m}$, $S = 16Y/\lambda$, and $\ell = \log(en)$. We treat $Y > 0$; zero labels give the stationary zero predictor. For each full or cavity trajectory write $\rho = \|r\|_2/\sqrt{m}$, also at complex times using the Euclidean norm of the complex residual vector. Write $W_0^{(j)} = W^{(j)}(0)$ for initialized matrices. Put

$$b = \max_j |\phi_j(0)|, \quad s = \max\{1, \sup_{j, |\operatorname{Im} z| \leq a/2} |\phi_j'(z)|\}, \quad t_2 = \max\{1, \sup_{j, |\operatorname{Im} z| \leq a/2} |\phi_j''(z)|\}.$$

Thus $1 + b, s, t_2, 16/a \leq \beta$, and the bound used for activation values is $|\phi_j(z)| \leq b + s|z|$, not boundedness. Write $v = x/\sqrt{d}$. Backward fields at a passive input are defined by the same algebraic recursion as at a training input,

$$k^{(L)}(t, v) = w(t), \quad \delta^{(j)}(t, v) = \phi_j'(z^{(j)}(t, v)) \odot k^{(j)}(t, v), \quad k^{(j)} = W^{(j+1)\top} \delta^{(j+1)}.$$

This definition uses no passive label and does not add a training term.

1. Local coefficient estimates and working domains. The following scalars are bounds, not coordinates of a model:

$$\begin{aligned} H_1 &= \max(1, b + 20s), & H_j &= \max(1, b + 10sH_{j-1}), \\ k_L &= H_L, & k_j &= 10sk_{j+1}, & \tau_j &= sk_j, \\ P_1 &= 3, & P_j &= H_{j-1} + 10sP_{j-1} + 1, & f_j &= sP_j, \\ A_* &= 2sP_L + 2s^2 \sum_{j=2}^L k_j P_{j-1}, & D_* &= t_2 \sum_{j=1}^L P_j^2, & H_* &= A_* + t_2 \sum_{j=1}^L P_j^2 k_j. \end{aligned} \quad (29)$$

Set $f_* = \max_j f_j$, $k_* = \max_j k_j$, $H_{\max} = \max_j H_j$, and define

$$\begin{aligned} E_* &= t_2 \sum_{j=1}^L (10s)^{2(j-1)} k_j, & T_* &= 2H_*^2 + 4A_*^2(e^2 - 1), \\ D_0 &= (1 + b + s)(1 + T_* + E_* + s^2 k_*^2), & D_1 &= 576e^3(1 + b + s)D_*^3, \\ C_F &= s(8f_*^2 + H_{\max}^2 + 1), & C_{\text{abs}} &= 8(D_0 + 1), \\ V_1^0 &= \tau_1, & V_j^0 &= \tau_j H_{j-1}^2 + 10sV_{j-1}^0, \\ G_L^0 &= sH_L + 14t_2 V_L^0 + s, & G_j^0 &= 10sG_{j+1}^0 + s^3 k_{j+1}^2 H_j + 14t_2 V_j^0 + s, \\ W_G &= 128(1 + H_{\max} + s \max_j V_j^0 + \max_j G_j^0). \end{aligned} \quad (30)$$

Choose the exponential-budget constants in this order:

$$\eta_* = \min\{1, (1024C_{\text{abs}}W_G)^{-1}\}, \quad \mathcal{B} = 1024e^2L, \quad \Lambda_* = \log(e + \mathcal{B}) + \log(1/\eta_*).$$

$$\begin{aligned} S_* &= \min \left\{ 1, (4A_*)^{-1}, \sqrt{\frac{\eta_*}{4000D_*}}, \sqrt{\frac{\eta_*}{16eD_*\sqrt{2\mathcal{B}}}}, \right. \\ &\quad \left. \sqrt{\frac{\eta_*}{\Lambda_*}}, \left(\frac{\eta_*^3}{D_1\mathcal{B}} \right)^{1/4}, (8D_0C_F)^{-1/2} \right\}. \end{aligned} \quad (31)$$

These quantities depend only on the activation bounds and depth. Further put

$$\begin{aligned}
C_G &= 32 \max(1, H_{\max}, \max_j \tau_j), & K_{\text{src}} &= 16C_{\text{abs}}(1 + C_G), \\
g_* &= \tau_1 + \sum_{j=2}^L \tau_j H_{j-1}, & r_j^* &= P_j g_*, & q_j^* &= f_j g_*, \\
j_j^* &= 20(10s)^{j-1}, & b_j^* &= s j_j^*, \\
e_1 &= t_2 r_1^* P_1, & e_j &= t_2 r_j^* P_j + s(q_{j-1}^* + \tau_j H_{j-1} f_{j-1} + 10e_{j-1}), \\
a_1^* &= t_2 j_1^* P_1 + 2s, & a_j^* &= t_2 j_j^* P_j + s(b_{j-1}^* + 10a_{j-1}^*), \\
T_Q &= 8f_* \max_j (f_j H_* + e_j), & T_J &= 8f_* \max_j a_j^*.
\end{aligned} \tag{32}$$

For a Gaussian multiplier G , define

$$\begin{aligned}
U_1(G) &= 4sK_{\text{src}}, \\
U_j(G) &= 2\{sK_{\text{src}}(H_{j-1}^2 + f_{j-1}^2 + ST_Q + S^2 H_{j-1} q_{j-1}^*) + Gq_{j-1}^* + 1\}, \\
V_1(G) &= 2(2G + 2sK_{\text{src}}S^2 + 1), \\
V_j(G) &= 2\{Gb_{j-1}^* + sK_{\text{src}}S^2(T_J + H_{j-1}b_{j-1}^*) + 1\}.
\end{aligned} \tag{33}$$

Use $U = \max_j U_j(16\sqrt{d+3})$, $V = \max_j V_j(16\sqrt{d+3})$ for the sphere, and $U_{\text{fin}} = \max_j U_j(64)$ for a fixed finite list of inputs. Finally let

$$\mathcal{K} = H_L^2 + S^2 \left(\tau_1^2 + \sum_{j=2}^L \tau_j^2 H_{j-1}^2 \right), \quad D_W = \max\{\tau_1, \max_{j \geq 2} \tau_j H_{j-1}\}. \tag{34}$$

Claim 1 (Local power bounds). *If $Y/\lambda \leq \beta^{-30L}$, then $S \leq \beta^{-26L} \leq S_*$. Moreover*

$$H_j \leq 3\beta^{2j}, \quad P_j \leq \beta^{3j-2}, \quad k_j \leq 3\beta^{4L-2j}, \quad \tau_j \leq 3\beta^{4L-2j+1}, \quad K_{\text{src}} \leq \beta^{21L}, \tag{35}$$

$$U \leq \beta^{26L}\sqrt{d+3}, \quad V \leq \beta^{2L}\sqrt{d+3}, \quad U_{\text{fin}} \leq \beta^{26L}, \quad \mathcal{K} \leq \beta^{14L}. \tag{36}$$

Proof. Unroll the affine recurrences using $10s \leq \beta^2$, $4j \leq \beta^j$, $L, L+1 \leq \beta^{L-1}$, and $\sum_{j=1}^L \beta^{cj} \leq \beta^{cL}/(1 - \beta^{-c})$. They give $H_j \leq 22\beta(10\beta)^{j-1} \leq 3\beta^{2j}$ and $P_j \leq 4j(10\beta)^{j-1} \leq \beta^{3j-2}$. Substitution gives the following bounds; both columns are included to account for the numerical factors.

quantity	intermediate bound	power bound
A_*	$7\beta^{5L-3}$	β^{5L-2}
D_*	$2\beta^{6L-3}$	β^{6L-2}
H_*	$4\beta^{8L-3}$	β^{8L-2}
E_*	$4\beta^{6L-3}$	β^{6L-2}
T_*	$3\beta^{16L-4}$	β^{16L-3}
D_0	$8\beta^{16L-3}$	β^{16L-2}
D_1	$9216e^3\beta^{18L-8}$	β^{18L-2}
C_F	$9\beta^{6L-1}$	β^{6L}
C_{abs}	$65\beta^{16L-3}$	β^{16L-1}

For instance, the summands of A_* 's mixed part, H_* 's curvature part, and E_* are at most $6\beta^{4L+j-3}$, $3\beta^{4L+4j-3}$, $3\beta^{4L+2j-3}$. The inner sum in D_0 is at most $1 + 3\beta^{16L-4} + 4\beta^{6L-3} + 9\beta^{8L-2} \leq 4\beta^{16L-4}$, and $9216e^3 < \beta^6$. Unrolling the two activity recurrences gives

$$V_j^0 \leq 27j\beta^{4L+2j-3}, \quad \max_j V_j^0 \leq \beta^{7L-2}, \quad G_j^0 \leq 66L\beta^{8L-2j-1}, \quad W_G \leq \beta^{9L+1}.$$

In the reverse recurrence the propagated terminal, mixer and curvature contributions are bounded respectively by $380L\beta^{8L-2j-2}$, $27L\beta^{8L-2j-1}$, $379L\beta^{8L-2j-6}$; the constant contribution is at most $2\beta^{2L-2j-1}$. Hence $\eta_*^{-1} \leq \beta^{25L+4}$, $\mathcal{B} \leq \beta^{L+3}$, and $\Lambda_* \leq \beta^{2L}$. The six nonconstant entries of (31) are at least β^{-c} , with exponents

$$5L-1, \quad (31L+6)/2, \quad (63L+12)/4, \quad (27L+4)/2, \quad (94L+13)/4, \quad (22L-1)/2.$$

All are at most $26L$. Also $16\beta^{-30L} \leq \beta^{-26L}$. The remaining recurrences give

$$\begin{aligned} g_* &\leq \beta^{5L-1}, & r_j^* &\leq \beta^{5L+3j-3}, & q_j^* &\leq \beta^{5L+3j-2}, \\ e_j &\leq 2\beta^{5L+6j-4}, & a_j^* &\leq 3\beta^{5j-2}, & T_Q &\leq \beta^{14L-3}, & T_J &\leq \beta^{8L-1}. \end{aligned}$$

For e_j , its three forcing exponents are at most $5L+6j-4$, $5L+3j-4$, $4L+3j-4$, with coefficient nine in the last term; geometric propagation by β^2 gives the displayed factor two. The same calculation gives the factor three for a_j^* . Further $K_{\text{src}} \leq 1552\beta^{20L-2} \leq \beta^{20L+2}$. Now $ST_Q, S^2H_{j-1}q_{j-1}^* \leq 1$, and the bracket in (33) is at most $11\beta^{6L-8}$. Thus for $j \geq 2$

$$U_j(16\sqrt{d+3}) \leq 22\beta^{26L-5} + 32\sqrt{d+3}\beta^{8L-5} + 2, \quad U_j(64) \leq 22\beta^{26L-5} + 128\beta^{8L-5} + 2.$$

The first-layer bound is $4\beta^{20L+3}$. These give the asserted bounds on U, U_{fin} . In V_j , the term containing S^2 is at most one, giving $V_j \leq 64\sqrt{d+3}\beta^{2L-2} + 4$ for $j \geq 2$; the first value is at most $67\sqrt{d+3}$. Finally substitute (35) in (34); $S \leq 1$ and the same geometric sum give $\mathcal{K} \leq \beta^{14L}$. Every comparison above holds already at $L=2, \beta=10$, and improves as either increases. $\square$

Working whole-sphere estimates. The estimates below in fact require only the conclusion of the fitting lemma and $S \leq S_*$, both already implied by the headline assumptions. For the remainder of this proof redefine the working radii by

$$T = 32\ell/\lambda, \quad c_t = \frac{a}{64YSU}, \quad c_q = \min\{1/8, a/(8V)\}, \quad r_t = c_t/\sqrt{\ell}, \quad r_q = c_q/\sqrt{\ell}.$$

For every fixed confidence, eventually with that confidence all forward and backward fields extend holomorphically to a neighborhood of

$$([-r_t, T+r_t] + i[-r_t, r_t]) \times \mathcal{Q}_{r_q}, \quad \mathcal{Q}_r = \{v \in \mathbb{C}^d : v^\top v = 1, \|\text{Im } v\|_2 \leq \sinh r\}.$$

For $d=1$, replace $\mathcal{Q}_r$ by the two points $\{-1, 1\}$. Holomorphy in the query is relative to the complex quadric. Normalized first-layer and hidden operator norms are at most ten, and

$$\|h^{(j)}\|_2/\sqrt{n} \leq H_j, \quad \|k^{(j)}\|_2/\sqrt{n} \leq Sk_j, \quad \|\delta^{(j)}\|_2/\sqrt{n} \leq S\tau_j.$$

For training points, $\max |z_{a,i}^{(j)}| \leq K_{\text{src}}\sqrt{\ell}$ and $\max |k_{a,i}^{(j)}| \leq SK_{\text{src}}\sqrt{\ell}$. In mobility coordinates $\Theta = (W^{(1)}, \sqrt{n}W^{(2)}, \dots, \sqrt{n}W^{(L)}, w)$, put $F_a = nf_n(v_a)$. Then

$$\max_{a,j,i,v} |(D_\Theta z^{(j)}(v) \nabla_\Theta F_a)_i| \leq SU\sqrt{\ell}, \quad \max_{j,i} |\partial_\zeta z_i^{(j)}(u \cos \zeta + v \sin \zeta)| \leq V\sqrt{\ell}$$

for real orthonormal frames (u, v) in the query tube. All four source families

$$h^{(j)}, \quad W_0^{(j)}h^{(j-1)}, \quad \delta^{(j)}, \quad W_0^{(j+1)\top}\delta^{(j+1)},$$

omitting boundary images, have coordinate modulus at most $M_0\sqrt{n}$, where $M_0 = 10 \max_j (H_j, \tau_j)$. The real fitting trajectory and its endpoint are those of Lemma 2.1. The proof below also establishes the finite-query and expanding-domain variants. Its probability step is shared; the domain verifications are given last.

2. Independent stops and deterministic differentiation. Delete a fixed number u of neurons at one layer, preserving the original normalization n in every rectangular matrix. For each deleted interior neuron, its initialized incoming row transpose y_i and outgoing column x_i are independent $N(0, I/n)$, independent of the retained initialization and other omitted pairs. At the first layer the incoming row instead has covariance I_d and evaluates only its own initial preactivation; at the top there is no outgoing root. Use a separate initial test for each cavity, and set its reference coefficient paths identically zero if it fails. All such paths are defined before integrating an omitted root.

For a nested domain D_s , stop each cavity at its own first exit level. Stop every training sample's budget

$$\mathcal{H}_a = \frac{1}{n} \sum_{j,i} \exp\{\eta_*(Z_{a,i}^{(j)} + K_{a,i}^{(j)}/S)\}, \quad Z_{a,i}^{(j)} = \sup_{D_s} |z_{a,i}^{(j)}|, \quad K_{a,i}^{(j)} = \sup_{D_s} |k_{a,i}^{(j)}|, \quad (37)$$

at $\mathcal{B}$ for the full network and $2\mathcal{B}$ for cavities. Also stop operators at ten, contour activity at $2 \int \rho|dt| = S$, and imaginary preactivations at $3a/8$ and $7a/16$, respectively. Impose provisional coordinate and response maxima twice the claimed full bounds and four times the cavity bounds, as well as the passive size and row bounds used below. These have strict margins on the successful prefix.

For rectangles use $D_s = sD_1$ and coordinate clipping P_s . For the query tube also clip its imaginary great-circle coordinate, retaining every real frame. If a cavity stops at s_c , extend a coefficient by $b_c(P_{s_c}z)$. With $a_0(z) = \text{clip}_{[0,T]} \text{Re } z$,

$$P_{s_c}a_0(z) = a_0(P_{s_c}z), \quad b_c(P_{s_c}z) - b_c(P_{s_c}a_0(z)) = b_c(P_{s_c}z) - b_c(a_0(P_{s_c}z)). \quad (38)$$

Clipping is measurable in retained initialization, is Lipschitz, and fixes the successful prefix. Its real anchors traverse a monotone interval and then freeze. Different cavities are extended separately before subtraction. The expanding domain below has the same properties with a uniformly bounded Lipschitz constant. These extensions need not be holomorphic outside their own domains.

Write $\Theta_0 = \Theta(0)$, $\bar{u} = (\Theta - \Theta_0)/S$, $\bar{F}_a = F_a(\Theta_0 + S\bar{u}, e_a)/S$, $\bar{r}_a = \bar{F}_a/n - y_a/S$, and put bars on backward fields divided by S . Here e_a is an additive forward port. The two actual ports of a deletion are $e_a = \sum_i W_{:,i}^{(j+1)} h_{a,i}^{(j)}$ and $\bar{q}_a = \sum_i W_{i,:}^{(j)\top} \bar{\delta}_{a,i}^{(j)}$. The exact retained equation is

$$\dot{\bar{u}} = -\frac{2}{m} \sum_a \bar{r}_a \left(\nabla_{\bar{u}} \bar{F}_a + D_{\bar{u}} h_a^{(j-1)\top} \bar{q}_a \right). \quad (39)$$

At the top add $n^{-1} \sum_i \bar{u}_{w,i} h_{a,i}^{(L)}$ to $\bar{r}_a$, multiplying both terms. The reverse port does not enter the forward residual. The gradient blocks of $\bar{F}_a$ are $S\bar{\delta}_a^{(1)} v_a^\top$, $S\bar{\delta}_a^{(j)} h_a^{(j-1)\top}/\sqrt{n}$, and $h_a^{(L)}$.

The physical recurrences and linear activation growth give (29)'s RMS bounds whenever the operator and activity stops hold. For augmented mobility/port directions U , write $Z_a^{(j)}U = D_{z_a}^{(j)}[U]$. The exact derivative recursion is

$$\begin{aligned} Z_a^{(1)}U &= U_{W^{(1)}}v_a + U_{e^{(1)}}, \\ Z_a^{(j)}U &= U_{\sqrt{n}W^{(j)}}h_a^{(j-1)}/\sqrt{n} + W^{(j)}\text{diag}(\phi'_{j-1})Z_a^{(j-1)}U + U_{e^{(j)}}. \end{aligned} \quad (40)$$

It is bounded by P_j ; its activated version is bounded by f_j . The Hessian, with all activations

evaluated at the reference state, is

$$\begin{aligned}
D^2 F_a[U, V] &= U_w^\top \text{diag}(\phi'_L) Z_a^{(L)} V + V_w^\top \text{diag}(\phi'_L) Z_a^{(L)} U \\
&\quad + \sum_j (Z_a^{(j)} U)^\top \text{diag}(k_a^{(j)} \phi_j'') Z_a^{(j)} V \\
&\quad + \sum_{j \geq 2} \frac{\delta_a^{(j)\top}}{\sqrt{n}} \{U_{\sqrt{n}W^{(j)}} \text{diag}(\phi'_{j-1}) Z_a^{(j-1)} V + (U \leftrightarrow V)\}.
\end{aligned} \tag{41}$$

For a rectangular matrix let $\|B\|_{p,n} = n^{-1/p} (\sum_i \sigma_i(B)^p)^{1/p}$. The budget implies $\|k_a^{(j)}\|_p / n^{1/p} \leq (S/\eta_*) p (2\mathcal{B})^{1/p}$, because $x^p \leq (p/e)^p e^x$. Each noncurvature Hessian block factors through at most n coordinates, and each curvature block contains one carrier diagonal. Matrix ideal inequalities in (41) therefore give

$$\begin{aligned}
\|D^2 F_a\|_{p,n} &\leq A_* + (SD_*/\eta_*) p (2\mathcal{B})^{1/p}, & \|D^2 F_a\|_{\text{HS}}/\sqrt{n} &\leq H_*, \\
\|D^2 F_a\|_{\text{op}} &\leq A_* + (SD_*/\eta_*) \log(2n\mathcal{B}).
\end{aligned} \tag{42}$$

These bounds also cover $D_\Theta \delta$, since $\partial_{e^{(j)}} F_a = \delta_a^{(j)}$. Residual mixtures preserve the bounds by $\sum_a |r_a|/(m\rho) \leq 1$.

Suppose for now that products of the negative-Gram propagators on ordered disjoint contour pieces have norm at most two, and $\rho/S \leq \lambda/8$. The variational generator is

$$-\frac{2}{mn} \sum_a g_a g_a^\top - \frac{2}{m} \sum_a r_a D^2 F_a, \quad g_a = \nabla_\Theta F_a.$$

Its full propagator obeys

$$\|J(t, s)\| \leq 2 \exp\{SA_* + (S^2 D_*/\eta_*) \log(2n\mathcal{B})\} \leq Cn^\kappa, \quad \kappa = 1/4000. \tag{43}$$

Here and only in qualitative local estimates, C and powers of ℓ may depend on the fixed problem and deletion count. The coefficient of $\log n$ is bounded by (31); it has not been hidden in a width threshold.

Claim 2 (Local maps, remainders, and passive observables). *On the independent stops just specified, with contours of length $O(\ell)$, all first insertion maps have norm $n^\kappa \text{poly}(\ell)$. Their dependence on the supremum norm of scalar controls is Lipschitz with that bound, and their raw terminal/time/frame derivatives are bounded by $\sqrt{n} n^\kappa \text{poly}(\ell)$. On the uniform Gaussian event proved below, the retained full/cavity differences in preactivations, scaled carriers, responses and angular derivatives are at most $n^{-1/30}$ per coordinate and at most $2n^{1/100}$ in Euclidean norm, for every fixed deletion count.*

Proof. For deterministic controls $\alpha_{a,i}, b_{a,i}$, use ports $e_a = \sum_i x_i \alpha_{a,i}$, $\bar{q}_a = \sum_i y_i b_{a,i}$. The three port derivatives of (39) at zero ports are

$$-\frac{2}{mn} \nabla_{\bar{u}} \bar{F}_a \bar{\delta}_a^{(j+1)\top}, \quad -\frac{2}{m} \bar{r}_a (D_{\bar{u}} \bar{\delta}_a^{(j+1)})^\top, \quad -\frac{2}{m} \bar{r}_a (D_{\bar{u}} h_a^{(j-1)})^\top.$$

The first term accounts for the changed residual. Integrating them against (43), using gradient norm $O(\sqrt{n})$, response RMS, and contour length $O(\ell)$, gives an insertion-map bound $\sqrt{un}^\kappa O(\ell^2)$.

The differentiated forward/backward recursions are exactly

$$\begin{aligned} z_{[1]}^{(j)} &= SV_{\sqrt{n}W^{(j)}} h^{(j-1)}/\sqrt{n} + W^{(j)} h_{[1]}^{(j-1)} + e_{[1]}^{(j)}, \\ h_{[1]}^{(j)} &= \phi'_j \odot z_{[1]}^{(j)}, \\ \bar{k}_{[1]}^{(j)} &= W^{(j+1)\top} \bar{\delta}_{[1]}^{(j+1)} + SV_{\sqrt{n}W^{(j+1)}} \bar{\delta}_{[1]}^{(j+1)}/\sqrt{n} + \bar{q}_{[1]}^{(j)}, \\ \bar{\delta}_{[1]}^{(j)} &= \phi'_j \odot \bar{k}_{[1]}^{(j)} + \phi''_j \odot \bar{k}_{[1]}^{(j)} \odot z_{[1]}^{(j)}. \end{aligned}$$

At the first layer replace the matrix product by $SV_{W^{(1)}}v$; at the top $\bar{k}_{[1]} = V_w$. A reference carrier costs $O(\ell)$ once in the inhomogeneous last term. Homogeneous propagation costs $10s$. Induction gives the asserted map bounds; subtracting controls gives their Lipschitz bounds because the linear variational equation is linear in these controls. Differentiating these displayed recursions introduces only the terms $\phi''z_t$, $\phi'''z_t z_{[1]} \bar{k} + \phi''z_{[1]} \bar{k}_t + \phi'' \bar{k} z_{[1],t}$, and ordinary product derivatives. There is at most one raw reference coordinate derivative, bounded by $\sqrt{n} \text{poly}(\ell)$. This proves the derivative claim without a width power depending on depth. Cauchy's formula on the strict strip bounds ϕ''' by $4\beta^2$.

Here are the nonlinear estimates needed to justify this linearization. Decompose the augmented displacement into $(V, e) + (U, 0)$, and let N bound its linear Euclidean variations, d_0 their coordinate maxima, and u_0 bound $\|U\|$. Put

$$R = d_0 N + d_0 u_0 + u_0^2, \quad P = (N + u_0)^2 / \sqrt{n}.$$

Assume $N \geq 1$, $d_0, u_0 \leq 1$, $N + u_0 \leq \sqrt{n}$, and $u_0 + R + P \leq 1$. If $E_g = g_1 - g_0 - Dg_0(V + U, e)$ and $g_i = \phi'(z_i)$ in the third line below, the exact identities are

$$\begin{aligned} E_{z^{(j)}} &= W_1^{(j)} E_{h^{(j-1)}} + S \Delta \bar{u}_{\sqrt{n}W^{(j)}} h_{[1]}^{(j-1)} / \sqrt{n}, \\ E_{\bar{k}^{(j)}} &= W_1^{(j+1)\top} E_{\bar{\delta}^{(j+1)}} + S \Delta \bar{u}_{\sqrt{n}W^{(j+1)}} \bar{\delta}_{[1]}^{(j+1)} / \sqrt{n}, \\ E_{\bar{\delta}} &= g_1 \odot E_{\bar{k}} + (g_1 - g_0) \odot \bar{k}_{[1]} + \bar{k}_0 \odot (g_1 - g_0 - \phi''(z_0) z_{[1]}). \end{aligned} \tag{44}$$

The first and top boundary remainders vanish. Forward and backward first derivatives are bounded by $J_f = \beta^{10L}$ and $J_b = \beta^{20L}(1 + M)$, where $M \geq 1$ is a common provisional bound for the absolute preactivations and scaled carriers of the reference and full paths, including the deleted-neuron values that define the scalar controls. Such an envelope is $\text{poly}(\ell)$ under the stated stops. The localized product estimate is $\|z_{[1]}^{\odot 2}\|_2 \leq 3J_f^2 R$. Insert $z_0 + z_{[1]}$ into activation subtraction. Taylor's formula and (44), summed by geometric propagation, give

$$\max(\|E_z\|_2, \|E_h\|_2) \leq \beta^{30L}(R + P), \quad \max(\|E_{\bar{k}}\|_2, \|E_{\bar{\delta}}\|_2) \leq \beta^{60L}(1 + M)(R + P). \tag{45}$$

Indeed the two gate cross terms cost at most $\beta[3J_f J_b R + \beta^{30L}(R + P)(d_0 + J_b u_0)]$ and $M[\beta\|E_z\| + \beta^2\|z_{[1]}^{\odot 2}\|/2]$; the matrix cross term costs $J_b P$. No changed carrier is used as a presumed reference bound.

The hidden gradient remainder is

$$\frac{S}{\sqrt{n}} [E_{\bar{\delta}} h_1^\top + \bar{\delta}_{[1]} (h_1 - h_0)^\top + \bar{\delta}_0 E_h^\top],$$

bounded by $\beta^{70L}(1 + M)(R + P)$. The Hessian identity along partial joining segments bounds the residual difference by $\beta^{13L}(N + u_0)/\sqrt{n}$ and its linearization error by $\beta^{87L}(1 + M)(N + u_0)^2/(2n)$. For a product of residual and gradient, its exact remainder is $\bar{r}_0 E_g + (\Delta \bar{r} - D\bar{r}_0 \Delta) g_0 + \Delta \bar{r} (g_1 - g_0)$.

For the reverse force one additionally needs a lower adjoint probe. For an omitted root y_i , include all derivatives of $y_i^\top h^{(j-1)}$ in the Gaussian coordinate event. For $q = \sum_i y_i b_{a,i}$, let $Q = 2u\beta M$, $P_q = u\beta M d_0$. Subtracting its adjoint recursion gives

$$\max_j \|k_{q,1}^{(j)} - k_{q,0}^{(j)}\|_2 \leq \beta^{20L}(P_q + Q/\sqrt{n})(N + u_0),$$

since the changed gate multiplies the reference probe maximum, the changed matrix costs $Q(N + u_0)/\sqrt{n}$, and propagation costs $11s$. Integrating the corresponding Hessian gives

$$\|(Dh_1 - Dh_0)^\top q\|_2 \leq \beta^{50L}(P_q + Q/\sqrt{n})[N + u_0 + (N + u_0)^2].$$

Combining these estimates in (39) yields the vector-field remainder bound

$$\beta^{240L}(1+u)(1+M)\{\bar{\rho}[R + d_0(N^2 + Nu_0 + u_0^2)] + (1+\bar{\rho})P\}, \quad \bar{\rho} = \rho/S. \quad (46)$$

This also holds on the complex joining segments: the sufficient gate $d_0 + \beta^{10L}u_0 + \beta^{30L}(R+P) < a/32$ keeps the segments and auxiliary Taylor points inside $15a/32$. A first-exit argument proves the gate's use before Taylor expansion; the factor β^{240L} pays for the Cauchy third-derivative bound. Scaling introduced no inverse power of S : in the effective network with readout $\bar{u}_w$, the affine change of variables multiplies hidden directions by $S \leq 1$.

For every existing outgoing column, and for an incoming row at a layer $j \geq 2$, integrating the omitted-parameter equations gives

$$\|\Delta W_{:,i}^{(j+1)}\| \leq S^2(\max_r \tau_r)(b + sM)/\sqrt{n}, \quad \|\Delta W_{i,:}^{(j)}\| \leq S^2 H_{\max} sM/\sqrt{n}.$$

At the first layer the incoming-row estimate is instead $\|\Delta W_{i,:}^{(1)}\| \leq sS^2M$. There is no retained reverse port below that layer: this row's motion is included in the deleted neuron's scalar controls, rather than in the learned retained-port remainder. Thus every existing learned port entering the retained equation still induces a force at most $S^2\beta^{260L}(1+u)^2(1+M)^3(1+\bar{\rho})/\sqrt{n}$. The top residual offset is at most $uM(b+sM)/n$; its product with both retained terms has the same enclosing bound.

It remains to specify all passive quantities in the local claim. For evaluated input q and training driver a , let $C_q^{(j)} = D_{\Theta} h_q^{(j)}$, $\bar{R}_{qa}^{(j)} = S^{-1} D_{\Theta} z_q^{(j)} g_a$, and $\bar{Q}_{qa}^{(j)} = \phi'_j \odot \bar{R}_{qa}^{(j)}$. Then

$$\bar{R}_{qa}^{(1)} = \bar{\delta}_a^{(1)} v_a^\top q, \quad \bar{R}_{qa}^{(j)} = \bar{\delta}_a^{(j)} c_{aq}^{(j-1)} + W^{(j)} \bar{Q}_{qa}^{(j-1)}, \quad c_{aq}^{(j-1)} = h_a^{(j-1)\top} h_q^{(j-1)}/n. \quad (47)$$

Angular fields satisfy $J^{(1)} = W^{(1)} q'$, $J^{(j)} = W^{(j)} I^{(j-1)}$, $I = \phi' \odot J$. Differentiate these product formulas: the four response terms are $\bar{\delta}_{[1]} c + \bar{\delta} c_{[1]}$, $SV_{\sqrt{n}W} \bar{Q}/\sqrt{n}$, and $W \bar{Q}_{[1]}$, together with the omitted port $\sum_i x_i \bar{Q}_{qa,i}$. The activated variation adds $\phi'' \bar{R} z_{[1]}$. Angular variations have the same matrix and gate terms, with omitted port $\sum_i x_i I_i$. Below a deletion there is also the direct reverse observable

$$S^{-1}(Q_{qa,\text{full}} - C_q g_a) = C_q C_a^\top \bar{q}_a. \quad (48)$$

Include the forward images of $C_a^{0\top} y_i$ in the Gaussian event; their coefficient maps are independent of that root.

Product subtraction gives

$$|c_{[1]}| \leq \beta^{10L}(N + u_0)/\sqrt{n}, \quad |E_c| \leq \beta^{40L}[(R + P)/\sqrt{n} + (N + u_0)^2/n].$$

The exact response remainder is $E_{\bar{\delta}} c_1 + \bar{\delta}_{[1]}(c_1 - c_0) + \bar{\delta}_0 E_c + W_1 E_{\bar{Q}} + \Delta W \bar{Q}_{[1]} + E_{\text{ports}}$; its gate remainder is the third line of (44) with $\bar{k}$ replaced by $\bar{R}$. The angular case uses that same identity

with J . Reference response RMS cancels the $1/\sqrt{n}$ in E_c . For $\xi = C_a^{0\top} y_i$, forward directional subtraction gives $\|(C_q^1 - C_q^0)\xi\| \leq \beta^{50L}(d_0 + n^{-1/2})(N + u_0)$. The other changed factor in (48) uses the adjoint-probe estimate above. Their sum is bounded by

$$\beta^{70L}(1+u)(1+M)(d_0 + n^{-1/2})[N + u_0 + (N + u_0)^2].$$

Missing feature products cost $u(b + sM)^2/n$, and learned ports cost $n^{-1/2} \text{poly}(\ell)$. These enumerate all additional remainders. Their time and frame derivatives have the same product structure already checked. The probabilistic and first-exit conclusion follows in the next paragraph. $\square$

The uniform Gaussian event and closing the local comparison. For fixed controls, a pairing of an omitted root with a linear variation has the form $g^\top Rg$, where $R = (E_i^\top L + L^\top E_i)/2$. Consequently $\|R\| \leq \|L\|$, and only its same-root diagonal block has a mean $\text{tr } R/n$. Cross-root forms are centered. On $\|g\| \leq 2\sqrt{2u}$, interpolation of a centered form costs at most $10u\|\Delta R\|$, including its trace. All stopped scalar controls have polynomial-logarithmic amplitudes and speeds $\sqrt{n} \text{poly}(\ell)$; this follows by differentiating the forward, backward and response recursions above. A scalar control with amplitude M_c , speed $D_c\sqrt{n}$ on a contour of length T_c has an $n^{-1/8}$ -net of logarithmic cardinality at most

$$(2 + 8T_c D_c n^{5/8}) \log(1 + 8M_c n^{1/8}).$$

Indeed sample it at spacing $n^{-1/8}/(8D_c\sqrt{n})$, round on mesh $n^{-1/8}/4$, and interpolate. Apply this separately to the finitely many real and imaginary controls. They are deterministic one-dimensional histories on a prescribed contour; no arbitrary two-dimensional analytic continuation of controls is assumed.

After eventual absorption of the fixed logarithmic factors, all linear and quadratic maps have norm at most $n^{1/200}$. A linear coordinate has variance at most $n^{-99/100}$. For a real symmetric form, diagonalization and

$$\mathbb{E} e^{t(G^\top R G - \text{tr } R)} = \prod_i e^{-t\lambda_i} (1 - 2t\lambda_i)^{-1/2}, \quad -\log(1 - v) - v \leq \frac{v^2}{2(1 - |v|)}$$

give, for $g \sim N(0, I/n)$,

$$\Pr\{|g^\top Rg - \text{tr } R/n| > t\} \leq 2 \exp \left[-c \min \left\{ \frac{n^2 t^2}{\|R\|_{\text{HS}}^2}, \frac{nt}{\|R\|_{\text{op}}} \right\} \right]$$

with an absolute $c > 0$, by applying exponential Markov with $|2t\lambda_i| \leq 1/2$ and optimizing the parameter. Here $\|R\|_{\text{HS}} \leq \sqrt{2un}\|R\|$. At threshold $n^{-1/10}/8$, real/imaginary splitting gives $4e^{-n^{.78}}$ eventually. Root-norm failure is at most e^{-un} . The control interpolation error is $n^{-1/8+1/200} \text{poly}(\ell) = o(n^{-1/10})$. A terminal/time/frame mesh of spacing $n^{-2}/\text{poly}(\ell)$ has polynomial size; the sphere frame has $2d$ ambient coordinates. Together with all coordinates, probes and at most uLn^u deletion sets, the logarithmic union cost is eventually less than $n^{.65}$. Thus the uniform event fails with probability at most $e^{-n^{.7}}$. Only after this event is established are the actual adaptive controls substituted. Root-norm bounds also make the total linear vector variation at most $n^{1/100}$, distributing a fixed share among the finitely many components.

Let U be the retained scaled displacement minus its linear variation. It satisfies Duhamel's formula with (46), the learned ports, and the top offset. Set $N = n^{1/100}$, $d_0 = n^{-1/10}$, $u_0 = n^{-1/25}$. On a first-exit prefix $\|U\| \leq u_0$,

$$\|U\| \leq n^\kappa \text{poly}(\ell)[n^{-.08} + u_0^2 + n^{-.48} + n^{-1/2}] = o(u_0). \quad (49)$$

The activity integral bounds the residual-weighted terms and the contour length bounds the others. This strictly improves the stop. The explicit remainders then give coordinate discrepancies $n^{-1/30}$ and vector discrepancies $2n^{1/100}$, including passive responses and angular fields. The joining-strip gate also holds eventually. For every deterministic frozen control history, the retained linear variations are centered Gaussian images conditional on the cavity. After substituting adaptive controls, we use only the uniform bounds already proved; we assert neither Gaussianity nor centering of the resulting variations. Pairing a frozen-control variation again with an omitted root produces a quadratic form. Its nonzero trace is retained before adaptive substitution, as treated next. This completes Claim 2.

3. Trace absorption and Gaussian moments.

Every noncentered contraction. Throughout this paragraph, conditional means and centered forms refer to deterministic frozen controls, conditional on the cavity. The resulting trace bounds hold uniformly over those controls and are then evaluated at the actual adaptive controls. For one interior omitted neuron work in unscaled mobility coordinates, and set $C_a = D_{\Theta} h_a^{(j-1)}$, $B_a = D_{\Theta} \delta_a^{(j+1)}$, $E_a = D_{e_a^{(j+1)}} \delta_a^{(j+1)}$, all at the zero-source cavity. With $\alpha_b = h_{b,i}^{(j)}$, $b_b = \delta_{b,i}^{(j)}$, and $d_b = \delta_b^{(j+1)}$, the linear parameter variation is

$$V(t) = -\frac{2}{m} \sum_b \int_0^t J(t, s) \{g_b(s) d_b(s)^\top x_i \alpha_b(s)/n + r_b(s) B_b(s)^\top x_i \alpha_b(s) + r_b(s) C_b(s)^\top y_i b_b(s)\} ds.$$

The scalar expansions are

$$z_{a,i} = y_i^\top h_a^0 + y_i^\top C_a V + (\Delta W_{i,:}) h_a + o(1), \quad k_{a,i} = x_i^\top \delta_a^0 + x_i^\top B_a V + x_i^\top E_a x_i \alpha_a + (\Delta W_{:,i})^\top \delta_a + o(S).$$

The errors follow in scaled coordinates from Claim 2. The four nonzero conditional means are

$$\begin{aligned} & -\frac{2}{m} \sum_b \int r_b b_b \frac{\text{tr}(C_a J C_b^\top)}{n} ds, & -\frac{2}{m} \sum_b \int r_b \alpha_b \frac{\text{tr}(B_a J B_b^\top)}{n} ds, \\ & \alpha_a \text{tr}(E_a)/n, & -\frac{2}{mn^2} \sum_b \int \alpha_b d_b^\top B_a J g_b ds. \end{aligned} \quad (50)$$

All other forms are centered same-root or independent-root forms covered by the uniform Gaussian event. The last displayed mean is $n^{-1+\kappa} \text{poly}(\ell) = o(1)$, since $\|d_b\|, \|g_b\| = O(\sqrt{n})$.

Expand J around its negative-Gram base. A term with h residual-Hessian insertions has ordered activity integral at most $S^h/h!$. The product of its base propagators costs at most two. Normalized Schatten Hölder, with exponent $h+2$ on both endpoint Hessian blocks and the h insertions, gives

$$\frac{2S^h}{h!} [A_* + (SD_*/\eta_*)(h+2)(2\mathcal{B})^{1/(h+2)}]^{h+2}.$$

The zero-insertion term is at most $2H_*^2$. Split the other powers by $(x+y)^{h+2} \leq 2^{h+1}(x^{h+2} + y^{h+2})$. The bounded part sums to $4A_*^2(e^{2SA_*} - 1)$; the carrier part is

$$\frac{8e^2 D_*^2 S^2 \mathcal{B}}{\eta_*^2} \sum_{h \geq 1} (2e D_* S^2 / \eta_*)^h (h+2)^2 \leq \frac{576e^3 D_*^3 \mathcal{B} S^4}{\eta_*^3}.$$

Indeed $h! \geq (h/e)^h$, $(1+2/h)^h \leq e^2$, and $\sum_{h \geq 1} q^h (h+2)^2 \leq 36q$ for $q \leq 1/2$, by differentiating the geometric series. The allowance gives this last condition. For forward endpoints give one exponent

two, the other infinity, and the h Hessians exponent $2h$. The two series are bounded by $e^{2SA^*} - 1$ and $\sqrt{2\mathcal{B}} \sum_{h \geq 1} (4eD_*S^2/\eta_*)^h$, each at most one by (31). Including the zero term,

$$|\text{tr}(C_a J C_b^\top)|/n \leq 8f_*^2. \quad (51)$$

Reciprocal exponents sum to one, giving normalization exactly n^{-1} , independent of parameter-space dimension. The direct port Hessian is $E_a = \sum_{r>j} T_{r,j+1}^\top \text{diag}(k_a^{(r)} \phi_r'') T_{r,j+1}$, with $\|T_{r,j+1}\| \leq (10s)^{r-j-1}$; hence carrier RMS gives $|\text{tr } E_a|/n \leq SE_*$.

Let $G_{h,a,i}$, $G_{\delta,a,i}$ denote suprema of the corresponding cavity Gaussian pairings. Set $Z_i = (m^{-1} \sum_a Z_{a,i}^2)^{1/2}$ and $U_i = (m^{-1} \sum_a (K_{a,i}/S)^2)^{1/2}$. At every driving time Cauchy–Schwarz gives $m^{-1} \sum_b |r_b h_{b,i}| \leq \rho(b + sZ_i)$ and $m^{-1} \sum_b |r_b \delta_{b,i}| \leq \rho s S U_i$. Integrating the rank-one updates and pairing retained factors in RMS, the learned outgoing column costs $s^2 k_*^2 S^3$ times its driving amplitude, and the learned incoming row costs $s S^2 U_i H_{j-1}^2$; at the first layer the latter is $s S^2 U_i$. Thus (50) gives

$$\begin{aligned} K_{a,i}/S &\leq G_{\delta,a,i}/S + (D_0 + D_1 \mathcal{B} S^4/\eta_*^3)(1 + Z_{a,i} + Z_i) + o(1), \\ Z_{a,i} &\leq G_{h,a,i} + C_F S^2 U_i + o(1). \end{aligned} \quad (52)$$

The direct external trace keeps the evaluated sample's amplitude $Z_{a,i}$; only integrated terms use the driving-sample RMS. At the first layer $G_{h,a,i} = |W_{0,i}^{(1)} v_a|$; at the top $\sup |w_i|/S \leq b + sZ_i$ and there is no outgoing root. These endpoint formulas obey the same enclosing inequalities.

Put $\overline{G}_h = (m^{-1} \sum_a G_{h,a,i}^2)^{1/2}$ and $\overline{G}_\delta = (m^{-1} \sum_a (G_{\delta,a,i}/S)^2)^{1/2}$. The backward coefficient is at most $2D_0$ and $D = C_F S^2 \leq 1/(8D_0)$. Taking sample RMS gives

$$U_i \leq \overline{G}_\delta + 2D_0(1 + 2Z_i) + o(1), \quad Z_i \leq \overline{G}_h + D U_i + o(1).$$

Since $4D_0 D \leq 1/2$, solve these two inequalities and substitute back sample by sample to obtain

$$Z_{a,i} + K_{a,i}/S \leq C_{\text{abs}}(1 + G_{h,a,i} + G_{\delta,a,i}/S + \overline{G}_h + \overline{G}_\delta) + o(1). \quad (53)$$

This coefficient is independent of sample and deletion counts.

Real reference moments. Parameterize a real stopped cavity by its own normalized activity $u = 2 \int_0^t \rho/S \in [0, 1]$, freezing after its endpoint. For $v = |u - u'|$, integrating the updates gives

$$\begin{aligned} \|\Delta w\|_2/\sqrt{n} &\leq H_L S v, & \|\Delta W^{(1)}\|_F/\sqrt{n} &\leq \tau_1 S^2 v, \\ \|\Delta W^{(j)}\|_F &\leq \tau_j H_{j-1} S^2 v, & \|\Delta z_a^{(j)}\|_2/\sqrt{n} &\leq V_j^0 S^2 v. \end{aligned}$$

The feature difference has the last bound multiplied by s . Split a reference carrier at SR . Its high part satisfies $\|k\mathbf{1}_{|k|>SR}\|_2/\sqrt{n} \leq (4S/\eta_*)\sqrt{2\mathcal{B}}e^{-\eta_* R/4}$, by the budget and $x^2 e^{-x/2} \leq 16$. Choose $R = (4/\eta_*) \log[8\sqrt{2\mathcal{B}}/(\eta_* v)]$ when $v > 0$. The high changed-gate part is at most $s S v$; the low part is at most $t_2 V_j^0 S^3 R v$. Since $R \leq \eta_*^{-1}(10\Lambda_* + 4 \log(1/v))$, $S^2 \Lambda_*/\eta_* \leq 1$, and $v \log(1/v) \leq 2\sqrt{v}/e$, the low part is bounded by $14t_2 V_j^0 S \sqrt{v}$. At the top add $s H_L S v$; below it add the changed-matrix term $s^3 k_{j+1}^2 H_j S^3 v$ and propagated difference. This proves

$$\|\delta_a^{(j)}(u) - \delta_a^{(j)}(u')\|_2/(S\sqrt{n}) \leq G_j^0 \sqrt{|u - u'|}. \quad (54)$$

The top carrier was included; no coordinate readout bound was used.

Conditioned on the cavity, a Gaussian reference has bounded initial variance and this square-root activity modulus. Grids of spacing 4^{-q} have at most $2 \cdot 4^q$ increments of standard deviation at most $2G2^{-q}$. Gaussian tails at thresholds proportional to $2^{-q}\sqrt{z^2 + 4(q+1)}$, summed over levels and the initial value, give

$$\Pr\{X > W_G(1+z) \mid \text{cavity}\} \leq 4e^{-z^2}, \quad \mathbb{E}[e^{X^2/(16W_G^2)} \mid \text{cavity}] < 2.$$

The factor 128 in W_G bounds the geometric threshold sums. For the moment, $(1+z)^2 \leq 2(1+z^2)$ and tail integration give $\mathbb{E}e^{Z^2/8} \leq 1 + 4/7$, with $e^{1/8}(1 + 4/7) < 2$. Jensen gives the same bound for sample RMS: $e^{\bar{X}^2/(16W_G^2)} \leq m^{-1} \sum_a e^{X_a^2/(16W_G^2)}$, without sample independence. Completing the square and Hölder for the four terms in (53) give the one-neuron moment base

$$2 \exp\{qC_{\text{abs}} + 64q^2C_{\text{abs}}^2W_G^2\} < 4, \quad 0 \leq q \leq 8\eta_*. \quad (55)$$

The chosen η_* verifies the last inequality.

Claim 3 (Local Gaussian rectangle bound). *Let a deterministic complex coefficient map on a rectangle of sides at most H satisfy $\sup_z \|b_z\| \leq D$ and $\|b_z - b_{z'}\| \leq K\|z - z'\|$. For a standard real Gaussian vector G , put $X = \sup_z |G^\top b_z|$. Then*

$$M = 256D\sqrt{\log(e + HK/D)}, \quad \log \mathbb{E}e^{qX} \leq qM + 2q^2D^2 \quad (q \geq 0).$$

For sample RMS with the same bounds, replace M by $M + 2D$. If $D = 0$, the process is zero. For covariance I/n , divide coefficient norms by $\sqrt{n}$.

Proof. Use dyadic coefficient meshes with spacing proportional to $D2^{-j}/K$, having at most $4(1 + HK/D)^2 4^j$ points. Connect each point to the preceding mesh; increment standard deviations are at most $3D2^{-j}$. Gaussian exponential moments give $\mathbb{E} \max_{r \leq N} |Z_r| \leq \sigma \sqrt{2 \log(2N)}$ when standard deviations are at most σ . Summing uses $\sum_{j \geq 1} 2^{-j} = 1$, $\sum_{j \geq 1} 2^{-j} \sqrt{j} \leq 2$. A factor 128 covers the real process, and real/imaginary splitting gives 256.

An a -Lipschitz Gaussian function satisfies $\log \mathbb{E}e^{t(F - \mathbb{E}F)} \leq a^2 t^2/2$ and $\text{Var } F \leq a^2$. Here is a derivation. For $P_t g(x) = \mathbb{E}g(e^{-t}x + \sqrt{1 - e^{-2t}}G)$, differentiating invariant entropy and Gaussian integration by parts give

$$\text{Ent}_\gamma(g) = \int_0^\infty \mathbb{E} \frac{|\nabla P_t g|^2}{P_t g} dt \leq \frac{1}{2} \mathbb{E} \frac{|\nabla g|^2}{g}.$$

Use $\nabla P_t g = e^{-t} P_t \nabla g$ and weighted Cauchy–Schwarz for the inequality. Apply it to $g = e^{uF}$ and integrate the resulting derivative inequality for $u^{-1} \log \mathbb{E}e^{uF}$. Truncation and smoothing extend the calculation to Lipschitz functions; their linear growth supplies exponential integrability. Differentiating at zero gives the variance. Here X is at most $2D$ -Lipschitz, proving the estimate. The sample RMS is also $2D$ -Lipschitz; its mean is at most $M + 2D$ by the variance bound and Cauchy–Schwarz. Independence among sample processes is unnecessary. $\square$

Initial cavities and maximum improvements. The initialization event of Lemma 2.1 applies jointly to fixed-size rectangular cavities. To see why weak covariance estimates need not be union-bounded over deletion sets, zero-embed their features. On the full initial operator event, conditional fresh-row Gaussian tails give initial training maxima $Z_0 = C\sqrt{\ell}$ with probability tending to one. Deleting u coordinates changes each later initialized feature vector by at most

$$D_{\text{init}} = (8s)^L \sqrt{u}(b + sZ_0) = O_u(\sqrt{\ell}).$$

Normalized feature and singular-value changes therefore vanish uniformly over deletion sets, transferring the strict fitting margins. For coordinate localization above a deletion, project the complete lower feature difference onto this deterministic ball before pairing with the independent current row. At threshold $n^{-1/10}$, the tail $2\exp[-n^{4/5}/(2D_{\text{init}}^2)]$ pays for the $O_u(n^{u+1})$ tests. A cavity's covariance retains normalization n ; it is not identified with the full covariance. Initial budgets are below $\mathcal{B}/2$ with probability tending to one: their conditional means are below $2e^{\eta_*^2 H_{\max}^2/2} < 3$, their second exponential moments are bounded, and conditional Chebyshev applies layer by layer. The coordinate-small deletion comparison transfers this event to all cavities. Their own initial tests remain part of each globally extended reference's definition.

Under the provisional stops, Gaussian coefficient RMS is bounded by $H_{\max}$ or $S \max_j \tau_j$. A mesh of spacing n^{-2} has polynomial size, with vanishing off-grid error from the normalized coefficient derivative bounds. Complex Gaussian splitting has tail $4e^{-u^2/(4\sigma^2)}$. The multiplier C_G , followed by (53), improves the full training stops to

$$\max |z_{a,i}^{(j)}| \leq K_{\text{src}} \sqrt{\ell}, \quad \max |k_{a,i}^{(j)}| \leq SK_{\text{src}} \sqrt{\ell}. \quad (56)$$

There are at most n^7 tests eventually on the original rectangle and n^{10} on the expanding finite-panel rectangle; the factor 32 leaves strict margin. This step uses Gaussian tails on a mesh, not already removed budgets or Gaussian supremum moments.

For response and angular stops, differentiate $Q_{qa} = C_q g_a$: $DQ_{qa} = C_q Dg_a + (DC_q)[\cdot]g_a$. The first term has normalized Hilbert–Schmidt norm at most $f_j H_*$. Curvature, changed matrix, changed feature and propagation in the second give respectively $t_2 r_j^* P_j$, sq_{j-1}^* , $s\tau_j H_{j-1} f_{j-1}$, $10se_{j-1}$. These are the recurrence for e_j . Angular fields give the recurrence for a_j^* , including the first-layer map $U_{W(1)} q'$ of normalized norm at most two. Apply (51) with these endpoints. The four noncentered response terms are exactly the direct feature pairing, the direct reverse term (48), the integrated mixed endpoint, and the learned incoming row. After division by S , their coefficients are

$$sK_{\text{src}} H_{j-1}^2, \quad sK_{\text{src}} f_{j-1}^2, \quad sK_{\text{src}} ST_Q, \quad sK_{\text{src}} S^2 H_{j-1} q_{j-1}^*.$$

The centered incoming pairing contributes Gq_{j-1}^* . Angular terms are Gb_{j-1}^* and $sK_{\text{src}} S^2 (T_J + H_{j-1} b_{j-1}^*)$. Thus the factor two in (33) strictly improves these stops. For sphere frames, at most n^{4d+10} mesh points require $G = 16\sqrt{d+3}$; polar normalization of nearby frames gives bounded frame derivatives for interpolation. A fixed finite input list has no frame mesh and at most n^{10} tests eventually; $G = 64$ gives failure $4n^{10}e^{-1024\ell}$. The number of passive inputs affects only the qualifying width.

Use the provisional passive-preactivation cap $2(GH_{\max} + 1 + U_0 S^2 + a)\ell^2$, where U_0 is the applicable response bound. Its initial maximum is $(GH_{\max} + 1)\sqrt{\ell}$, by the same fresh-row argument; real activity adds at most $U_0 S^2 \sqrt{\ell}$. The domain checks below control the complex displacement and improve this stop without a passive exponential budget.

Claim 4 (Local time derivative under provisional stops). *Let $U_0 = U$ or U_{fin} , as appropriate, and put $N_* = \max_j \{r_j^*, 4U_0\}$, $B_n = 1 + (S^2/\eta_*) \log(e + \ell)$. Define*

$$J_L = 2sH_L + 4et_2 N_*, \quad J_j = 10sJ_{j+1} + 2s^3 k_{j+1}^2 H_j + 4et_2 N_*, \quad J_* = \max_j J_j.$$

For $\ell \geq \max(e^2, 2\mathcal{B})$, on each independently stopped cavity,

$$\|\dot{h}_a^{(j)}\|_2/\sqrt{n} \leq 2\rho S q_j^*, \quad \|\dot{\delta}_a^{(j)}\|_2/\sqrt{n} \leq J_* B_n \rho.$$

These bounds use the provisional response stops, not their improvement.

Proof. The exact response identity gives $\|\dot{z}\|_2/\sqrt{n} \leq 2\rho S r_j^*$, $\|\dot{z}\|_\infty \leq 8\rho S U_0 \sqrt{\ell}$. Interpolate to exponent $2v/(v-2)$ and use the carrier v -norm supplied by (37). For $v \geq 4$,

$$\|k \odot \dot{z}\|_2/\sqrt{n} \leq (2\rho S^2 N_*/\eta_*)v(2\mathcal{B}\ell)^{1/v}.$$

Choose $v = \max(4, \log(2\mathcal{B}\ell))$. The exponential factor is at most e , and $v \leq 2\log(e + \ell)$. Backward differentiation now gives the readout term $2sH_L$, propagated derivative, changed-matrix term $2s^3 k_{j+1}^2 H_j$, and changed-gate term $4et_2 N_*$. They are the displayed recursion. Forward differentiation gives the first bound. This growing real exponent is a deterministic budget inequality and does not use growing deletion sets. $\square$

Removing budgets on complete independently stopped paths. The domain checks below will supply a bounding rectangle of side $O(\ell)$, uniformly Lipschitz retractions commuting with real anchors, and normalized complex-minus-real coefficient radius $D_n = O(B_n/\sqrt{\ell})$ with global modulus $O(B_n)$. Claim 3 then gives mean $O((\log(e + \ell))^{3/2}/\sqrt{\ell})$ and vanishing tail radius. Every fixed exponential moment of these corrections tends to one. Fixed squared-exponential moments do also, by integrating the sub-Gaussian tail; sample RMS uses Jensen. Notice that the modulus here is polylogarithmic, not an arbitrary polynomial in n .

For singleton/common-cavity differences, first extend each path by its own retraction, then project their difference onto the deterministic ball of radius $4n^{1/100}$. Both paths omit the paired root. Projection preserves this independence and equals the actual difference on the successful prefix. Its normalized Gaussian radius is $4n^{-49/100}$, and its modulus is at most a fixed power of n , by the already proved derivative bounds. The Gaussian rectangle claim again gives moments tending to one at every fixed order. The two coefficient paths need not be independent of each other.

Fix a positive integer moment order u , a sample and a layer. Distinct roots in an empirical u th moment are independent given their common cavity. Apply Cauchy–Schwarz to separate products of main real references from products of corrections, and then Hölder over the finitely many roots and four reference families for the corrections. Main moments use exponent at most $8\eta_*$ in (55); corrections use fixed orders proportional to u and tend to one. The limiting per-root base is therefore below sixteen. Drop full-event indicators only after dominating by these nonnegative, globally defined cavity bounds. A tuple with $k < u$ distinct roots has normalized multiplicity $O_u(n^{k-u})$. On (56), repeated weights cost at most $e^{2u\eta_* K_{\text{src}}\sqrt{\ell}} = n^{o(1)}$, so collision contributions vanish. Local exceptional events are superpolynomially small, and the stopped budgets are bounded.

Cavities cannot stop first on the successful full prefix: their coordinate differences are $n^{-1/30}$, maxima and poles have strict transferred margins, and

$$\mathcal{H}_a^{\text{cavity}} \leq e^{2\eta_* n^{-1/30}} \mathcal{H}_a + O_u(1/n) < 2\mathcal{B}.$$

Operator and activity margins are established in the domain checks. If a full budget is hit, some sample/layer mean is at least $\mathcal{B}/L$. Markov’s inequality and the fixed sample/layer union give

$$\limsup_{n \rightarrow \infty} \Pr\{\text{a full budget is hit}\} \leq mL(16L/\mathcal{B})^u. \quad (57)$$

Take the width limit at each fixed u , then the infimum over positive integers. Since $16L/\mathcal{B} < 1$, the result is zero. The label allowance precedes the moment order and does not shrink with confidence or sample count. This is the shared probability step.

4. Verification of the working domains.

Verification of the working whole-sphere estimates. Lemma 2.1 independently supplies the real trajectory, strict operator caps below nine, and real activity at most $S/2$. The same holds for the independently tested cavities. Impose the eventual deterministic gates

$$n^{-1} \leq \min(Y, S, 1), \quad \ell \geq \max(e^2, 2\mathcal{B}), \quad \sqrt{\ell} \geq c_t \max\{8, \lambda, 4\mathcal{K}/\log 2, 32YSD_W\}. \quad (58)$$

Follow the real solution to its nearest anchor in $[0, T]$, then at most two short contour pieces of total length $2r_t$. The algebraic residual Gram and the negative-Gram variational generator, divided by their factor two, have norm at most $\mathcal{K}$, by the explicit gradient blocks. No complex positivity is used. Residual growth and products of the base propagators cost at most $e^{4\mathcal{K}r_t} \leq 2$. Additional activity is at most $8Yr_t \leq S/2$; normalized hidden increments are at most $8YSD_W r_t \leq 1/4$. Thus operators have strict margins inside ten and $\rho/S \leq \lambda/8$. The total contour length is $O(\ell)$.

These are the premises of the local insertion and trace argument, which improves maxima and response/angular caps. The exact velocity $\partial_t z^{(j)} = -(2/m) \sum_a r_a R_a^{(j)}$ now bounds the complex-time preactivation displacement by $8c_t Y S U = a/8$. Every query in the intrinsic tube has the form $u \cosh h + iv \sinh h$ for a real orthonormal frame and $|h| \leq r_q$. Its additional displacement is at most $c_q V \leq a/8$. Thus the full pole stop improves strictly; the local coordinate comparison transfers this margin to cavities. The passive size stop also improves as already shown.

Under the provisional response stops, Claim 4 bounds normalized complex-minus-real reference radii by $Cr_t B_n = O(B_n/\sqrt{\ell})$, and their global time moduli by CB_n . This holds on complete independently stopped paths: the clipped arguments commute with the real-anchor map as in (38). Their parameter rectangle has side $O(\ell)$. The budget-removal argument therefore applies, with probability of failure tending to zero. The logical order is provisional deterministic bounds, local insertion, maximum/response improvement, strict pole/cavity transfer, and then complete-path Gaussian moments and budget removal. None of these steps conditions an omitted Gaussian root on full-network survival.

Before each stop, finite-width bounded parameters and strict separation from the activation singularities give local holomorphic ODE continuation. Compactness and uniqueness glue the local continuations on the nested rectangles. At a first exit the strict derivative strip allows one-sided estimates; the strict improvements continue the solution across the closed boundary. Query holomorphy follows by forward composition on the quadric. Passive backward fields are then holomorphic by their finite downward recursion, with RMS bounds $Sk_j, S\tau_j$, using the same readout and operator bounds; they require no new probabilistic budget. Initialized operator norm eight proves the four-family coordinate bound. All failures tend to zero, hence each prescribed fixed confidence is attained eventually. $\square$

Claim 5 (Working finite-query time domain and all-time carriers). *Under $Y/\lambda \leq \beta^{-30L}$, for any fixed finite list of real unit inputs, eventually with every fixed confidence its predictions are holomorphic on a neighborhood of*

$$[-r, T+r] + i[-r, r], \quad T = 32\ell/\lambda, \quad r = 1/(\lambda\sqrt{\ell}),$$

and $|f_n(t, v)| \leq SH_L^2$ there. Two independent copies thus satisfy $|f_n - \tilde{f}_n| \leq 32\beta^{6L}Y/\lambda$ on their common event. In particular the Bernstein ellipse of parameter two on $[0, r]$ is contained in this rectangle. On the real all-time trajectory, also

$$\sup_{t \in [0, \infty]} \max_{a, j, i} |k_{a, i}^{(j)}(t)| \leq 2SK_{\text{src}}\sqrt{\ell}. \quad (59)$$

The fitted endpoint is included.

Proof. Use the same proof without angular fields or the frame mesh, replacing the Gaussian multiplier by 64. The finite-list response bound is $SU_{\text{fin}}\sqrt{\ell}$. The time-radius coefficient may be any $c \leq a/(64YSU_{\text{fin}}) = a/(4\lambda S^2U_{\text{fin}})$. By Claim 1,

$$\frac{a}{4S^2U_{\text{fin}}} \geq \frac{\beta^{34L-1}}{64} > 1,$$

so $c = 1/\lambda$ is permitted. All other rectangle gates are eventual. Cauchy-Schwarz gives $|f_n| \leq SH_L^2$; use $H_L \leq \beta^{3L}$ for the difference bound. The parameter-two ellipse on $[0, r]$ has real range $[-r/8, 9r/8]$ and imaginary height $3r/8$, hence lies inside the stated rectangle eventually.

It remains to extend the carrier maximum beyond T ; no complex late-time extension is required. Let $\|\Delta\theta\|$ denote the physical parameter norm with blocks $(W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$. Forward subtraction in the real fitting tube gives feature RMS changes bounded by $P_j\|\Delta\theta\|$. Backward subtraction then gives

$$\max_{a,j} \|k_a^{(j)}(t) - k_a^{(j)}(T)\|_\infty \leq C_k n \|\theta(t) - \theta(T)\|,$$

where $C_L^k = 1$, $C_j^k = S\tau_{j+1} + 10sC_{j+1}^k + 10t_2SP_{j+1}k_{j+1}$, and $C_k = \max_j C_j^k \leq \beta^{20L}$. The terms are the changed mixer, propagated upper difference, and changed activation gate; RMS-to-coordinate conversions account for the two factors $\sqrt{n}$. The real segment has bounded operators and real activations, so this subtraction needs no complex strip hypothesis. Lemma 2.1 gives $\|\theta(t) - \theta(T)\| \leq 2Ye^{-16\ell}/\sqrt{\lambda}$. Consequently the last display is eventually at most $SK_{\text{src}}\sqrt{\ell}$, including at $t = \infty$. Add (56) at T . $\square$

Claim 6 (Working residual-adapted finite-panel domain). *Fix the training inputs and any finite list of passive inputs before initialization. Use the same allowance as the working whole-sphere estimates, and let*

$$\begin{aligned} \nu &= \lambda/4, \quad T = 32\ell/\lambda, \quad r_n = \frac{a}{64YSU_{\text{fin}}\sqrt{\ell}}, \\ h(t) &= \frac{1}{2\mathcal{K}} \log(1 + 2\mathcal{K}r_n e^{\nu t}), \quad d_h = \frac{\nu}{2\mathcal{K}}, \quad T_+ = T + h(T) + r_n. \end{aligned}$$

Eventually with every fixed confidence, the required forward and backward fields are holomorphic on a neighborhood of

$$D_1 = \{x + iy : -h(0) \leq x \leq T_+, \quad |y| \leq h(\max(x, 0))\},$$

with the same operator, RMS, training-maximum, and four-family coordinate bounds as in the working whole-sphere estimates; the response bound is $SU_{\text{fin}}\sqrt{\ell}$. In particular it suffices to retain forward families on all declared inputs and backward families only on training inputs. For $0 \leq t \leq T$, the disk of radius $\varrho(t) = h(t)/(2(1 + d_h))$ about t lies in D_1 . The adaptive subdivision $t_0 = 0$, $t_{b+1} = \min(T, t_b + \varrho(t_b)/2)$ has at most

$$J \leq 1 + 40960 \frac{U_{\text{fin}}}{a} \left(\frac{Y}{\lambda}\right)^2 \sqrt{\ell} + 80 \frac{\mathcal{K}}{\lambda} \log\left(1 + \frac{4\ell}{\log 2}\right) \quad (60)$$

panels. Here $U_{\text{fin}}/a \leq \beta^{27L}$ and $\mathcal{K} \leq \beta^{14L}$, independently of the input dimension and the size of the fixed passive list.

Proof. Population RMS and the Gram gap give $\lambda \leq H_L^2 \leq \mathcal{K}$. Thus $0 \leq h' \leq d_h \leq 1/8$, $h(0) \leq r_n$, and $h(t) \leq r_n + d_h t$. Define the nested domains by

$$D_s = \{x + iy : -sh(0) \leq x \leq sT_+, \quad |y| \leq sh(\max(x, 0))\}, \quad 0 \leq s \leq 1.$$

For $X_s(x) = \text{clip}_{[-sh(0), sT_+]} x$, their retractions are

$$P_s(x + iy) = X_s(x) + i \text{clip}_{[-sh(X_s(x)_+), sh(X_s(x)_+)]} y.$$

With $a_0(z) = \text{clip}_{[0, T_+]} \text{Re } z$, $P_s a_0 = a_0 P_s$, and $\text{Lip}(P_s) \leq 1 + d_h$. Indeed $|\Delta X_s| \leq |\Delta x|$ and the imaginary increment is at most $|\Delta y| + d_h |\Delta x|$; the shear has norm at most $1 + d_h$. Integrate derivatives along the image under P_s of a straight segment to get global coefficient Lipschitz bounds. Convexity of D_s is unnecessary. The real anchors remain monotone and freeze, as required by the moment proof.

We verify the base-propagator premise before Gaussian insertion. Under each cavity's provisional response stop, use Claim 4 with $U_0 = U_{\text{fin}}$. For $G = (mn)^{-1/2} [\nabla_{\Theta} F_1, \dots, \nabla_{\Theta} F_m]$, the explicit gradient blocks give

$$\|G\| \leq \sqrt{\mathcal{K}}, \quad \|\dot{G}\| \leq C_g B_n \rho, \quad C_g = J_* \left(1 + \sum_{j=2}^L H_{j-1} \right) + 2S^2 \sum_{j=2}^L \tau_j q_{j-1}^* + 2S q_L^*. \quad (61)$$

For the second estimate differentiate each column block $m^{-1/2} \delta_a^{(1)} v_a^\top / \sqrt{n}$, $m^{-1/2} \delta_a^{(j)} h_a^{(j-1)\top} / n$, and $m^{-1/2} h_a^{(L)} / \sqrt{n}$, then bound Frobenius norm. The division by $\sqrt{m}$ prevents a sample factor.

The exact residual equation is $\dot{r} = -2G^\top G r$. On a vertical segment based at real $t \geq 0$, norm differentiation and real fitting give

$$\rho(t \pm iv) \leq Y e^{-\nu t} e^{2\mathcal{K}v}, \quad \int_0^{h(t)} \rho(t \pm iv) dv \leq Y r_n. \quad (62)$$

The second bound is the defining identity for $h(t)$. It also gives $\rho \leq Y + 2\mathcal{K}Y r_n \leq 2Y$ eventually. The base variational equation is $\dot{P} = -2GG^\top P$, with algebraic transposes. At a real anchor, $A_t = 2G(t)G(t)^\top$ is real symmetric, so $-iA_t$ generates unitary motion. Freeze this generator on the vertical segment and use (61):

$$\|2G(t \pm iv)G(t \pm iv)^\top - A_t\| \leq 4\sqrt{\mathcal{K}}C_g B_n \int_0^v \rho(t \pm iu) du.$$

Conjugating by the frozen unitary bounds its propagator by

$$\exp \left(4\sqrt{\mathcal{K}}C_g B_n \int_0^{h(t)} (h(t) - v) \rho(t \pm iv) dv \right) \leq \exp \left(\frac{2C_g}{\sqrt{\mathcal{K}}} B_n Y r_n \right).$$

For the last inequality, integrate the exponential bound in (62); the double integral is at most $Y r_n / (2\mathcal{K})$. Products on ordered disjoint subsegments obey the same bound, since their growth integrals add. Forward real segments contract. The negative-real corner has total exceptional length at most $2r_n$, costing at most $e^{4\mathcal{K}r_n}$. Hence impose the directly checkable eventual gate

$$4\mathcal{K}r_n + (2C_g / \sqrt{\mathcal{K}}) B_n Y r_n < \log(3/2). \quad (63)$$

It supplies the factor-two base premise, without treating a complex Gram matrix as positive. Its left side tends to zero because $B_n r_n = O(\log(e + \ell) / \sqrt{\ell})$.

Also impose (58) with $c_t = a/(64YSU_{\text{fin}})$. Additional activity is at most $8Yr_n$, and normalized parameter increments are at most $8YSD_W r_n \leq 1/4$. Thus operators, activity and $\rho/S \leq \lambda/8$ meet the shared premises. All canonical anchor contours have length $O(\ell)$, and the bounding rectangle $[-h(0), T_+] + i[-h(T_+), h(T_+)]$ has sides $O(\ell)$. The local map/remainder argument is therefore unchanged: the one-dimensional control entropy is $n^{5/8} \text{poly}(\ell)$, endpoint meshes are polynomial, and the same n^{78} Gaussian tail dominates them. After the improved response bound, vertical displacement is at most $2YSU_{\text{fin}}\sqrt{\ell}r_n = a/32$; the negative corner costs at most $a/8$. The pole and passive-size stops improve, and cavity transfer is strict.

Finally Claim 4, integrated over these segments, gives the complete-path normalized correction radius

$$D_n = 4Yr_n \max_j \{2S \max_j q_j^*, J_* B_n / S\} = O(\log(e + \ell)/\sqrt{\ell}).$$

The commuting retractions preserve this estimate and give global modulus $O(B_n)$. Thus the shared Gaussian moment and budget argument applies. Local holomorphic continuation glues because the domain is vertically fibered, retracts onto its real interval, and is simply connected; strict stopped margins continue across its compact boundary. This proves the source event and its confidence.

For the disk claim, if its real coordinate is nonnegative, then $h(x) \geq h(t) - d_h \varrho(t) \geq \varrho(t)$. If it is negative, $t < \varrho(t)$ and $h(t) \leq h(0) + d_h t$ imply $\varrho(t) < h(0)$. The right endpoint is below $T + h(T) < T_+$. The radius is nondecreasing with Lipschitz constant $C_r = d_h/[2(1 + d_h)]$. Each complete adaptive panel contributes at least $1/(2 + C_r)$ to $\int dt/\varrho(t)$, by bounding the radius on that panel by $(1 + C_r/2)\varrho(t_b)$. Consequently $J \leq 1 + 5 \int_0^T dt/h(t)$. Split at $2Kr_n e^{\nu t} = 1$. Before this point use $\log(1 + x) \geq x/2$; afterward the logarithm grows at least with slope $\nu/2$. This bounds the integral by

$$\frac{2}{\nu r_n} + \frac{4K}{\nu} \log \left(1 + \frac{\nu T}{2 \log 2} \right).$$

Substitute $\nu = \lambda/4$, $S = 16Y/\lambda$ and T to obtain (60). The coefficient bounds follow from Claim 1 and $a^{-1} \leq \beta/16$. $\square$

5. Reduction to the public bounds. We now compare the stronger working estimates with the statement, so no coefficient from this proof is needed by a later construction. The working whole-sphere time radius is

$$\frac{a}{64YSU\sqrt{\ell}} = \frac{a\lambda}{1024Y^2U\sqrt{\ell}}.$$

Since $U \leq \beta^{26L}\sqrt{d+3}$ and $a^{-1} \leq \beta/16$,

$$\frac{1024U}{a} \leq 64\beta^{26L+1}\sqrt{d+3} \leq \beta^{30L}\sqrt{d+3}.$$

The last inequality uses $64 \leq \beta^{4L-1}$, valid for $L \geq 2$, $\beta \geq 10$. This gives at least the public time radius. Likewise $V \leq \beta^{2L}\sqrt{d+3}$ gives

$$\max\{8, 8V/a\} \leq \max\{8, \frac{1}{2}\beta^{2L+1}\sqrt{d+3}\} \leq \beta^{3L}\sqrt{d+3},$$

so the working query tube contains the stated tube.

The local power bounds yield, for every layer,

$$H_j \leq 3\beta^{2j} \leq \beta^{3L}, \quad \max(k_j, \tau_j) \leq 3\beta^{4L-1} \leq \beta^{4L}.$$

Multiplication by $S = 16Ym/\gamma$ gives the backward bounds. The four-family coordinate envelope obeys

$$M_0 = 10 \max_j (H_j, \tau_j) \leq 30\beta^{4L-1} \leq \beta^{6L}.$$

For all real times, the fitting statement bounds normalized operators by nine and $\|w\|_2/\sqrt{n} \leq 2Y/\sqrt{\lambda}$. To check the remaining population comparison directly, Minkowski and linear activation growth give $\sqrt{Q_{11}^{(j)}} \leq b + s\sqrt{Q_{11}^{(j-1)}}$, starting at $Q_{11}^{(0)} = 1$. The local recursion for H_j dominates this recursion, so $\lambda \leq Q_{11}^{(L)} \leq H_L^2$. Hence $2Y/\sqrt{\lambda} \leq SH_L$. The same forward and backward recursions with operator bound nine give $H_j, Sk_j, S\tau_j$ on the whole real trajectory. Their parameter limits give the endpoint bounds as well. This proves (25) in both stated domains.

The working finite-query claim has exactly the public time radius. Its prediction envelope is $SH_L^2 \leq 16\beta^{6L}Ym/\gamma$, proving (26). Its all-time training-carrier envelope is $2SK_{\text{src}}\sqrt{\ell} \leq 32\beta^{21L}(Ym/\gamma)\sqrt{\ell}$. Substituting the explicit definition of $k^{(j)}$ proves (27), without exporting that auxiliary notation.

For the finite panel, choose $r_b = \varrho(t_b)$ in the working domain's adaptive partition. Its disk containment and half-radius intervals are exactly those in (iii), and its source envelopes have just been bounded in canonical quantities. Its defining coefficients depend only on the width, m, γ, Y , activations and depth, so these choices are deterministic. Finally

$$40960 \frac{U_{\text{fin}}}{a} \leq 40960\beta^{27L} \leq \beta^{30L}, \quad 80\mathcal{K} \leq 80\beta^{14L} \leq \beta^{16L},$$

because $40960 \leq \beta^{3L}$ and $80 \leq \beta^{2L}$. Applying these inequalities to (60) proves (28). For any requested finite collection of the source domains, split the fixed failure allowance and intersect their events. This supplies the single event asserted in the proposition and completes its proof. $\square$

4 Legendre history compression

Lemma 4.1 (Projection identities). *Let $A > 0$ and let $q \geq 1$ be an integer. Set $p_j^A(\xi) = P_j(2\xi/A - 1)$, where P_j is the Legendre polynomial, and let Π_q^A project onto degrees below q in $L^2([0, A])$. For an absolutely continuous Hilbert-valued history u ,*

$$\|(I - \Pi_q^A)u\|_{L^2}^2 \leq \frac{1}{q(q+1)} \int_0^A \xi(A - \xi) \|u'(\xi)\|^2 d\xi, \quad (64)$$

$$\|u(A) - (\Pi_q^A u)(A)\| \leq \sqrt{A/(3q)} \|u'\|_{L^2}, \quad \|(\Pi_q^A u)(A)\| \leq 64\sqrt{q} \|u\|_{L^\infty}. \quad (65)$$

For a fixed absolutely continuous history g , the growing-interval error obeys

$$\frac{d}{dA} \int_0^A \|g - \Pi_q^A g\|^2 d\xi = \|g(A) - (\Pi_q^A g)(A)\|^2. \quad (66)$$

Proof. Rodrigues' formula and integration by parts give $\int_0^A p_i^A p_j^A = A\mathbf{1}_{i=j}/(2j+1)$ and $-\xi(A - \xi)(p_j^A)' = j(j+1)p_j^A$. The derivative polynomials are orthogonal for the weight $\xi(A - \xi)$. Bessel's inequality followed by the ordinary expansion therefore proves (64); polynomial approximation and increasing finite-dimensional projections give the stated absolutely continuous, Hilbert-valued version. Integration against each lower polynomial also gives $P_j' = \sum_{i < j, j-i \text{ odd}} (2i+1)P_i$. Thus the endpoint kernel is

$$K_q^A(A, \xi) = \frac{1}{A} \sum_{j < q} (2j+1)p_j^A(\xi) = \frac{1}{2}[(p_q^A)'(\xi) + (p_{q-1}^A)'(\xi)].$$

Its integral is one, and integration by parts gives $u(A) - (\Pi_q^A u)(A) = \frac{1}{2} \int_0^A (p_q^A + p_{q-1}^A) u'$. The squared L^2 norm of this multiplier is $A[(2q+1)^{-1} + (2q-1)^{-1}]/4 \leq A/(3q)$.

For completeness the needed kernel L^1 estimate has a short one-dimensional proof. Put $v(\theta) = \sqrt{\sin \theta} P_k(\cos \theta)$ and $a_k = (k+1/2)^2 + (4 \sin^2 \theta)^{-1}$. The Legendre equation implies $v'' + a_k v = 0$, whence $\partial_\theta(v^2 + v'^2/a_k) = -a'_k v'^2/a_k^2 \geq 0$ on $(0, \pi/2)$. At the midpoint use $P_{2j}(0) = (-1)^j \binom{2j}{j}/4^j$, $P'_{2j+1}(0) = (2j+1)P_{2j}(0)$, and $(\binom{2j}{j}/4^j)^2 \leq 1/(j+1)$; the last inequality follows by induction. The midpoint energy is at most $4/k$. Differentiating $P_k = v/\sqrt{\sin \theta}$ now yields

$$|\partial_\theta P_k(\cos \theta)| \leq 3\sqrt{k}(\sin \theta)^{-1/2} + 2k^{-1/2}(\sin \theta)^{-3/2}.$$

On $[1/k, \pi/2]$ its integral is at most $3\pi\sqrt{k} + 8$, using $\sin \theta \geq 2\theta/\pi$. On $[0, 1/k]$ use $|P'_k| \leq k(k+1)/2$ to obtain at most $1/2$. This derivative bound follows from the derivative expansion and $|P_j| \leq 1$, the latter following from the integral formula $P_j(\cos \theta) = \pi^{-1} \int_0^\pi (\cos \theta + i \sin \theta \cos \psi)^j d\psi$ (expand its generating function). Reflection gives total variation at most $36\sqrt{k}$. The endpoint kernel consequently has L^1 norm at most $36\sqrt{q} \leq 64\sqrt{q}$; $q = 1$ is the averaging kernel. Finally differentiation of the squared projection error proves (66): the interior term vanishes because $\partial_A \Pi_q^A g$ is still a polynomial of degree below q . $\square$

Proposition 4.2 (Legendre approximation at the prescribed size). *Eventually at each fixed confidence, there is an initialization-only Legendre model with*

$$q = \left\lceil C \left(\frac{m}{\gamma} \right)^3 n^{1/4} [\log(en)]^2 e^{\sqrt{\log(en)}/2} \right\rceil \quad (67)$$

such that

$$\sup_{t \in [0, \infty], \|x\| = \sqrt{d}} |f_{\text{Leg}}(t, x) - f_n(t, x)| \leq \frac{CY}{\sqrt{n} [\log(en)]^3}. \quad (68)$$

Its moving storage is at most

$$Cm(m/\gamma)^3 n^{5/4} [\log(en)]^2 e^{\sqrt{\log(en)}/2};$$

the fixed storage is the additional $(L-1)n^2$.

Proof. Local notation and construction. All auxiliary quantities below belong to this proof. Put

$$\lambda = \gamma/m, \quad H = \max_{0 \leq j \leq L} \sqrt{Q_{11}^{(j)}}, \quad s = \max \left\{ 1, \max_j \sup_{|\text{Im } z| \leq a/2} |\phi'_j(z)| \right\}, \quad b = \max_j |\phi_j(0)|,$$

and define $D_j = s(9s)^{L-j}$. These definitions are made here; no coefficient from the proof of Lemma 2.1 is imported. In particular $H \leq (2\beta)^L$, by the affine population-RMS recursion, and $s, b \leq \beta$.

The Legendre model constructed here shares the realized dense initialization and physical time. Here the whole-sphere version of (6) is abbreviated by

$$\|g\|_* = \sup_{t \in [0, \infty], \|x\| = \sqrt{d}} |g(t, x)|,$$

including the fitted limit at $t = \infty$. Write $\hat{f} = f_{\text{Leg}} = n^{-1} \hat{w}^\top \hat{h}^{(L)}$, $\hat{r}_a = \hat{f}(x_a) - y_a$ and $\hat{\rho} = \|\hat{r}\|_2 / \sqrt{m}$. We first construct the model for an arbitrary integer $q \geq 1$ and then take the order in the statement.

At each hidden interface $\ell = 2, \dots, L$, sample a , and order $j = 0, \dots, q-1$, retain $\bar{h}_{a,j}^{(\ell-1)}, \bar{\delta}_{a,j}^{(\ell)} \in \mathbb{R}^n$. Together with $\widehat{W}^{(1)}$, $\widehat{w}$, and a scalar τ , these are the moving state. Reconstruct

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{mn\tau} \sum_{a=1}^m \sum_{j=0}^{q-1} (2j+1) \bar{\delta}_{a,j}^{(\ell)} \bar{h}_{a,j}^{(\ell-1)\top}. \quad (69)$$

Use these matrices in the ordinary forward pass and in $\widehat{k}_a^{(L)} = \widehat{w}$, $\widehat{\delta}_a^{(\ell)} = \phi'_\ell(\widehat{z}_a^{(\ell)}) \odot \widehat{k}_a^{(\ell)}$, $\widehat{k}_a^{(\ell)} = \widehat{W}^{(\ell+1)\top} \widehat{\delta}_a^{(\ell+1)}$. The autonomous physical-time equations are

$$\begin{aligned} \dot{\widehat{W}}^{(1)} &= -\frac{2}{m} \sum_a \widehat{r}_a \widehat{\delta}_a^{(1)} v_a^\top, & \dot{\widehat{w}} &= -\frac{2}{m} \sum_a \widehat{r}_a \widehat{h}_a^{(L)}, & \dot{\tau} &= \widehat{\rho}, \\ \dot{\bar{h}}_{a,j}^{(\ell-1)} &= \widehat{\rho} \widehat{h}_a^{(\ell-1)} - \frac{\widehat{\rho}}{\tau} \left(j \bar{h}_{a,j}^{(\ell-1)} + \sum_{i < j} (2i+1) \bar{h}_{a,i}^{(\ell-1)} \right), \\ \dot{\bar{\delta}}_{a,j}^{(\ell)} &= \widehat{r}_a \widehat{\delta}_a^{(\ell)} - \frac{\widehat{\rho}}{\tau} \left(j \bar{\delta}_{a,j}^{(\ell)} + \sum_{i < j} (2i+1) \bar{\delta}_{a,i}^{(\ell)} \right). \end{aligned} \quad (70)$$

Initially $\tau = 1$, $\widehat{W}^{(1)} = W_0^{(1)}$, $\widehat{w} = 0$, $\bar{h}_{a,0}^{(\ell-1)} = h_{0,a}^{(\ell-1)}$, and all other moments vanish. No equation divides by a residual. At $\widehat{\rho} = 0$ all velocities vanish, and the vector field is locally Lipschitz for $\tau > 0$. The exact inventory is

$$S_{\text{moving}} = n(d+1) + 1 + 2(L-1)mnq, \quad S_{\text{fixed}} = (L-1)n^2. \quad (71)$$

The latter entries are the initialized mixers. Data storage is additional; reconstructed matrices need not be retained independently.

Moment identities and the physical defect. For use in this proof, let $p_j^A(\xi) = P_j(2\xi/A - 1)$ and let Π_q^A be the degree-below- q orthogonal projection in $L^2([0, A])$, as in Lemma 4.1. The histories in the proof are specified by $h_a(\tau(t)) = \widehat{h}_a(t)$ and $b_a(\tau(t)) = (\widehat{r}_a/\widehat{\rho})\widehat{\delta}_a(t)$. On the prefix $[0, 1]$ they equal the initialized forward feature and zero, respectively. The backward history is continuous at the join because the initial readout is zero. The moments in (70) are exactly $\int_0^\tau p_j^\tau h_a$ and $\int_0^\tau p_j^\tau b_a$; differentiating these integrals proves the equations, including their factors j and $2i+1$. Differentiating their bilinear projection pairing gives the exact defect

$$\dot{\widehat{W}}^{(\ell)} = -\frac{2}{mn} \sum_a \widehat{r}_a \widehat{\delta}_a^{(\ell)} \widehat{h}_a^{(\ell-1)\top} + \mathcal{E}_\ell, \quad \mathcal{E}_\ell = \frac{2\widehat{\rho}}{mn} \sum_a e_{b,a}^{(\ell)} e_{h,a}^{(\ell-1)\top}, \quad (72)$$

where $e_g = g(\tau) - (\Pi_q^\tau g)(\tau)$. Indeed the derivative of the pairing of the two projected histories equals the full endpoint product minus the product of their endpoint errors, by orthogonality. These identities hold until a residual zero and extend there by stationary continuation. Histories are proof objects, not extra state.

Claim 7 (Order-independent physical bounds). *On (16), under $Y \leq \lambda\beta^{-30L}$, every finite order is globally defined, has convergent physical parameters and moments, and fits the labels. Its operator and feature bounds are those of (17), and*

$$\widehat{\rho}(t) \leq Y e^{-\lambda t/4}, \quad \tau(t) \leq 1 + 4Y/\lambda, \quad \|\widehat{w}(t)\|/\sqrt{n} \leq 4Y/\sqrt{\lambda}. \quad (73)$$

Verification of the local claim. Let $\sigma = 8Y/\lambda$, $R = \sigma\sqrt{\lambda}$, and $\kappa = \lambda/4$. Define, within the present construction proof,

$$C_1 = 4s^2 D_1^2, \quad C_\ell = 3s^2(64H^4 D_\ell^2 + 82C_{\ell-1}), \quad E_* = 130 \sum_{\ell=2}^L D_\ell \sqrt{C_{\ell-1}}.$$

It suffices that

$$\sigma \leq \min \left\{ 1, \frac{1}{64H^2 D_1}, \frac{1}{\sqrt{8\sqrt{C_L}}}, \left(\frac{1}{8H^2 D_1 E_*} \right)^{2/7} \right\}. \quad (74)$$

The present label cap implies this: $D_\ell \leq \beta^{2L}$, $\sqrt{C_\ell} \leq \beta^{8L}$, and $E_* \leq \beta^{12L}$. For the middle estimate unroll $C_\ell = 192s^2 H^4 D_\ell^2 + 246s^2 C_{\ell-1}$, bound $192s^2, 246s^2 \leq \beta^5$, $H^4 D_\ell^2 \leq \beta^{10L}$, and sum at most L terms. For the last use $130L \leq (\beta^L)^2$. Substitution gives a lower bound β^{-8L} for the minimum in (74), whereas $\sigma \leq 8\beta^{-30L}$.

Stop at clock activity σ , readout RMS R , operator norm nine, or feature RMS $2H$. Before the stop, $\|\hat{\delta}_a^{(\ell)}\|/\sqrt{n} \leq D_\ell R$. Projection contraction, sample Cauchy–Schwarz, and the zero backward prefix give

$$\|\widehat{W}^{(\ell)} - W_0^{(\ell)}\|_F \leq 4HD_\ell R \sqrt{(1+\sigma)\sigma}, \quad \|\widehat{W}^{(1)} - W_0^{(1)}\|_F / \sqrt{n} \leq 2D_1 R \sigma.$$

These are strictly below one by (74) and $\sqrt{\lambda} \leq H$. Let Z_ℓ be the supremum, over queries and stopped horizons, of $n^{-1} \int_1^{\tau(t)} \|\partial_\xi h^{(\ell)}(\xi, v)\|^2 d\xi$. The polynomial endpoint norm on $L^2([0, A])$ is $q/\sqrt{A}$, since the sum of squared normalized basis values at A is $\sum_{j < q} (2j+1)/A = q^2/A$. Thus the aggregate backward endpoint error is at most $(q+1)D_\ell R$. By (66), (64), and (72),

$$\int_0^t \hat{\rho} \|\mathcal{E}_\ell / \hat{\rho}\|_F^2 du \leq 2(1+\sigma)^2 D_\ell^2 R^2 Z_{\ell-1}.$$

Here the forward prefix error is zero, $\xi(A - \xi) \leq A^2/4$, and $(q+1)/q \leq 2$. The first-layer chain rule and the three later derivative terms (gradient update, defect, and propagated feature derivative) now give

$$Z_1 \leq 4\sigma s^2 D_1^2 R^2, \quad Z_\ell \leq 3s^2 \{64\sigma H^4 D_\ell^2 R^2 + [81 + 32H^2 D_\ell^2 R^2] Z_{\ell-1}\} \leq C_\ell \sigma R^2.$$

The last step uses $32H^2 D_\ell^2 R^2 < 1$. Consequently every feature moves in RMS by at most $\sqrt{\sigma Z_\ell} \leq \sqrt{C_L} \sigma^2 \sqrt{\lambda} \leq \sqrt{\lambda}/8$. This excludes the feature boundary and preserves the top Gram gap $\lambda/4$ by singular-value perturbation.

For the pointwise defect use the sharper bounds (65). The two endpoint errors are at most $65\sqrt{q}D_\ell R$ and $\sqrt{(1+\sigma)/(3q)}\sqrt{Z_{\ell-1}}$, respectively. Hence

$$\sum_{\ell=2}^L \|\mathcal{E}_\ell\|_F \leq E_* \sqrt{\sigma} R^2 \hat{\rho}. \quad (75)$$

The prediction Jacobian applied to this hidden defect has sample RMS at most $2HD_1 R \sum \|\mathcal{E}_\ell\|_F \leq \lambda \hat{\rho}/4$, by the last entry in (74). Let J be the Jacobian of the training prediction vector in mobility coordinates $(W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$, evaluated at the reconstructed state, and put $\Gamma = JJ^\top/m$. The mean full tangent Gram then gives $\dot{\hat{r}} = -2\Gamma \hat{r} + J\mathcal{E}$ with $\Gamma \succeq \lambda I/4$. Thus $\dot{\hat{\rho}} \leq -\kappa \hat{\rho}$ and $\int_t^T \hat{\rho} \leq \hat{\rho}(t)/\kappa$ on every stopped interval. The clock activity is at most $\sigma/2$.

Let $\mathcal{F}$ be the dense gradient vector field at the reconstructed parameters, in the norm of Lemma 2.1. Its squared norm is $4\hat{r}^\top \Gamma \hat{r}/m \geq \lambda \hat{\rho}^2$, and the defect estimate implies $-\partial_t \hat{\rho}^2 \geq \|\mathcal{F}\|_{\text{par}}^2/2$. Multiplying by $e^{\kappa t}$, integrating by parts, and using $\hat{\rho} \leq Y e^{-\kappa t}$ gives $\int e^{\kappa t} \|\mathcal{F}\|_{\text{par}}^2 dt \leq 4Y^2$. Weighted Cauchy–Schwarz then gives $\int \|\mathcal{F}\|_{\text{par}} \leq 2Y/\sqrt{\kappa} = R/2$. The readout has no defect, so its boundary is excluded as well. Every stopped inequality improves strictly. The integral formulas bound all moments at finite n, q , proving global continuation. Uniqueness excludes a finite first residual zero when $Y > 0$, since zero residual is an equilibrium of the entire stored-state ODE. The integrable gradient and defect lengths yield physical limits; the moment equations also have integrable velocities. This proves (73) and fitting at the limit. This finishes the local fitting claim. The coefficients C_j just defined remain private to this construction proof.

Comparison with dense training. Intersect the initialization event and the all-time training-carrier bound (27) of Proposition 3.1. Set $X = \beta^L$, $z = Y/\lambda$, $\chi = 1 + \lambda^{-1}$, $u = \sqrt{\log(en)}$, and $M = 32X^{21}zu$. Write $\rho_D = \|r\|_2/\sqrt{m}$ for the dense residual RMS. For the backward comparison at any parameter state, define its pre-gated training responses by $k_a^{(L)} = w$ and $k_a^{(j)} = W^{(j+1)\top} \delta_a^{(j+1)}$ for $j < L$. Only actual dense training carriers, not closure carriers or passive query carriers, are bounded by M . Throughout this proof $z \leq X^{-30}$, $X \geq 100$. Use the Euclidean mobility coordinates $(W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$, whose Euclidean norm is the parameter norm of Lemma 2.1. Let e be the Euclidean discrepancy and D the sum of its $L+1$ block norms, so $D \leq \sqrt{L+1}e$.

We first verify the signed comparison with no reciprocal-gap factor inserted into its Jacobian coefficient. Put $T_0 = 9s$, $R = 8z\sqrt{\lambda}$, $B_\delta = sT_0^{L-1}R$, and define

$$\begin{aligned} Q_1 &= b + 9s, \quad Q_\ell = b + 9sQ_{\ell-1}, \quad Q = \max(1, Q_1, \dots, Q_L), \\ F_{\text{seg}} &= QT_0^{L-1}, \quad P = 1 + Q(L-1), \\ d_0 &= sT_0^{L-1}(1 + B_\delta), \quad d_1 = Lt_2 F_{\text{seg}} T_0^{L-1}, \\ K_0 &= \sqrt{L+1}\{Pd_0 + [1 + (L-1)B_\delta]sF_{\text{seg}}\}, \quad K_1 = \sqrt{L+1}Pd_1, \end{aligned}$$

where $t_2 = \max\{1, \max_\ell \sup_{|\text{Im } z| \leq a/2} |\phi_\ell''(z)|\} \leq \beta$. On the straight parameter segment from dense to closure, matrix and readout norms keep their bounds by convexity; linear growth gives feature RMS at most Q . Forward subtraction therefore bounds the preactivation difference at segment fraction $\vartheta \in [0, 1]$ by $F_{\text{seg}}\vartheta D$. In backward subtraction use

$$\delta_\vartheta - \delta_D = \phi'(z_\vartheta) \odot (k_\vartheta - k_D) + [\phi'(z_\vartheta) - \phi'(z_D)] \odot k_D.$$

Propagation costs T_0 , a changed mixer acts on a response of RMS at most B_δ , and the changed gate uses only the dense carrier M . Induction gives response difference at most $(d_0 + d_1 M)\vartheta D$. Writing $f_a(\theta) = f_\theta(x_a)$, the gradient blocks of a training output are $\delta_a^{(1)}v_a^\top/\sqrt{n}$, $\delta_a^{(\ell)}h_a^{(\ell-1)\top}/n$, and $h_a^{(L)}/\sqrt{n}$. Subtract each hidden product as $(\delta_\vartheta - \delta_D)h_\vartheta^\top + \delta_D(h_\vartheta - h_D)^\top$. Their block bounds give $\|\nabla f_a(\theta_\vartheta) - \nabla f_a(\theta_D)\| \leq (K_0 + K_1 M)\vartheta e$. Segment integration proves both hypotheses of Lemma 2.2 with $K = K_0 + K_1 M$. The residual integrals are at most $2z$ and $4z$, so for every finite horizon

$$\sup_{t \leq T} D(t) \leq \mathcal{A}_n \mathcal{D}_T, \quad \mathcal{A}_n = \sqrt{L+1}e^{14(K_0 + K_1 M)z}, \quad \mathcal{D}_T = \int_0^T \sum_{\ell=2}^L \|\mathcal{E}_\ell\|_F dt. \quad (76)$$

In particular the carrier contribution is additive in K , not multiplied once per layer.

Here are coefficient bounds needed to control the forcing. Their dependence on the gap and labels is displayed before any simplification. Put $a_0 = 4z$, $\kappa = \lambda/4$, $F_z = 2HT_0^{L-1}$, $P_0 = 1 + 2H(L-1)$,

retain the local coefficients C_ℓ defined above, and define

$$\begin{aligned}
B_d &= sT_0^{L-1}(1 + B_\delta) + Lt_2F_zT_0^{L-1}M, \\
V_z &= 2F_zB_\delta P_0, \\
V_\delta &= T_0^{L-1}[4sH + 4sH(L-1)B_\delta^2 + Lt_2MV_z], \\
T_1 &= 2D_1R, \quad T_\ell = 4HD_\ell R + 130D_\ell\sqrt{8zC_{\ell-1}R^2}, \\
V_1 &= sT_1, \quad V_\ell = s(2HT_\ell + 9V_{\ell-1}), \quad V_h = \max_\ell V_\ell, \\
C_c &= 4\left(4H^2 + D_1^2R^2 + 4H^2R^2\sum_{\ell=2}^L D_\ell^2\right) + \lambda/2, \\
F_h &= V_h a_0 \sqrt{(1 + a_0)/2}, \\
b_0 &= \sqrt{1 + a_0}(Y/\kappa)V_\delta + 2B_\delta\sqrt{a_0}, \quad b_1 = 2\sqrt{1 + a_0}C_cB_\delta/\kappa, \\
C_f &= 2H + RsF_z.
\end{aligned}$$

The meanings follow directly from the recursions: B_dD bounds the response discrepancy between the two actual trajectories; $V_z\rho_D$ and $V_\delta\rho_D$ bound dense preactivation and response speeds; $V_h\hat{\rho}$ bounds closure feature speeds, using (75). Finally $\|\partial_t(\hat{r}/\hat{\rho})\|_m \leq C_c$ follows by differentiating the normalized residual and bounding its tangent Gram by its trace and its defect by $\lambda\hat{\rho}/4$.

We spell out sufficient power bounds to fix the exponential rate. The preceding finite recursions imply

$$\begin{aligned}
K_0 &\leq X^9, \quad K_1 \leq X^{13}, \quad 14(K_0 + K_1M)z \leq X^{10}z + X^{36}z^2u, \\
V_h &\leq X^{20}R, \quad C_c \leq X^{11}, \quad B_d \leq X^{33}(1 + zu), \quad V_\delta \leq X^{40}(1 + zu), \\
F_h/Y &\leq X^{22}z\sqrt{\chi}, \quad b_0 \leq X^{42}z(1 + zu), \quad b_1 \leq X^{16}z\sqrt{\chi}, \quad C_f \leq X^7.
\end{aligned} \tag{77}$$

To check the first line, $Q \leq X^2$, $F_{\text{seg}} \leq X^4$, $d_0 \leq 2X^3$, $d_1 \leq X^8$, $P \leq X^4$ give the first two bounds; insert $M \leq 32X^{21}zu$ and use $448 \leq X^2$. For the second line, $R \leq X^{-27}$, $\sqrt{C_\ell} \leq X^8$, and $D_\ell \leq X^2$ give $T_\ell \leq X^{12}R$. The V_ℓ recursion has forcing at most $2X^{15}R$, propagation at most $T_0^L \leq X^2$, and at most $L \leq X$ terms. The trace defining C_c gives its stated bound. Also $F_z \leq X^5$, $P_0 \leq X^4$, and $V_z \leq 2X^{12}R$. The non-carrier and carrier terms of V_δ are at most $4LX^5$ and $64LX^{36}Rzu$, respectively; those of B_d are at most $2X^3$ and $32LX^{29}zu$. For the final line retain the normalization: $F_h/Y \leq 32X^{20}z/\sqrt{\lambda}$, $b_1 \leq 96X^{14}z/\sqrt{\lambda}$, and $b_0 \leq 6X^{40}z(1 + zu) + 32X^5z^{3/2}$. These inequalities imply (77) because $X \geq 100$, $z \leq 1$, and $\lambda^{-1/2} \leq \sqrt{\chi}$. In particular

$$\mathcal{A}_n \leq eXe^u, \quad \log \mathcal{A}_n = O_{\text{problem}}(\sqrt{\log(en)}). \tag{78}$$

The universal coefficient of u follows from $X^{36}z^2 \leq 1$; no extra power of m/γ occurs in that exponent.

By (64), the forward history tail on $[0, A]$, $A \leq 1 + a_0$, has aggregate RMS L^2 norm at most F_h/q : use its clock derivative bound V_h and $\int_1^A \xi(A - \xi)d\xi \leq A(A - 1)^2/2$. For the backward history compare it with $\tilde{b}_a(\tau(t)) = (\hat{r}_a/\hat{\rho})\delta_{D,a}(t)$, with zero prefix. Its physical derivative has aggregate RMS at most $C_cB_\delta + V_\delta\rho_D$. Freeze it after $t_q = 2\log q/\kappa$ (or the current horizon if earlier). Since $A - \tau(t) \leq \hat{\rho}(t)/\kappa$, the weighted derivative energy cancels the inverse clock speed and is at most

$$\frac{1 + a_0}{\kappa} \left[2t_q C_c^2 B_\delta^2 + 2V_\delta^2 \int_0^{t_q} \rho_D(t)^2 dt \right].$$

The frozen tail is therefore at most $(b_1\sqrt{\log q} + \sqrt{1+a_0}(Y/\kappa)V_\delta)/q$. The freezing error is at most $2B_\delta\sqrt{a_0}e^{-\kappa t_q/2} = 2B_\delta\sqrt{a_0}/q$. The difference between closure and dense response histories contributes at most $B_d\sqrt{a_0}\sup_{t\leq T} D(t)$; the normalized residual has sample norm one. All aggregate history norms here use the factor $(mn)^{-1/2}$.

Use Cauchy–Schwarz in samples and then in $dA = \hat{\rho}dt$ in (72). The initial projection errors vanish, so (66) gives

$$\mathcal{D}_T \leq 2(L-1)F_h \left[\frac{b_0 + b_1\sqrt{\log q}}{q^2} + \frac{B_d\sqrt{a_0}\sup_{t\leq T} D(t)}{q} \right]. \quad (79)$$

For $q \geq q_{\text{abs}} := 4(L-1)F_h B_d\sqrt{a_0}\mathcal{A}_n$, (76) absorbs the last term with factor at most one half. Whole-sphere forward/readout subtraction costs $C_f D$. The coefficients do not depend on T , so increasing horizons and the already constructed physical limits yield

$$\|f_{\text{Leg}} - f_n\|_* \leq \frac{4(L-1)C_f\mathcal{A}_n F_h}{q^2} (b_0 + b_1\sqrt{\log q}).$$

By the ledger its numerator divided by Y is at most $eX^{75}z^2\chi(1+zu)e^u\sqrt{\log(eq)}$. The cap and $\lambda \leq H^2 \leq X^4$ imply $Y \leq X^{-26} \leq 1$, so $z \leq \lambda^{-1} \leq \chi$. Consequently

$$\|f_{\text{Leg}} - f_n\|_* \leq CY\chi^6 e^{\sqrt{\log(en)}} \frac{\sqrt{\log(en)\log(eq)}}{q^2} \quad (q \geq q_{\text{abs}}). \quad (80)$$

This enlargement preserves the displayed sample/gap power without spending the label cap to reduce it. All coefficients except $\mathcal{A}_n$ grow at most polynomially in u , hence $q_{\text{abs}} = n^{o(1)}$ for the fixed problem. The prescribed order (67) eventually exceeds it. Since $m/\gamma \geq 1$, $\chi \leq 2m/\gamma$, and $\log(eq) \leq C\log(en)$ eventually, substitution proves (68). The count follows from (71); nd and rounding are absorbed only after taking sufficiently large width for the fixed problem. One event supports every order used here, so no order-dependent failure union is needed. All initialization data are the specified dense initialization and initialized training features. $\square$

5 An actual dense-trajectory fluctuation

Lemma 5.1 (Derivative to trajectory). *Let $r > 0$ and $M \geq 0$. If the scalar function g is holomorphic on a neighborhood of the filled parameter-two Bernstein ellipse for $[0, r]$ and bounded there by M , then for every integer $N \geq 1$,*

$$\sup_{0 \leq t \leq r} |g(t)| \geq \frac{r}{2N^2} |g'(0)| - 24M2^{-N}. \quad (81)$$

Proof. Set $t = r(x+1)/2$. The Joukowski map $x = (z+z^{-1})/2$ and Cauchy’s coefficient formula give Chebyshev coefficients $|a_k| \leq 2M2^{-k}$. The degree- N truncation has uniform tail at most $2M2^{-N}$; its physical derivative tail at zero is at most $4Mr^{-1}2^{-N}(N^2 + 4N + 6)$, by summing $\sum_{k>N} k^2 2^{-k}$. For a degree- N polynomial, including complex coefficients, $|P'(-1)| \leq N^2\|P\|_{[-1,1]}$. Indeed interpolate at $x_j = \cos(j\pi/N)$: the endpoint derivative weights have alternating signs, and their absolute sum equals $|T'_N(-1)| = N^2$ by evaluating on $T_N(x_j) = (-1)^j$. The sign assertion follows directly from the products defining the Lagrange basis; the endpoint weight is $\sum_{j<N} (-1-x_j)^{-1} < 0$. Apply this inequality to the truncation, include the physical factor $2/r$, and add the two tails. Rearranging gives error coefficient $4 + 8/N + 12/N^2 \leq 24$, as required. $\square$

Proposition 5.2 (Positive-time dense variability). *Under the assumptions of Section 1, for every fixed $0 < \delta < 1$, there is a constant $c_{\phi,L,\delta} > 0$, depending only on the activations, depth and this confidence, such that all sufficiently large individual widths satisfy*

$$\Pr \left\{ \|f_n - \tilde{f}_n\|_{\mathcal{X}} \geq c_{\phi,L,\delta} \frac{Y\sqrt{\gamma}}{\sqrt{n}[\log(en)]^{5/2}} \right\} \geq 1 - \delta. \quad (82)$$

This holds for either the whole sphere or any declared panel containing the training inputs. The witness is at a positive training time; no endpoint lower bound is asserted.

Proof. All moments and fluctuation variables introduced here are private to this proof. Let $G \sim N(0, 1)$, define

$$\mu_{2,j} = Q_{11}^{(j)} \quad (j = L-1, L), \quad \mu_4 = \mathbb{E}\phi_L(\sqrt{\mu_{2,L-1}}G)^4,$$

and write Φ for the standard normal distribution function. Let $Z \sim N(0, Q^{(L-1)})$, $U_a = \phi_L(Z_a)$ and $Q = \mathbb{E}UU^\top = Q^{(L)}$. We first establish the innovation and conditional fluctuation needed below, then convert an initial derivative into a positive-time trajectory separation.

Claim 8 (A final-layer innovation). *For $m \geq 2$ and $Y > 0$, some deterministic training index a_* satisfies*

$$\text{Var}\{U_{a_*}(U^\top y)\} \geq \frac{\gamma^3 \mu_{2,L}}{16\mu_4} Y^2 > 0. \quad (83)$$

Verification. All fourth moments are finite by linear growth. The positive gap gives $\mu_{2,L} > 0$ and $\mu_4 > 0$. Put $c = Qy$, $A = \|c\|$, $T = \text{tr } Q$, $M_4 = \mathbb{E}\|U\|^4$, and $D = T - c^\top Qc/A^2 > 0$. The last inequality holds because $m \geq 2$ and $Q \succeq \gamma I$. For $V = \mathbb{E}\|U(U^\top y) - c\|^2$, the squared-area identity gives

$$A^2\|U\|^2 - (c^\top U)^2 \leq \|U(U^\top y) - c\|^2\|U\|^2.$$

Splitting at $\|U\| = R_0$ yields $A^2D \leq R_0^2V + A^2M_4/R_0^2$. Choose $R_0^2 = 2M_4/D$ to obtain $V \geq [y^\top Q^2(TI - Q)y]^2/(4M_4\|Qy\|^2)$. Here $M_4 \leq m^2\mu_4$ and $T = m\mu_{2,L}$. Every eigenvalue x of Q lies in $[\gamma, T - (m-1)\gamma]$. Thus, for $N = y^\top Q^2(TI - Q)y$,

$$N \geq (m-1)\gamma\|Qy\|^2, \quad N \geq \gamma^2(T - \gamma)\|y\|^2.$$

The second uses $x(T-x) - \gamma(T-\gamma) = (x-\gamma)(T-\gamma-x) \geq 0$ and then $x \geq \gamma$. Multiplying these lower bounds gives

$$V \geq \frac{(m-1)\gamma^3(m\mu_{2,L} - \gamma)}{4m^2\mu_4} \|y\|^2 \geq \frac{\gamma^3\mu_{2,L}}{16\mu_4} \|y\|^2,$$

because $\gamma \leq \mu_{2,L}$ and $(m-1)^2/m^2 \geq 1/4$. At least one of the m coordinate variances is at least V/m , which proves (83).

Claim 9 (Conditional scalar fluctuation). *For the index in the preceding local claim, put $g_n(t) = f_n(t, x_{a_*}) - \tilde{f}_n(t, x_{a_*})$, where the two dense initializations are independent. There is a fixed $s_* > 0$ with*

$$s_*^2 \geq \frac{\gamma^3\mu_{2,L}Y^2}{2m^2\mu_4}, \quad \liminf_{n \rightarrow \infty} \Pr\{\sqrt{n}|g'_n(0)| \geq us_*\} \geq 2[1 - \Phi(u)] \quad (u > 0), \quad (84)$$

where Φ is the standard normal distribution function.

Verification. Let $\mathcal{F}_n$ contain both initializations through layer $L - 1$. Their training empirical Grams converge to $Q^{(L-1)}$ by (15) of Lemma 2.1, including when this matrix is singular. Conditionally on $\mathcal{F}_n$, the last-layer preactivation rows of each run are independent Gaussian vectors. Applying ϕ_L gives independent feature vectors, generally non-Gaussian; the two groups are also independent. For either run write $U_{i,a} = h_{0,i}^{(L)}(v_a)$ for its initialized top-layer feature entries. The zero readout gives exactly

$$\dot{f}_n(0, x_a) = \frac{2}{mn} \sum_{i=1}^n U_{i,a} (U_i^\top y).$$

Write $v_* = \text{Var}[U_{a*}(U^\top y)] > 0$ and $s_*^2 = 8v_*/m^2$. Conditional variances of the row products converge to v_* : couple their Gaussian vectors with the continuous covariance square root and dominate the fourth-degree products by Gaussian polynomials. Their centered absolute third moments are uniformly bounded on bounded covariance sets, since linear growth requires only Gaussian sixth moments.

For a centered row product X , Taylor's remainder gives $|\mathbb{E}(e^{itX/\sqrt{n}} \mid \mathcal{F}_n) - 1 + t^2\mathbb{E}(X^2 \mid \mathcal{F}_n)/(2n)| \leq |t|^3\mathbb{E}(|X|^3 \mid \mathcal{F}_n)/(6n^{3/2})$. Raise this identity to the n th power in each of the two independent groups. It proves that the conditional law of $Z_n = \sqrt{n}(g'_n(0) - \mathbb{E}[g'_n(0) \mid \mathcal{F}_n])$ converges weakly in probability to $N(0, s_*^2)$. To justify this random-law statement, every subsequence has a further subsequence on which both covariance matrices converge almost surely; the preceding characteristic-function calculation and the continuity theorem then apply deterministically on almost every such outcome. For a continuous limiting CDF, weak convergence is uniform in its argument: choose finite grids with small Gaussian CDF increments and small tails, and use monotonicity between grid points. Therefore the conditional Kolmogorov distance d_n tends to zero in probability, and $\mathbb{E}d_n \rightarrow 0$ since $0 \leq d_n \leq 1$.

For any $\mathcal{F}_n$ -measurable shift b_n , a centered Gaussian has its largest interval probability at the interval centered at zero. Consequently

$$\Pr\{|Z_n + b_n| \leq us_* \mid \mathcal{F}_n\} \leq 2\Phi(u) - 1 + 2d_n.$$

The Gaussian assertion follows by differentiating the interval mass with respect to its center and using the even, decreasing Gaussian density. Take $b_n = \sqrt{n}\mathbb{E}[g'_n(0) \mid \mathcal{F}_n]$ and average. This proves (84); previous layers may produce an arbitrary shift and cannot increase the limiting small-ball probability. No recursive matrix central limit theorem is needed.

From the derivative to the trajectory. Use the finite-query conclusion (26) of Proposition 3.1 for the two runs and the finite training query list. Put $\lambda = \gamma/m$, $\ell = \log(en)$ and $r = 1/(\lambda\sqrt{\ell})$. Each run is bounded there by $16\beta^{6L}Y/\lambda$, so their difference has twice this bound. The parameter-two ellipse of $[0, r]$ has real projection $[-r/8, 9r/8]$ and imaginary projection $[-3r/8, 3r/8]$, and hence lies inside the supplied rectangle. On its event, g_n is bounded by $32\beta^{6L}Y/\lambda$. Choose $N = \lceil 2\ell/\log 2 \rceil \leq 4\ell$. Lemma 5.1 gives

$$\sup_{0 \leq t \leq r} |g_n(t)| \geq \frac{|g'_n(0)|}{32\lambda\ell^{5/2}} - \frac{768\beta^{6L}Y}{\lambda n^2}.$$

For fixed $u > 0$, the second term is eventually at most half the first on $|g'_n(0)| \geq us_*/\sqrt{n}$. The two source failures tend to zero; no independence from the derivative event is required. Thus the conditional fluctuation claim gives a limiting lower probability at least $2[1 - \Phi(u)]$ for threshold $us_*/(64\lambda\sqrt{n}\ell^{5/2})$. Insert the lower bound on s_* , use $\lambda = \gamma/m$ and $1/(64\sqrt{2}) \geq 1/128$, and choose $u = \Phi^{-1}(1/2 + \delta/4)$. The limiting probability is at least $1 - \delta/2$, leaving a strict margin to obtain $1 - \delta$ eventually. Since $g_n(0) = 0$, a positive threshold on the compact witness interval requires a strictly positive time. That training query belongs to each promised domain. At its fitted endpoint

the predictions equal the same label, consistently with the asserted scope. The asserted constant can therefore be chosen as the following local quantity:

$$c_{\phi,L,\delta} = \frac{\Phi^{-1}(1/2 + \delta/4)}{128} \sqrt{\frac{\mu_{2,L}}{\mu_4}} > 0.$$

Its moment ratio depends only on the activations and depth, as required. $\square$

6 One selected nonlinear dynamics for both source constructions

The sparse-metric lemma first supplies exact geometry on any finite source space. The transfer proposition then constructs and analyzes one autonomous selected model, with its optimizer and comparison coefficients local to that proof. Two later propositions supply its source spaces from whole-sphere polynomials or finite-panel Taylor expansions. They use only the transfer statement's error and storage interface.

Lemma 6.1 (Sparse coordinates with an exact source metric). *Let $E \subset \mathbb{R}^n$ be a linear subspace containing $\mathbf{1}_n$, with $r = \dim E$. There are a set I of at most $9r$ coordinates, a positive diagonal matrix D on I , and a positive definite matrix M such that*

$$u_I^\top M v_I = u^\top v / n \quad (u, v \in E), \quad D/4 \preceq M \preceq D, \quad \mathbf{1}^\top M \mathbf{1} = 1, \quad \mathbf{1}^\top D \mathbf{1} \leq 4.$$

Write $\|u\|_M = (u^\top M u)^{1/2}$. Consequently a coordinate multiplication operator B satisfies $\|B\|_{M \rightarrow M} \leq 2 \max_i |B_{ii}|$, and an activation with $|\phi'| \leq s$, $|\phi(0)| \leq b$ satisfies $\|\phi(u) - \phi(v)\|_M \leq 2s\|u - v\|_M$ and $\|\phi(u)\|_M \leq 2b + 2s\|u\|_M$.

Proof. We give the finite sparsification argument. For vectors $v_i \in \mathbb{R}^r$ with $\sum_i v_i v_i^\top = I$, start with $A = 0$, lower barrier $l = -3r$ and upper barrier $u = 6r$. Maintain

$$lI \prec A \prec uI, \quad \Phi_l(A) = \text{tr}(A - lI)^{-1} \leq \frac{1}{3}, \quad \Phi^u(A) = \text{tr}(uI - A)^{-1} \leq \frac{1}{6}.$$

Set $l' = l + 1$, $u' = u + 2$, and define

$$U(v) = \frac{v^\top (u'I - A)^{-2} v}{\Phi^u(A) - \Phi^{u'}(A)} + v^\top (u'I - A)^{-1} v, \quad L(v) = \frac{v^\top (A - l'I)^{-2} v}{\Phi_l(A) - \Phi_{l'}(A)} - v^\top (A - l'I)^{-1} v.$$

The lower potential implies that every eigenvalue is at least three above l , so both inverse matrices and both denominators are positive. The identity $(B + t v v^\top)^{-1} = B^{-1} - B^{-1} v v^\top B^{-1} / (t^{-1} + v^\top B^{-1} v)$ shows that $U(v) \leq t^{-1} \leq L(v)$ preserves the two potential inequalities at the shifted barriers. For the upper potential, the positive first term of $U(v)$ also preserves the strict spectral barrier.

Such a nonzero v_i exists. If a_j are the eigenvalues of A , then $(u - a_j)^{-1} - (u' - a_j)^{-1} \geq 2(u' - a_j)^{-2}$, hence $\sum_i U(v_i) \leq 1/2 + \Phi^{u'}(A) \leq 2/3$. For the lower sum put $a'_j = (a_j - l')^{-1}$, $b'_j = (a_j - l)^{-1} = a'_j / (1 + a'_j)$ and $\Delta = \sum_j a'_j b'_j = \Phi_{l'} - \Phi_l$. Weighted Cauchy–Schwarz gives

$$\Delta^2 \leq \left(\sum_j (a'_j)^2 b'_j \right) \sum_j b'_j \leq \sum_j (a'_j)^2 - \Delta.$$

Therefore $\sum_i L(v_i) \geq 1 - \Phi_l(A) \geq 2/3$. Choose $t > 0$ with reciprocal in the resulting nonempty interval and add $t v_i v_i^\top$. After $9r$ iterations the barriers are $6r, 24r$. Dividing all accumulated weights by $6r$ gives nonnegative weights s_i , supported on at most $9r$ indices, with $I \preceq \sum_i s_i v_i v_i^\top \preceq 4I$.

Choose U with range E and $U^\top U/n = I$, apply this result to $v_i = U_{i,:}^\top/\sqrt{n}$, and set $P = U_I$, $D = \text{diag}(s_i/n)$, $G = P^\top D P$ and $Z = D^{1/2} P$. Define

$$M = D^{1/2} \{ Z G^{-2} Z^\top + I - Z G^{-1} Z^\top \} D^{1/2}. \quad (85)$$

On $\text{ran } Z$ the middle matrix has eigenvalues those of G^{-1} , and on its orthogonal complement it is the identity. Thus $D/4 \preceq M \preceq D$, while multiplication gives $P^\top M P = I$. The constant source proves the two mass assertions. Finally $\|Bu\|_M \leq \|Bu\|_D \leq \max_i |B_{ii}| \|u\|_D \leq 2 \max_i |B_{ii}| \|u\|_M$. Apply this to coordinate divided differences and use $\|\mathbf{1}\|_M = 1$ for the activation claims. $\square$

6.1 A portable source-to-runtime construction

Proposition 6.2 (Source-to-runtime transfer). *Work on the fitting and analytic-source event of Lemma 2.1 and proposition 3.1. Write $W_0^{(j)} = W^{(j)}(0)$ for the initialized matrices in this statement. Fix*

$$T = 32 \frac{m}{\gamma} \log(en), \quad 0 < \eta \leq \min\{1, Y, 16Ym/\gamma\}, \quad q \geq \max\{m, d\}.$$

For each layer let $E_j \subset \mathbb{R}^n$ have dimension at most q and contain the constant, exact initialized training features and their initialized forward images when present, and (at the first layer) the columns of $W_0^{(1)}$. Suppose these spaces approximate the families (24) to coordinate accuracy η on $[0, T]$, with forward queries in $\mathcal{X}$ and backward queries at training inputs. Require exact initialized-image pairing: each image approximant is the corresponding initialized map applied to its base approximant. There is an autonomous selected nonlinear predictor satisfying

$$\begin{aligned} \|f_{\text{model}} - f_n\|_{\mathcal{X}} &\leq C \frac{Ym}{\gamma} \left(1 + \sqrt{\frac{m}{\gamma}}\right) (1 + \sqrt{\log(en)}) e^{\sqrt{\log(en)}} \eta \\ &\quad + C \frac{Ym}{\gamma} \left(1 + \sqrt{\frac{m}{\gamma}}\right) e^{-\gamma T/(4m)}. \end{aligned} \quad (86)$$

It fits, has a fitted limit, and retains at most

$$1020(L+1)q^2 + 36q + 10m(d+1) \quad (87)$$

real coordinates, including data, selected indices, metrics, initial copies and working arrays. Here $C \geq 1$. In particular, the permissible choice

$$\eta = \min \left\{ 1, Y, \frac{16Ym}{\gamma}, \frac{\gamma}{2Cmn(1 + \sqrt{m/\gamma})(1 + \sqrt{\log(en)})e^{\sqrt{\log(en)}}} \right\} \quad (88)$$

gives error at most Y/n eventually. Source bases, dense coefficients and compilation scratch are discarded after the selected initial arrays are formed.

Proof. All auxiliary quantities in this proof belong to this construction. Put $\lambda = \gamma/m$, $z = Y/\lambda$, $\alpha = Y/\sqrt{\lambda}$, $S = 16z$, and $\ell = \log(en)$, so $0 < \lambda \leq 1$ and $0 < z \leq \beta^{-30L}$. For activation estimates use the envelopes $b = \beta - 1$ and $s = t_2 = \beta$; thus $|\phi_j(0)| \leq b$, $|\phi'_j| \leq s$ and $|\phi''_j| \leq t_2$ on the real line. Write $c_n = (y_a - f_n(t, x_a))_{a=1}^m$ for the dense training deficits and $\rho_n = \|c_n\|_2/\sqrt{m}$ for their RMS. The public source bounds permit the local envelopes

$$u_0 = \beta^{3L}, \quad \tau = \beta^{4L}, \quad K_{\text{src}} = \beta^{21L}.$$

Dense feature norms are at most u_0 and dense training response norms at most $S\tau$ in neuron RMS. Dense training carriers on $[0, T]$ are the local fields $k_{n,a}^{(L)} = w$ and $k_{n,a}^{(j)} = W^{(j+1)\top} \delta_{n,a}^{(j+1)}$; by (27) they are bounded coordinatewise by $32zK_{\text{src}}\sqrt{\ell}$. Dense vector norms below are normalized by $\sqrt{n}$; selected norms are specified by their metrics. Every query v denotes $x/\sqrt{d}$ for an input x in the promised domain.

Autonomous selected model. Apply Lemma 6.1 to obtain (I_j, M_j, D_j) and let $q_j = |I_j| \leq 9q$. The selected moving variables are $W_C^{(1)} \in \mathbb{R}^{q_1 \times d}$, $B_C^{(j)} \in \mathbb{R}^{q_j \times q_{j-1}}$ ($2 \leq j \leq L$), $w_C \in \mathbb{R}^{q_L}$ and $c_C \in \mathbb{R}^m$. For $v = x/\sqrt{d}$ the forward pass is

$$z_C^{(1)} = W_C^{(1)}v, \quad z_C^{(j)} = B_C^{(j)}h_C^{(j-1)}, \quad h_C^{(j)} = \phi_j(z_C^{(j)}).$$

All layer norms and adjoints below use M_j ; in particular $(B_C^{(j)})^* = M_{j-1}^{-1}B_C^{(j)\top}M_j$. Let V_C have columns $h_C^{(L)}(v_a)/\sqrt{m}$ and define

$$Q_C = V_C^*V_C, \quad \hat{w}_C = w_C + V_CQ_C^{-1}\{(y - c_C)/\sqrt{m} - V_C^*w_C\}, \quad f_C(v) = \langle \hat{w}_C, h_C^{(L)}(v) \rangle_{M_L}. \quad (89)$$

The statement's predictor is $f_{\text{model}}(t, x) = f_C(t, x/\sqrt{d})$; physical time is suppressed in the local forward formulas. Thus $f_C(v_a) = y_a - c_{C,a}$ exactly. Define training signals by $k_{C,a}^{(L)} = \hat{w}_C$, $\delta_{C,a}^{(j)} = \phi'_j(z_{C,a}^{(j)}) \odot k_{C,a}^{(j)}$ and $k_{C,a}^{(j)} = (B_C^{(j+1)})^*\delta_{C,a}^{(j+1)}$. The unnormalized sample matrix is

$$\begin{aligned} K_{C,ab} &= \langle h_{C,a}^{(L)}, h_{C,b}^{(L)} \rangle_{M_L} + \langle \delta_{C,a}^{(1)}, \delta_{C,b}^{(1)} \rangle_{M_1} v_a^\top v_b \\ &\quad + \sum_{j=2}^L \langle \delta_{C,a}^{(j)}, \delta_{C,b}^{(j)} \rangle_{M_j} \langle h_{C,a}^{(j-1)}, h_{C,b}^{(j-1)} \rangle_{M_{j-1}}. \end{aligned} \quad (90)$$

The autonomous optimizer is

$$\begin{aligned} \dot{W}_C^{(1)} &= \frac{2}{m} \sum_a c_{C,a} \delta_{C,a}^{(1)} v_a^\top, & \dot{B}_C^{(j)} &= \frac{2}{m} \sum_a c_{C,a} \delta_{C,a}^{(j)} h_{C,a}^{(j-1)\top} M_{j-1}, \\ \dot{w}_C &= \frac{2}{m} \sum_a c_{C,a} h_{C,a}^{(L)}, & \dot{c}_C &= -2K_C c_C / m. \end{aligned} \quad (91)$$

These are prescribed directions; coordinate gates need not be self-adjoint in M_j , and no gradient interpretation of the corrected predictor is used.

For bases U_j of E_j with $U_j^\top U_j / n = I$, set $P_j = (U_j)_{I_j}$. Initialize

$$W_C^{(1)}(0) = (W_0^{(1)})_{I_1}, \quad w_C(0) = 0, \quad c_C(0) = y, \quad B_C^{(j)}(0) = P_j \frac{U_j^\top W_0^{(j)} U_{j-1}}{n} P_{j-1}^\top M_{j-1}. \quad (92)$$

Since $P_j^\top M_j P_j = I$, these operators have norms at most eight. They reproduce an initialized forward image exactly whenever both members belong to the source spaces; their metric adjoints reproduce paired reverse images. The exact additions therefore preserve the entire initialized training forward pass and Gram. All bases, dense arrays and source coefficients are discarded after these finite initial arrays are formed.

Claim 10 (Independent fitting and endpoint tails). *With the notation of this construction, the following local bounds hold. The selected dynamics exist for every $t \geq 0$, have $Q_C \succeq \lambda I/4$, fit the training labels and converge in parameter space. Write*

$$H_1^c = 2b + 16s, \quad H_j^c = 2b + 18sH_{j-1}^c, \quad H_c = H_L^c, \quad d_j^c = 2s(18s)^{L-j}, \quad F_c = (d_1^c)^2 + 4H_c^2 \sum_{j=2}^L (d_j^c)^2.$$

For the selected parameter tuple $\theta_C = (W_C^{(1)}, (B_C^{(j)})_{j=2}^L, w_C)$ use

$$\|\theta_C\|_{\text{par}}^2 = \text{tr}(W_C^{(1)\top} M_1 W_C^{(1)}) + \sum_{j=2}^L \|M_j^{1/2} B_C^{(j)} M_{j-1}^{-1/2}\|_F^2 + \|w_C\|_{M_L}^2.$$

Then $\|h_C^{(j)}\| \leq 2H_c$, $\|\hat{w}_C\| \leq 5\alpha$, $\rho_C := \|c_C\|/\sqrt{m} \leq Y e^{-\lambda t/2}$ and $\int_0^\infty \|\dot{\theta}_C\|_{\text{par}} \leq 2\alpha$. There is a coefficient $\mathcal{D} \leq Cz(1 + \lambda^{-1/2})$ such that comparison on $[0, T]$ extends to all times and the fitted endpoint with additional error at most $\mathcal{D} e^{-\lambda T/4}$. Here and below C depends only on activations and depth.

Proof of the claim. Stop at operator norm nine, feature norm $2H_c$, or Gram margin $\lambda/4$. Expanding the velocities, with all cross-sample terms, gives

$$-\frac{d}{dt}\rho_C^2 = 4c_C^\top K_C c_C / m^2 = \|\dot{\theta}_C\|_{\text{par}}^2, \quad K_C / m \succeq Q_C. \quad (93)$$

Thus $-\dot{\rho}_C \geq \lambda\rho_C/2$ and $\|\dot{\theta}_C\| \leq 2(-\dot{\rho}_C)/\sqrt{\lambda}$. Integration gives the asserted residual and path bounds, as well as $\int \rho_C \leq 2z$ and $\|w_C\| \leq 2\alpha$. With $T_C = V_C Q_C^{-1}$ and $P_C = T_C V_C^*$, the readout is the orthogonal sum $(I - P_C)w_C + T_C(y - c_C)/\sqrt{m}$. Its squared norm is at most $4\alpha^2 + 16\alpha^2$.

Set $R_c = 5\alpha$, $U_1^c = d_1^c$, $U_j^c = 2H_c d_j^c$ for $j \geq 2$, and $F_1^c = 2sU_1^c$, $F_j^c = 2s(2H_c U_j^c + 9F_{j-1}^c)$. The latter recurrence has $F_L^c = F_c$. Backward propagation bounds $\|\delta_C^{(j)}\| \leq d_j^c R_c$, and integration bounds each hidden parameter displacement by $20U_j^c Y^2/\lambda^{3/2}$ and each feature displacement by $20F_j^c Y^2/\lambda^{3/2}$. Under the stated cap, $H_c \leq \beta^{3L}$, $F_c \leq \beta^{15L}$ and $z \leq (16H_c \sqrt{F_c})^{-1}$; hence the last bound is at most $5\sqrt{\lambda}/(64H_c^2) < \min\{1/(8H_c), \sqrt{\lambda}/8\}$. The operator and feature stops improve strictly. The normalized training feature matrix starts with smallest singular value at least $\sqrt{\lambda}/2$ and moves by less than $\sqrt{\lambda}/8$, so the Gram stop improves too. Smoothness on this strict Gram-positive domain and bounded parameters give global existence. Finite path length and (89) give convergence; $c_C \rightarrow 0$ gives fitting.

For quantitative tails, differentiating the inverse gives

$$\dot{T}_C = (I - P_C)\dot{V}_C Q_C^{-1} - T_C \dot{V}_C^* T_C, \quad \dot{P}_C = (I - P_C)\dot{V}_C T_C^* + T_C \dot{V}_C^* (I - P_C).$$

Thus $\|\dot{T}_C\| \leq 8\|\dot{V}_C\|/\lambda$, $\|\dot{P}_C\| \leq 4\|\dot{V}_C\|/\sqrt{\lambda}$ and $\|\dot{V}_C\| \leq 2R_c F_c \rho_C$. Put $G_c = 4H_c^2 + R_c^2 F_c$ and

$$B_w = 4H_c + 48R_c F_c z + 4G_c/\sqrt{\lambda}, \quad B_f = 2H_c B_w + 2R_c^2 F_c.$$

Differentiating the orthogonal readout formula proves $\|\dot{\hat{w}}_C\| \leq B_w \rho_C$ and $|\dot{f}_C(v)| \leq B_f \rho_C$ uniformly over unit inputs. Therefore $|f_C(\infty, v) - f_C(t, v)| \leq 2B_f z e^{-\lambda t/2}$. For the dense tail, the RMS bounds hold at every real time, not merely on the finite complex source domain. Indeed Lemma 2.1 bounds all normalized operators by nine and gives $\|w\|/\sqrt{n} \leq 2Y/\sqrt{\lambda} \leq S u_0$. The all-time RMS bounds of Proposition 3.1, or the same real forward/backward induction, give $\|h^{(j)}\|/\sqrt{n} \leq u_0$ and $\|\delta^{(j)}\|/\sqrt{n} \leq S\tau$ for all real times. The flow equations and sample Cauchy–Schwarz therefore imply $|\dot{f}_n(v)| \leq 2\mathcal{K}\rho_n$, where

$$\mathcal{K} = u_0^2 + S^2 \tau^2 [1 + (L - 1)u_0^2].$$

Comparing each flow with its value at T proves the claimed extension with $\mathcal{D} = z(16\mathcal{K} + 4B_f)$. The small cap gives $\mathcal{K} \leq C$, $B_f \leq C(1 + \lambda^{-1/2})$, which establishes the stated parameter dependence. $\square$

Comparison coefficients. Here is an explicit coefficient ledger for the comparison. Its size is independent of the number of sources. Use the local envelopes $u_0, \tau, K_{\text{src}}$ fixed at this proof's start, and put

$$\begin{aligned} H_r &= u_0 + 3, & H_C &= 2H_c, & P_h &= 6u_0 + 9, & P_\delta &= 6\tau + 9, \\ A_f &= 18 + S^2(\tau + 3)P_h, & A_b &= 18 + S^2H_rP_\delta. \end{aligned}$$

Define the forward-error recursion and the reference-response bounds by

$$\begin{aligned} F_1^z &= 1, & F_1^h &= 2s, \\ F_j^z &= 9F_{j-1}^h + H_r + A_f, & F_j^h &= 2sF_j^z \quad (j \geq 2), & F &= \max_j F_j^h, \\ d_j^E &= 2s(9s)^{L-j} + 3H_c, & J_C &= 5\sqrt{F_c}, \\ J_R &= \left[(d_1^E)^2 + H_r^2 \sum_{j=2}^L (d_j^E)^2 \right]^{1/2}, \\ D_s &= P_h + 16zP_\delta\{1 + (L-1)H_r^2\} + (L-1)(16z)^2\tau^2P_h. \end{aligned}$$

The backward-error recursion is

$$\begin{aligned} B_L &= 2s(1 + 6zF + 32zP_h) + 2t_2F_L^z, \\ B_j &= 18sB_{j+1} + 2s(d_{j+1}^E zH_C + A_b) + 2t_2F_j^z \quad (j < L), \\ B_h &= \sqrt{L}\{H_C \max_j B_j + (\max_j d_j^E)zH_C F\}, \\ K_1 &= 12F + 2B_h + 20z(J_C + J_R)B_h + 20D_s. \end{aligned}$$

Increasing any of these positive coefficients is harmless. Finally set

$$\begin{aligned} \mathcal{B}_n &= 400z^2F_c + 2zK_1 + 40zF + 64z^2K_1K_{\text{src}}\sqrt{\ell}, \\ \mathcal{A}_n &= [H_C(1 + 6zF) + 3\alpha F](1 + \lambda^{-1/2})(e^{\mathcal{B}_n} - 1) \\ &\quad + 6H_CzF + 32H_CzP_h/\sqrt{\lambda} + 3\alpha F + 16zP_h. \end{aligned} \tag{94}$$

Claim 11 (Finite-horizon comparison). *With the local coefficients (94),*

$$\sup_{0 \leq t \leq T} \sup_{x \in \mathcal{X}} |f_C(t, x/\sqrt{d}) - f_n(t, x)| \leq \mathcal{A}_n \eta.$$

Proof of the comparison claim. We record both the source-energy estimate and the cancellation that make the parameter dependence possible. For one layer abbreviate $E = E_j$, $I = I_j$ and $M = M_j$. If $p \in E$ approximates u to coordinate accuracy e_u , exact isometry and the mass bound give

$$\|u - p\|_n \leq e_u, \quad \|u_I - p_I\|_M \leq 2e_u, \quad \|u_I\|_M \leq \|u\|_n + 3e_u.$$

Expanding two inner products about $p, q \in E$ therefore gives

$$|\langle u_I, v_I \rangle_M - \langle u, v \rangle_n| \leq 3e_u\|v\|_n + 3e_v\|u\|_n + 9e_ue_v. \tag{95}$$

Thus feature-pair defects are at most $P_h\eta$, and response-pair defects at most $SP_\delta\eta$; here $\eta \leq 1$, S is used.

Let the proof reference have first layer and readout equal to the coordinate restrictions of the dense ones, and hidden operators

$$B_R^{(j)}(t) = B_C^{(j)}(0) + \frac{2}{m} \int_0^t \sum_a c_{n,a}(u) (\delta_{n,a}^{(j)}(u))_{I_j} (h_{n,a}^{(j-1)}(u))_{I_{j-1}}^\top M_{j-1} du.$$

Its preactivation, feature and response symbols below mean the restricted dense values, not a fresh forward pass; its carrier symbols mean $k_{R,a}^{(j)} = (k_{n,a}^{(j)})_{I_j}$ as well. The exact initialized pairing and operator bound eight make an initialized action defect at most 18η . In each learned action, subtract the dense outer-product integral and use (95) together with $\int \rho_n \leq 2z$, $\|\delta_n^{(j)}\|_n \leq S\tau$ and $\|h_n^{(j)}\|_n \leq u_0$. The resulting forward and reverse defects are at most $A_f\eta$ and $A_b\eta$. The identical calculation for the integrated raw readout gives

$$|\langle w_R, h_R^{(L)}(v) \rangle - f_n(v)| \leq 16zP_h\eta. \quad (96)$$

Write V_n, V_R for the dense and restricted top feature maps, with columns divided by $\sqrt{m}$. Their common approximants and exact isometry give, for every $a \in \mathbb{R}^m$, $\|V_R a\| \leq \|V_n a\| + 3\eta\|a\|$. For $\nu = \|V_R c_n / \sqrt{m}\|$, the dense energy identity gives

$$\int_0^T \nu \leq \alpha + 6\eta z \leq 2\alpha. \quad (97)$$

Here let $\theta_n = (W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$ with its Euclidean block norm. Indeed $2\|V_n c_n / \sqrt{m}\| \leq \|\dot{\theta}_n\|$, whose integral is at most 2α . Integrating source approximants for $\dot{w}_n = 2V_n c_n / \sqrt{m}$ gives a vector in E_L at coordinate error at most $4\eta z$. Consequently $\|w_R\| \leq 2\alpha + 12\eta z \leq 3\alpha$. Likewise the restricted responses have norm at most $d_j^E \alpha$. These arguments never differentiate an approximation error.

Use the hidden-parameter Hilbert norms of the preceding local fitting claim, and let $\theta_{h,C}, \theta_{h,R}$ denote the selected and reference tuples of first-layer and hidden-layer matrices. Let $\mathcal{J}_C, \mathcal{J}_R$ have sample columns, divided by $\sqrt{m}$, equal respectively to the tuples

$$(\delta^{(1)} v_a^\top, (\delta^{(j)} h_a^{(j-1)})^\top M_{j-1})_{j=2}^L.$$

Thus $K_C/m = V_C^* V_C + \mathcal{J}_C^* \mathcal{J}_C$ and $\dot{\theta}_{h,C} = 2\mathcal{J}_C c_C / \sqrt{m}$; the reference satisfies the latter identity with R, n . Moreover $\|\mathcal{J}_C\| \leq J_C \alpha$, $\|\mathcal{J}_R\| \leq J_R \alpha$. Put

$$a_h = \|\theta_{h,C} - \theta_{h,R}\|, \quad e = (c_C - c_n)/\sqrt{m}, \quad u = \|e\|, \quad z_w = w_C - w_R,$$

$$T_C = V_C Q_C^{-1}, \quad P_C = T_C V_C^*, \quad p = T_C e, \quad \zeta = z_w + p, \quad b_e = \|p\| + \|\zeta\|, \quad E = a_h + b_e.$$

All these error variables vanish at zero. Forward induction, first subtracting the reference action and then the nonlinear gate, gives

$$\|h_C^{(j)} - h_R^{(j)}\| \leq F_j^h(a_h + \eta), \quad \|V_C - V_R\| \leq F(a_h + \eta). \quad (98)$$

The first-layer preactivation error is a_h ; at layer j it is at most $9F_{j-1}^h(a_h + \eta) + H_r a_h + A_f \eta$, which proves the displayed recursion rather than presuming an ambient Lipschitz bound.

Set $d_R = V_R^* w_R - (y - c_n)/\sqrt{m}$ and $\Delta V = V_C - V_R$. Subtracting the exact readout formula yields

$$\hat{w}_C - w_R = (I - P_C)z_w - p - T_C \Delta V^* w_R - T_C d_R. \quad (99)$$

Since $p \in \text{ran } V_C$, $(I - P_C)z_w = (I - P_C)\zeta$. Equations (96)–(98) and $\|T_C\| \leq 2/\sqrt{\lambda}$ imply

$$\|\hat{w}_C - w_R\| \leq b_e + 6zF(a_h + \eta) + 32zP_h\eta/\sqrt{\lambda}, \quad (100)$$

$$|f_C(v) - f_n(v)| \leq H_C\{b_e + 6zF(a_h + \eta) + 32zP_h\eta/\sqrt{\lambda}\} + 3\alpha F(a_h + \eta) + 16zP_h\eta. \quad (101)$$

Let K_n be the dense training tangent Gram: it is the formula (90) with dense fields and the inner products $n^{-1}u^\top v$ in each neuron layer. The dense deficit therefore obeys $\dot{c}_n = -2K_n c_n/m$. The source-only Gram defect $D = V_R^*V_R + \mathcal{J}_R^*\mathcal{J}_R - K_n/m$ satisfies $\|D\| \leq D_s\eta$. To see this without an extra factor m , split each product of two pairings, use (95), and recall that the sample matrix entries carry the factor $1/m$; a matrix all of whose entries have magnitude at most a/m has operator norm at most a . With $\Delta\mathcal{J} = \mathcal{J}_C - \mathcal{J}_R$, write

$$\Delta\mathcal{K} = V_C^*\Delta V + \Delta V^*V_R + \mathcal{J}_C^*\Delta\mathcal{J} + \Delta\mathcal{J}^*\mathcal{J}_R + D.$$

The error equations are exactly

$$\begin{aligned} \dot{e} &= -2(Q_C + \mathcal{J}_C^*\mathcal{J}_C)e - 2\Delta\mathcal{K}c_n/\sqrt{m}, \\ \dot{z}_w &= 2V_C e + 2\Delta V c_n/\sqrt{m}, \\ \dot{\zeta} &= 2\Delta V c_n/\sqrt{m} + \dot{T}_C e - 2T_C \mathcal{J}_C^* \mathcal{J}_C e - 2T_C \Delta\mathcal{K} c_n/\sqrt{m}. \end{aligned}$$

The term $2V_C e$ cancels because $T_C Q_C = V_C$. This is why estimating $c_C - c_n$ by an unconstrained Gronwall inequality would lose the result.

Here are the needed norm estimates, including the dissipative one. The derivative formula for T_C , and $\|\dot{V}_C\| \leq 10\alpha F_c \rho_C$, give $\|\dot{T}_C e\| \leq 40zF_c \rho_C \|p\|$. In the derivative of $\|p\|^2/2$, the Q_C term is $-2u^2$. The possible absolute contribution from the hidden Gram is at most $8\|\mathcal{J}_C\|^2 u^2/\lambda \leq 200z^2 F_c u^2$; the label cap bounds this by $25u^2/(32H_c^2)$, leaving at least $39u^2/32$. Define

$$\mathfrak{f} = 2\|T_C \Delta\mathcal{K} c_n/\sqrt{m}\|, \quad I(t) = \int_0^t \{40zF_c \rho_C b_e + \mathfrak{f}\}.$$

Dividing the dissipative inequality by $\|p\|$ gives

$$\|p(t)\| + \int_0^t u^2/\|p\| \leq I(t), \quad \int_0^t u \leq 2I(t)/\sqrt{\lambda}.$$

At zeros use $(\|p\|^2 + \epsilon^2)^{1/2}$ and let $\epsilon \downarrow 0$; $p = 0$ implies $e = 0$ because Q_C is invertible. The second inequality also follows directly from $\|p\| \leq 2u/\sqrt{\lambda}$. Integrating the ζ equation bounds its hidden-Gram term by $200z^2 F_c I \leq 25I/32$. Therefore

$$b_e \leq 2F \int_0^t \rho_n(a_h + \eta) + 3I(t). \quad (102)$$

For completeness, the remaining response estimate can be checked layer by layer. The dense training preactivation carriers are bounded coordinatewise by $M = 1 + 32zK_{\text{src}}\sqrt{\ell}$ on $[0, T]$. At each gate split $\phi'(z_C)k_C - \phi'(z_R)k_R$ as $\phi'(z_C)(k_C - k_R) + (\phi'(z_C) - \phi'(z_R))k_R$. The first multiplication costs $2s$, and the second at most $2t_2 M F_j^z(a_h + \eta)$. Subtracting the initialized and learned reverse actions costs $A_b\eta$, and the hidden parameter difference costs $d_{j+1}^E \alpha a_h$. Starting with (100), the displayed B_j recursion, followed by the outer-product split, gives

$$\|\Delta\mathcal{J}\| \leq B_h\{b_e + M(a_h + \eta) + \eta/\sqrt{\lambda}\}.$$

Only the carrier factor contains M ; it is not multiplied once per layer. Applying this to the four terms of $\Delta\mathcal{K}$ gives

$$\begin{aligned} \mathfrak{f} &\leq 2F\rho_n(a_h + \eta) + 4F\nu(a_h + \eta)/\sqrt{\lambda} + 4D_s\eta\rho_n/\sqrt{\lambda} \\ &\quad + 4z(J_C + J_R)B_h\rho_n\{b_e + M(a_h + \eta) + \eta/\sqrt{\lambda}\}. \end{aligned}$$

The hidden-parameter equation, $\int u \leq 2I/\sqrt{\lambda}$ and $4zJ_C \leq 5/4$ similarly yield

$$a_h \leq \frac{5}{4}I + 2B_h \int_0^t \rho_n\{b_e + M(a_h + \eta) + \eta/\sqrt{\lambda}\}.$$

Combining with (102) proves the scalar inequality

$$E(t) \leq \int_0^t \{200zF_c\rho_C + K_1M\rho_n + 20F\nu/\sqrt{\lambda}\}\{E + \eta(1 + \lambda^{-1/2})\}.$$

The integral of its coefficient is at most $\mathcal{B}_n$, by $\int \rho_n, \int \rho_C \leq 2z$ and (97). The integrating-factor formula gives $E \leq \eta(1 + \lambda^{-1/2})(e^{\mathcal{B}_n} - 1)$. Substitution in (101) proves the finite-horizon claim with precisely (94). $\square$

Canonical error bound and retained inventory. The local fitting claim supplies the tails and endpoint. Thus

$$\|f_{\text{model}} - f_n\|_{\mathcal{X}} \leq \mathcal{A}_n\eta + \mathcal{D}e^{-\lambda T/4}.$$

We verify that the explicit ledger has the promised dependence. Put $X = \beta^L \geq 100$. The local envelopes and fitting bounds give $u_0, H_c \leq X^3$, $\tau \leq X^4$ and $K_{\text{src}} \leq X^{21}$. For example $16z \leq 16X^{-30}$ makes both additional terms in A_f, A_b less than one; hence $A_f, A_b \leq 19$. The forward recursion then gives $F \leq X^6$. For further detail, $d_j^E \leq X^4$, $H_C \leq 2X^3$ and $P_h, P_\delta \leq X^4, X^5$. In the backward recursion the additive term is at most X^7 , and $18s \leq \beta^3$. Geometric summation over at most L layers gives $\max_j B_j \leq X^{11}$ and then $B_h \leq X^{15}$. Also $J_C, J_R \leq X^9$ and $D_s \leq X^5$. Substitution gives $K_1 \leq X^{18}$; here the generous powers absorb $L \leq X$, all numerical factors, and $H_C = 2H_c$. The selected fitting recursion already gives $F_c \leq X^{15}$. Consequently

$$\mathcal{B}_n \leq 400X^{-45} + 2X^{-12} + 40X^{-24} + 64X^{-21}\sqrt{\ell} \leq 1 + \sqrt{\ell},$$

and $\mathcal{B}_n/z \leq C(1 + \sqrt{\ell})$ with C depending only on X . Since $e^u - 1 \leq ue^u$, (94) gives

$$\mathcal{A}_n \leq Cz(1 + \lambda^{-1/2})(1 + \sqrt{\ell})e^{\sqrt{\ell}}, \quad \mathcal{D} \leq Cz(1 + \lambda^{-1/2}),$$

after increasing the activation/depth constant to absorb e . These are exactly the canonical factors in (86). This is a small-cap proof, not a claim on a larger recurrence-defined label interval. Finally $T = 32\ell/\lambda$ makes the tail $\mathcal{D}e^{-8\ell}$. For fixed positive Y it is at most $Y/(2n)$ eventually; the other half is (88).

Retained inventory. Retain both moving and initial copies of the first matrix, mixers and raw readout; the selected indices; each M_j , its inverse or factorization, and D_j ; the training data and labels; current training forward and backward arrays; c_C, V_C, Q_C and its inverse or factorization; the corrected readout; direction arrays; and one-query forward workspace. Every selected width is at most $9q$. Each mixer or metric therefore costs at most $81q^2$, each first matrix at most $9qd$, each layer's training arrays at most a fixed multiple of $9mq$, and each sample matrix at most m^2 .

Since $m, d \leq q$, these arrays total at most $1020(L+1)q^2 + 36q + 10m(d+1)$, including simultaneous initial copies and working arrays. A nonprimitive activation evaluator is charged separately. Dense mixers, source bases, coefficient vectors and coefficient-compilation scratch are not retained by the runtime. The construction itself uses the finite source spaces only through its initial arrays; the separate compiler below will establish their initialization-only provenance for the two methods. $\square$

6.2 Finite initialized jets suffice

Proposition 6.3 (Initialization-only coefficient compiler). *On the initialized operator event of Lemma 2.1, fix $T, r_0, M > 0$. Let finitely many sources from Proposition 3.1 extend coordinatewise holomorphically near $[-r_0, T + r_0] + i[-r_0, r_0]$, bounded by M uniformly over their compact query domain. For sphere queries assume also the joint holomorphic extension of Proposition 3.1. Any finite list of their integrals against specified polynomial time–sphere bases, or derivatives at real anchors in $[0, T]$, can be approximated to arbitrary positive accuracy using finitely many derivatives of the dense ODE and passive recursions at time zero, with exact initialized-image pairing. No later dense state or trajectory sample is needed; no bound on initialization time or working storage is asserted.*

Proof. We first prove the continuation assertion that makes initialized jets sufficient. For any one scalar source g , use its common bound M on the closed rectangle $\{-r_0 \leq \Re t \leq T + r_0, |\Im t| \leq r_0\}$, with a slightly larger open domain. The source rectangle, or a finite closed-disk cover with the real interval strictly inside its union, supplies such a positive r_0 by compactness. Put

$$\chi = \frac{\pi T}{8r_0}, \quad b_0 = \tanh \chi, \quad b_1 = \tanh(\chi + \pi/4), \quad \theta = b_0/b_1, \quad \psi(\xi) = \frac{\xi - \theta}{1 - \theta\xi},$$

$$\mathfrak{t}(\xi) = \frac{T}{2} + \frac{2r_0}{\pi} \log \frac{1 + b_1\psi(\xi)}{1 - b_1\psi(\xi)}, \quad \xi_* = \frac{2\theta}{1 + \theta^2} < 1.$$

The logarithm is the branch equal to a real number on the real interval. For $|\xi| < 1$, $|\psi(\xi)| < 1$; the ratio in the logarithm has positive real part and modulus between $(1 - b_1)/(1 + b_1)$ and its reciprocal. Thus $|\Im \mathfrak{t}| < r_0$ and $-r_0 < \Re \mathfrak{t} < T + r_0$. Direct substitution gives $\mathfrak{t}(0) = 0$ and $\mathfrak{t}(\xi_*) = T$, and the derivative is nonzero on $[0, \xi_*]$. This proves continuation from the origin onto the entire physical interval, without stepping a dense trajectory.

The Taylor coefficients of $g \circ \mathfrak{t}$ are

$$a_j = \sum_{k=0}^j \frac{g^{(k)}(0)}{k!} [\xi^j] \mathfrak{t}(\xi)^k.$$

Cauchy’s coefficient formula gives $|a_j| \leq M$, so truncation at N has error at most $M\xi_*^{N+1}/(1 - \xi_*)$ on the whole interval. This bound is uniform over all passive inputs in the compact domain and over all source coordinates. Derivatives at finitely many later anchors are also recovered: choose $\xi_* < \rho_* < 1$ and $0 < \epsilon_* < \rho_* - \xi_*$. On $|\xi| \leq \rho_*$ the tail is $M\rho_*^{N+1}/(1 - \rho_*)$, hence its k th derivative at any real anchor is at most $k!M\rho_*^{N+1}/\{\epsilon_*^k(1 - \rho_*)\}$. Repeated application of $(\mathfrak{t})^{-1}\partial_\xi$ converts these to the finitely many required time derivatives. All its coefficients are bounded on that compact interval. A finite N therefore achieves every prescribed finite list of accuracies.

Let $\Theta = (W^{(1)}, \dots, W^{(L)}, w)$ be the dense parameter tuple and $\mathcal{F}$ its autonomous vector field. For the initialized ODE $\dot{\Theta} = \mathcal{F}(\Theta)$ the formal coefficients are computed successively by

$$\Theta_{j+1} = \frac{1}{j+1} [t^j] \mathcal{F} \left(\sum_{i=0}^j \Theta_i t^i \right).$$

The analytic local solution has precisely these coefficients, by coefficient comparison and uniqueness of the local ODE. The forward and passive backward recursions then give their source jets by finite addition, multiplication and composition at time zero. Any scalar activation derivatives required here are part of the fixed analytic activation evaluator; their description, evaluation and scratch are not asserted to have unit cost unless taken as primitives.

For time–sphere integral coefficients, insert the uniformly convergent continued polynomials into their defining integrals. Each integrand is continuous on the compact time interval times sphere. Finite partitions and weighted Riemann sums converge uniformly for the finite collection of integrands. A finite mesh can be chosen from their modulus of continuity: the source strip gives derivative bounds by Cauchy’s formula, and the finitely many basis functions are known polynomials. Thus the required integral approximations use only finitely many evaluations of the continued origin-jet polynomial. For $d = 1$ the spatial integral is simply the two-point average.

Finally perform every scalar linear coefficient operation on a base source and set its image coefficient equal to the initialized matrix times that vector. Linear operations commute with this fixed matrix. Since its Euclidean operator norm is at most eight, approximating a base coefficient in Euclidean norm to $\epsilon/8$ approximates both the base and its image to ϵ . There are only finitely many coefficients and a finite sum of their basis suprema; choose each coefficient tolerance smaller than the desired final error divided by this sum. This imposes exact pairing while retaining the chosen uniform source tolerance. No coefficient is retained after the selected initial arrays are formed. $\square$

6.3 Whole-sphere polynomial sources

Lemma 6.4 (Joint time–sphere approximation and its dimension). *Let $T, r_t, r_q, M_n > 0$ and $0 < \eta \leq \min\{1, M_n\}$, and put $\alpha_T = r_t/(4T)$. Suppose a real $\mathbb{R}^n$ -valued source on $[0, T] \times S^{d-1}$ extends coordinatewise holomorphically to a neighborhood of*

$$([-r_t, T + r_t] + i[-r_t, r_t]) \times \{v \in \mathbb{C}^d : v^\top v = 1, \|\operatorname{Im} v\|_2 \leq \sinh r_q\},$$

with coordinate modulus at most M_n on this set. Query holomorphy is relative to the quadric; for $d = 1$ use its two real points. For $d \geq 2$ put

$$b_d = 2d - 2, \quad D_d = 2^{d+1}d^{d-2}, \quad P_T = \frac{18D_db_d!2^{b_d+1}}{\alpha_T r_q^{b_d+1}}, \quad H_* = 2 \log \frac{16M_n P_T}{\eta}.$$

For $\alpha_T, r_q \leq 1$, a real expansion in Chebyshev time polynomials and real spherical harmonics with $\alpha_T k + r_q j \leq H_$ approximates the source to coordinate error $\eta/8$ using at most*

$$N \leq \frac{2\{H_* + \alpha_T + (d-1)r_q\}^d}{d! \alpha_T r_q^{d-1}} \tag{103}$$

coefficient vectors. For the dense sources of Proposition 3.1, Proposition 6.3 gives initialized-jet coefficients with total coordinate error at most η and exact initialized-image pairing. For $d = 1$, use the two points of S^0 and at most $2N_1$ vectors, where $N_1 = 1 + \lfloor H_1/\alpha_T \rfloor$ and $H_1 = \log(64M_n/(\alpha_T \eta))$.

Proof. Fix one coordinate and call its scalar source g . We supply the spatial decay argument. For $d \geq 2$ let $\mathcal{H}_j$ be the restrictions of homogeneous harmonic polynomials of degree j to S^{d-1} , with probability surface measure. Its dimension is

$$h_j = \binom{j+d-1}{d-1} - \binom{j+d-3}{d-1} \leq 2 \binom{j+d-2}{d-2},$$

where a binomial coefficient with negative upper index is zero. Indeed, applying the Laplacian to $|x|^{2k}H_l(x)$ gives $2k(2l+2k+d-2)|x|^{2k-2}H_l(x)$; induction decomposes every homogeneous polynomial into these summands and gives the dimension formula by subtraction. The spherical Laplacian eigenvalues $-j(j+d-2)$ give orthogonality of different degrees by integration by parts. Polynomials contain constants and separate points on the compact sphere, so their uniform density (the real Stone–Weierstrass theorem) proves completeness. For an orthonormal basis $Y_{j\nu}$, rotational invariance and integration give the addition identity $\sum_\nu Y_{j\nu}(v)^2 = h_j$; consequently $|Y_{j\nu}(v)| \leq \sqrt{h_j}$ and the projection Π_j has supremum operator norm at most h_j .

For fixed v , average an analytic function over the complex tangent circle $\cos \zeta v + \sin \zeta u$, $u \perp v$, $|u| = 1$. On $\mathcal{H}_j$ this averaging is scalar: rotations fixing v leave a one-dimensional zonal eigenspace, and its normalized eigenfunction solves $(1-x^2)p'' - (d-1)xp' + j(j+d-2)p = 0$, $p(1) = 1$. Coefficient comparison gives $p(x) = C_j^{(d-2)/2}(x)/C_j^{(d-2)/2}(1)$ for $d > 2$. For real ζ the averaging is self-adjoint because the joint law of its two endpoints is symmetric. Apply this to each harmonic coefficient and continue analytically in ζ ; it follows that the averaged analytic source has j th projection equal to $p(\cos \zeta)\Pi_j g$. At $\zeta = ir_q$ its arguments satisfy $z^\top z = 1$, $|\Im z| \leq \sinh r_q$, so its supremum is at most M_n . Thus $\|\Pi_j g\|_\infty \leq h_j M_n / p(\cosh r_q)$.

For $\nu = (d-2)/2 > 0$, the generating function $(1-2xw+w^2)^{-\nu} = \sum_j C_j^\nu(x)w^j$ at $x = \cosh r$, for any $r > 0$, factors as $(1-e^r w)^{-\nu}(1-e^{-r} w)^{-\nu}$ with nonnegative coefficients. It gives $C_j^\nu(\cosh r) \geq e^{rj}(\nu)_j/j!$ and $C_j^\nu(1) = (2\nu)_j/j!$. For $j \geq 1$ their ratio is $2 \prod_{i=1}^{j-1} (1 + \nu/(\nu+i))$. Its logarithm is at most $\log 2 + \nu \int_0^1 (\nu+x)^{-1} dx$. Since $\nu \geq 1/2$, the ratio is at most $2(2j)^\nu$, and hence at most $2^d(j+1)^d$; the $j=0$ inequality is immediate. Together with $h_j \leq 2d^{d-2}(j+1)^{d-2}$ this proves

$$\|\Pi_j g\|_\infty \leq M_n D_d (j+1)^{b_d} e^{-r_q j}. \quad (104)$$

For $d=2$, the basis is the constant and the sine–cosine pair at each positive integer, and the averaging multiplier is $\cosh(jr) \geq e^{jr}/2$; the same loose bound with $D_2 = 8$, $b_2 = 2$ holds directly.

Set $t = T(1 + \cos u)/2$. For $|\Im u| \leq \alpha_T \leq 1$ this lies inside the time rectangle, since its displacement from $[0, T]$ is at most $T\alpha_T$. The resulting even periodic function has cosine coefficients whose j th harmonic block is bounded by $2M_n D_d (j+1)^{b_d} e^{-\alpha_T k - r_q j}$: shift the Fourier contour to $\Im u = \pm \alpha_T$ and use (104) there. The series is absolutely uniformly convergent. Discarding indices with $\alpha_T k + r_q j > H_*$ bounds the tail by

$$2M_n D_d e^{-H_*/2} \sum_{k \geq 0} e^{-\alpha_T k/2} \sum_{j \geq 0} (j+1)^{b_d} e^{-r_q j/2} \leq M_n P_T e^{-H_*/2} \leq \eta/16.$$

Here $\sum_k e^{-\alpha_T k/2} \leq 3/\alpha_T$ and $\sum_j (j+1)^{b_d} e^{-r_q j/2} \leq 3b!(2/r_q)^{b+1}$, obtained by comparison with the gamma integral after shifting each unit interval.

The number of retained coefficients is $N = \sum_{r_q j \leq H_*} h_j (1 + \lfloor (H_* - r_q j)/\alpha_T \rfloor)$. Replace h_j by twice the number of weak compositions of j into $d-1$ parts. The corresponding nonnegative lattice points have weighted sum $\alpha_T k + r_q \sum l_i \leq H_*$. Their disjoint unit cubes lie in the simplex with right-hand side $H_* + \alpha_T + (d-1)r_q$, whose volume is exactly the right-hand side of (103). This proves the count. For the finite collection of coefficients, choose the Euclidean approximation tolerance in Proposition 6.3 so that its product with $8 \sum_{k,j,\nu} \|Y_{j\nu}\|_\infty$ is at most $\eta/16$. Both a base source and its paired image then have total error below η , without increasing the coefficient count.

If $d=1$, there is no spatial approximation: apply the preceding one-variable Fourier contour argument separately at the two points. The geometric tail after degree $\lfloor H_1/\alpha_T \rfloor$ is at most $6M_n e^{-H_1}/\alpha_T < \eta/8$; the same finite coefficient accuracy argument finishes this case. $\square$

Proposition 6.5 (Harmonic: whole-sphere accuracy and total state). *Eventually at each fixed confidence, the initialized-jet construction gives a selected nonlinear model with $\|f_{\text{Harm}} - f_n\|_{\mathcal{X}} \leq Y/n$ on the whole sphere and total retained real-coordinate count at most*

$$C(m+d)^2 + \left(\frac{C}{d}\right)^{d+1} \left(\frac{Ym}{\gamma}\right)^4 [\log(en)]^{3d+2}.$$

Proof. For this whole-sphere construction put $\lambda = \gamma/m$, $z = Y/\lambda$, $\ell = \log(en)$, and $T = 32\ell/\lambda$. Choose η by the canonical tolerance formula (88). The public source proposition supplies the radii and coordinate bound

$$r_t = \frac{\gamma}{\beta^{30L} Y^2 m \sqrt{(d+3)\ell}}, \quad r_q = \frac{1}{\beta^{3L} \sqrt{(d+3)\ell}}, \quad M_n = \beta^{6L} \sqrt{n}.$$

Set $\alpha_T = r_t/(4T)$. For $d \geq 2$, let N be the coefficient count in Lemma 6.4, with its explicitly defined P_T, H_* evaluated at these local parameters. For $d = 1$, let N_1 be the two-point time count defined in that same statement. Apply the approximation lemma to the four joint time-sphere families from Proposition 3.1, including its passive backward family; hence the same coefficient vectors serve all training inputs. In layer j include the approximate feature and response coefficients and the paired initialized forward and reverse image coefficients. Add $\mathbf{1}_n$, the at most $2m$ exact initialized training vectors needed for forward-image pairing, and the d first-layer columns. Each source space has rank at most

$$R = 2m + d + 1 + 4N \quad (d \geq 2), \quad R = 2m + d + 1 + 8N_1 \quad (d = 1).$$

The source contract is now satisfied. Invoke Proposition 6.2 with its source-dimension bound $q = R$; it selects at most $9R$ coordinates per layer and gives the claimed absolute error. All choices use initialized jets only.

We spell out the asymptotic count because dropping the label factor here would change the theorem. The exact source radius gives

$$\alpha_T^{-1} = \frac{4T}{r_t} = 128\beta^{30L} z^2 \sqrt{d+3} \ell^{3/2}, \quad r_q^{-1} = \beta^{3L} \sqrt{(d+3)\ell}. \quad (105)$$

For each fixed admissible problem, (86) and (88) imply $\log(1/\eta) \leq 2\ell$ eventually. Every other fixed parameter inside the logarithm defining H_* is additive, whereas $\log P_T = O_{d,\text{problem}}(\log \ell)$. Thus $H_* + \alpha_T + (d-1)r_q \leq 8\ell$ eventually. The dimension bound gives

$$N \leq C^d \beta^{CLd} \frac{(d+3)^{d/2}}{d!} z^2 \ell^{3d/2+1}.$$

For $d = 1$, the direct two-point count gives the same form with $N_1 \leq C\beta^{CL} z^2 \ell^{5/2}$.

The public inventory (87), with $q = R$, is $1020(L+1)R^2 + 36R + 10m(d+1)$, plus the separately charged activation evaluator. It already includes selected mixers, metrics, initial copies, data and working arrays. No dense mixer, source basis, initialized jet, quadrature table or coefficient vector is retained.

Finally, for $d \geq 2$, $R^2 \leq C(m+d)^2 + CN^2$ and $d! \geq (d/e)^d$ imply

$$\frac{(d+3)^d}{(d!)^2} \leq \frac{(8e^2)^d}{d^{d+1}}.$$

Indeed, for $d \geq 2$, $(1+3/d)^d \leq (5/2)^d$ and $d \leq 2^d$ give the stronger numerator $(5e^2)^d$. Absorb activation/depth powers into C^d to obtain the asserted count. For $d = 1$, use $R^2 \leq C(m+1)^2 + CN_1^2$ and the already proved bound on N_1 ; this gives the stated logarithmic exponent five. If a finite small width makes $9R \geq n$, retaining all coordinates is a valid fallback; the claimed polylogarithmic bound concerns the eventual range only. $\square$

6.4 Finite-panel sources on an adaptive time partition

Proposition 6.6 (Taylor: panel accuracy and total state). *On the declared $(m + p)$ -input panel of Section 1, eventually at each fixed confidence there is an initialization-only selected model with $\|f_{\text{Taylor}} - f_n\|_{\mathcal{X}} \leq Y/n$ and total state at most*

$$C(m + p)^2 \left[\left(\frac{Ym}{\gamma} \right)^4 [\log(en)]^3 + \left(\frac{m}{\gamma} \right)^2 [\log(en)]^2 \{\log(e + \log(en))\}^2 \right] + C(m + p)d.$$

Only training inputs enter the deficit equation and Gram inverse. Accuracy outside the panel is not asserted.

Proof. For this finite-panel construction put $\lambda = \gamma/m$, $z = Y/\lambda$, $\ell = \log(en)$ and $T = 32\ell/\lambda$. Use the public disk-and-partition interface of Proposition 3.1: there are $0 = t_0 < \dots < t_J = T$ and radii $r_b > 0$ such that each source is holomorphic near $|t - t_b| \leq r_b$ and $t_{b+1} - t_b \leq r_b/2$. Its count (28) gives

$$J \leq 1 + \beta^{30L} z^2 \sqrt{\ell} + \frac{\beta^{16L}}{\lambda} \log(1 + 4\ell/\log 2). \quad (106)$$

The centers and radii are deterministic functions of the problem parameters, as that public interface guarantees, so they require no later dense training data.

There are at most $2(2m + p)$ source curves per layer: forward features and their initialized images at all declared inputs, and backward responses and their initialized transpose images at training inputs. Each coordinate is bounded by $M_n = \beta^{6L} \sqrt{n}$ on its disks. Choose the source tolerance (88) and

$$K = \max\{0, \lceil \log_2(8M_n/\eta) \rceil\}.$$

On panel b , retain coefficient vectors $r_b^k u^{(k)}(t_b)/k!$ for $0 \leq k \leq K$, where u denotes one of the source curves. They have coordinate magnitude at most M_n . Since $|t - t_b|/r_b \leq 1/2$, the discarded Taylor tail is at most $M_n 2^{-K} \leq \eta/8$. The vectors from all panels span one time-independent space of dimension at most $J(K + 1)$ per curve. The local approximants may differ at a panel boundary: both approximate the same source there, and the comparison theorem uses values, never their derivatives or a boundary matching rule.

Apply Proposition 6.3 to approximate every scaled base coefficient in Euclidean norm to $\eta/128$, and form its initialized image exactly. The base reconstruction error is at most $(\eta/128) \sum_{k \geq 0} 2^{-k} = \eta/64$; initialized operator norm eight gives image error at most $\eta/8$. Adding their independently controlled Taylor tails leaves both below η . Include the constant, exact initial training vectors and paired images, and first-weight columns required by Proposition 6.2. Before input reduction the rank per layer is at most $2m + d + 1 + 2(2m + p)J(K + 1)$. This verifies the identical source contract, so the same selected runtime, fitting, cancellation, and endpoint proofs apply. In particular taking a maximum over the passive inputs does not introduce p into any comparison coefficient. No passive deficit or additional sample Gram is introduced.

There is an exact reduction of the first-layer dimension. Let $U_0 \in \mathbb{R}^{d \times k}$ have orthonormal columns spanning all declared normalized inputs, where $k \leq \min(d, m + p)$. Replace v_i by $U_0^\top v_i$ and the first matrix by $W^{(1)} U_0$. Every first-layer derivative is a sum of outer products with v_a , hence the component on the orthogonal complement of this input span is constant. Induction through the layers proves equality of the original and reduced coupled trajectories on the panel. The reduced inputs remain unit vectors; there is no new normalization factor. Each initialized Gaussian row multiplied by U_0 is exactly standard Gaussian in $\mathbb{R}^k$, and the sample inner products and population feature Gram are unchanged. Thus the same source event applies to the reduced system, whose

public panel-count coefficients depend only on activations and depth, not on k or d . The source rank is now

$$R = 2m + k + 1 + 2(2m + p)J(K + 1).$$

Invoke Proposition 6.2 with source-dimension bound $q = R$ and input dimension k . To its public inventory (87), add the passive inputs and output buffer and retain the input map. A safe explicit count is

$$1020(L + 1)R^2 + 36R + 10m(k + 1) + pk + (m + p) + 3(m + p)d,$$

plus the separately charged nonprimitive activation evaluator. This includes original and transformed data, the map, all metrics and inverse caches, initial and moving weights, direction arrays, training feature/response and sample-matrix buffers, and query workspace. Discard the dense arrays, source coefficients, origin jets, panel anchors and partition schedule. The optimizer is autonomous; a partition used to prove source approximation is not a retained forcing table.

For each fixed problem, (88) implies $K + 1 \leq 8\ell$ eventually. The coefficients β^{30L} and β^{16L} in (106) depend only on activations and depth. Squaring that bound with $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$, and using $k \leq m + p$, yields

$$R^2 \leq C(m + p)^2 \left[z^4 \ell^3 + \lambda^{-2} \ell^2 \{\log(e + \ell)\}^2 \right].$$

The second term absorbs the initial rank and plain ℓ^2 terms because $\lambda \leq 1$. Crucially, the z^2 in the first term of the panel count was retained before squaring, so its storage contribution remains $z^4 = (Ym/\gamma)^4$; it was not replaced by a label-independent constant. This proves the stated inventory and finishes the construction. $\square$

7 Nonlinearity and feature learning

We prove Theorem 1.2 using the real bounds already established in Lemma 2.1. All auxiliary notation below is local; constants and $O(\cdot)$ bounds may depend on the entire fixed problem, but not on width. The key simplification is to test feature increments against their initial adjoints, rather than follow every feature acceleration.

Lemma 7.1 (Nonlinear initial velocity). *Every $Q^{(\ell)}$, $\ell \geq 1$, is positive definite. The limiting initial prediction velocity is nonaffine on the sphere, witnessed by an affine-annihilating combination of four deterministic inputs chosen before initialization.*

Proof. Write $v = x/\sqrt{d}$. Gaussian covariance recursion gives $Q_{ab}^{(\ell)} = F_\ell(v_a^\top v_b)$, $F_0(s) = s$, with

$$F_\ell(s) = \sum_{k \geq 0} c_{\ell,k} s^k, \quad c_{\ell,k} \geq 0, \quad \sum_k c_{\ell,k} < \infty, \quad (107)$$

and unbounded positive coefficient support. Here is a direct justification. The orthonormal Gaussian Hermite expansion of $\phi_\ell(\sqrt{q}G)$ has coefficients α_k and covariance $\sum_k \alpha_k^2 r^k$ for Gaussian correlation r . The covariance identity follows by comparing coefficients in $\mathbb{E}[e^{zG - z^2/2} e^{wG' - w^2/2}] = e^{rzw}$. Completeness follows because a function orthogonal to all polynomials has entire transform $\mathbb{E}[h(G)e^{zG}]$ identically zero, hence zero Fourier transform. Linear growth gives Gaussian L^2 integrability. Finite Hermite support would make the activation a polynomial, hence affine by its bounded derivative. Composing the covariance series with $F_{\ell-1}(s)/F_{\ell-1}(1)$ proves (107): Tonelli at $s = 1$ gives summability, and a positive inner coefficient of degree $j \geq 1$ and outer degree k produces degree jk . These arguments allow nonzero means and justify uniform absolute convergence on $[-1, 1]$.

The tensor Grams $[(v_a^\top v_b)^k]$ are positive semidefinite and converge to I_m by the nonparallel-input hypothesis. Thus every $Q^{(\ell)} \succ 0$.

Let P be the orthogonal projection onto affine functions in spherical L^2 , with normalized surface measure σ : $Ph(v) = \int h d\sigma + d v^\top \int u h(u) d\sigma(u)$. Projecting the tensor feature in both variables gives the positive kernel

$$R_k(v, u) = (v^\top u)^k - a_k - b_k v^\top u, \quad a_k = \int (z^\top u)^k d\sigma(z), \quad b_k = d \int (z^\top u)^{k+1} d\sigma(z).$$

Here $a_k, b_k \rightarrow 0$ by dominated convergence, since $d \geq 2$, so $[R_k(v_a, v_b)] \rightarrow I_m$. The limiting velocity $g(v) = (2/m) \sum_a y_a F_L(v^\top v_a)$ is not affine: otherwise

$$0 = \sum_a y_a [(I - P)g](v_a) = \frac{2}{m} \sum_k c_{L,k} y^\top [R_k(v_a, v_b)] y > 0.$$

Every term is nonnegative and some large-degree term is positive. Uniform convergence justifies the projection and sums.

Some great circle witnesses this nonaffinity. Otherwise the antipodal average of g is constant, and the odd remainder extends homogeneously to a function linear on every plane, hence globally linear. On such a circle, interpolate three distinct points by an affine function and take a fourth where it fails. The resulting dependence can be signed and normalized so that, for unit vectors u_i ,

$$\sum_i |c_i| = 1, \quad \sum_i c_i = 0, \quad \sum_i c_i u_i = 0, \quad \sum_i c_i g(u_i) > 0. \quad (108)$$

These inputs depend on the fixed problem only. The finite-query Gram convergence proved in Lemma 2.1, applied to this augmented list, and $\dot{f}_n(0, x) = (2/mn) \sum_a y_a h_a^{(L)}(0)^\top h^{(L)}(0, x)$ give convergence of their initial velocities. $\square$

Lemma 7.2 (Initial backward fields and hidden forces). *The initialized forward fields and first backward derivatives, including the vectors before derivative gates, have deterministic joint empirical row-law limits with second moments. This includes the full first-layer weight and acceleration rows. Every hidden block's squared initial acceleration in inverse-mobility norm converges to a positive constant.*

Proof. All unmarked fields in this proof are at zero. Put $k = 2/m$ and define the pre-gated vectors P and their gated versions B by

$$P_a^{(L)} = \sum_b y_b h_b^{(L)}, \quad B_a^{(\ell)} = \phi'_\ell(z_a^{(\ell)}) \odot P_a^{(\ell)}, \quad P_a^{(\ell-1)} = W^{(\ell)\top} B_a^{(\ell)}.$$

Thus $\dot{\delta}_a^{(\ell)}(0) = k B_a^{(\ell)}$. Differentiating (3), since all hidden velocities initially vanish, gives

$$\ddot{W}^{(1)}(0) = k^2 \sum_a y_a B_a^{(1)} v_a^\top, \quad \ddot{W}^{(\ell)}(0) = \frac{k^2}{n} \sum_a y_a B_a^{(\ell)} h_a^{(\ell-1)\top}. \quad (109)$$

Only a forward pass and a backward pass are needed for their joint limits. For one hidden Gaussian matrix W , after revealing $WH = Z$, its conditional law is $Z(H^\top H)^{-1} H^\top + \widetilde{W}(I - \Pi_H)$, where $\Pi_H = H(H^\top H)^{-1} H^\top$ and $\widetilde{W}$ is fresh with entries $N(0, 1/n)$. This is orthogonal Gaussian conditioning on the linear constraint. For a reverse query U , known from the previous answers,

$$W^\top U \stackrel{d}{=} H Q_n^{-1} C_n + (I - \Pi_H) \Xi D_n^{1/2}, \quad Q_n = H^\top H/n, \quad C_n = Z^\top U/n, \quad D_n = U^\top U/n, \quad (110)$$

with independent standard Gaussian rows in Ξ . The removed projection has conditional squared RMS rank(H) $\text{tr}(D_n)/n$, hence vanishes in probability. Adaptive queries cause no extra dependence: conditional on the global transcript, the next query is fixed; revealing one Gaussian action leaves the other matrices' residuals independent. In forward-then-backward order no matrix is queried again after its reverse answer.

For precision, the induction uses empirical quadratic Wasserstein convergence. Appending independent Gaussian rows preserves a deterministic such limit: conditional variances for bounded tests are $O(1/n)$, and the appended second moment obeys the law of large numbers. Continuous coordinate maps of at most linear growth preserve this convergence, by an L^2 coupling and uniform integrability. These include $\phi(z)$ and $\phi'(z)u$. Same-layer products consequently converge; inverse Grams and positive covariance square roots converge when their limiting Grams are, respectively, positive definite or positive semidefinite. The vanishing projected term in (110) does not change the row-law limit. Starting with the full iid first-layer rows, these rules prove the forward limits and then the backward limits, including all pre-gated fields. The forward Grams being inverted are positive by Lemma 7.1; $Q^{(0)}$ is never inverted. The first-row acceleration is a linear combination of the B fields, so its joint limit is included.

Let $D^{(\ell)}$ be the limiting Gram of $B^{(\ell)}$. At the top it is positive definite: a null vector c would give

$$\left(\sum_b y_b \phi_L(z_b)\right) \left(\sum_a c_a \phi'_L(z_a)\right) = 0 \quad (z \in \mathbb{R}^m).$$

Indeed $L \geq 2$ gives full Gaussian support. The first factor is not identically zero since $y^\top Q^{(L)} y > 0$; analyticity forces the second to vanish everywhere. Varying each coordinate, the nonconstant derivative gives $c = 0$. Downward, the fresh reverse innovation in (110) is independent of the lower-layer fields, with covariance $D^{(\ell+1)}$, so

$$c^\top D^{(\ell)} c \geq \lambda_{\min}(D^{(\ell+1)}) \sum_a c_a^2 \mathbb{E}[\phi'_\ell(Z_a^{(\ell)})^2] > 0.$$

Each Gaussian marginal is nondegenerate and a nonzero analytic derivative has a zero set of measure zero. Finally, expanding (109), the layer- ℓ squared acceleration limit is $k^4 y^\top (Q^{(\ell-1)} \circ D^{(\ell)}) y > 0$: for $Q \succeq 0$, $D \succ 0$, $Q \circ D \succeq \lambda_{\min}(D) \text{diag}(Q)$ by decomposing Q into rank-one terms, and every relevant Q has positive diagonal. This also covers singular input Grams. $\square$

Lemma 7.3 (Adjoint test of the actual motion). *Every hidden layer satisfies (14). On the training set,*

$$y^\top (f_n(t) - f_{\text{NTK},n}(t)) = \frac{m}{3} t^3 \left[\frac{\|\ddot{W}^{(1)}(0)\|_F^2}{n} + \sum_{\ell=2}^L \|\ddot{W}^{(\ell)}(0)\|_F^2 \right] + o_*(t^3). \quad (111)$$

Here $o_*(t^j)$ means: for every $\varepsilon > 0$, some deterministic $T_\varepsilon > 0$ makes the probability that the normalized remainder exceeds ε anywhere on $0 < t \leq T_\varepsilon$ tend to zero.

Proof. Use $\|u\|_n = \|u\|_2/\sqrt{n}$ and the inverse-mobility norm $\|U\|_{\text{mob}}^2 = \|U^{(1)}\|_F^2/n + \sum_{\ell \geq 2} \|U^{(\ell)}\|_F^2$ on hidden matrices. On the event of Lemma 2.1, all relevant operators, feature RMS and residuals have deterministic bounds. The readout equation therefore gives $\|w(t)\|_n = O(t)$; backpropagation gives $\|\delta_a^{(\ell)}(t)\|_n = O(t)$; integration of the hidden updates and forward subtraction give, uniformly on the sphere for $0 \leq t \leq 1$,

$$\|W(t) - W(0)\|_{\text{mob}} = O(t^2), \quad \|\Delta z^{(\ell)}\|_n + \|\Delta h^{(\ell)}\|_n = O(t^2), \quad f_n(t, x) = t \dot{f}_n(0, x) + O(t^2). \quad (112)$$

Here Δ means the increment from zero. For the last assertion, the readout equation gives $\|\dot{w}(t) - \dot{w}(0)\|_n = O(t)$; integrate and use $f = w^\top h/n$. No new bootstrap is needed.

The remainder control uses one elementary estimate. For bounded Lipschitz g and an initial vector u ,

$$\|[g(z) - g(z_0)] \odot u\|_n^2 \leq \text{Lip}(g)^2 D^2 \|z - z_0\|_n^2 + 4\|g\|_\infty^2 \|u \mathbf{1}_{|u|>D}\|_n^2. \quad (113)$$

Split at $|u_i| = D$. The deterministic row-law limits with second moments in Lemma 7.2 give, for every $\eta > 0$, a fixed cutoff D for which the probability that any needed initial squared tail exceeds η tends to zero. Choose D first, then a small time in (112). Thus gate differences times initial fields are $o_*(1)$, uniformly also along the segment from z_0 to z .

For this proof define $k = 2/m$, $P_a^{(L)} = \sum_b y_b h_b^{(L)}(0)$, $B_a^{(\ell)} = \phi'_\ell(z_a^{(\ell)}(0)) \odot P_a^{(\ell)}$, and $P_a^{(\ell-1)} = W^{(\ell)}(0)^\top B_a^{(\ell)}$. Since $w(t)/t = \dot{w}(0) + O_{\|\cdot\|_n}(t)$, downward subtraction of the backward recursion yields

$$\delta_a^{(\ell)}(t) = ktB_a^{(\ell)} + o_{*,\|\cdot\|_n}(t), \quad \Delta W = \frac{1}{2}t^2 \ddot{W}(0) + o_{*,\|\cdot\|_{\text{mob}}}(t^2). \quad (114)$$

Indeed each backward difference is the next-layer error through bounded operators, an $O(t^2)$ operator increment acting on B , and a gate difference acting on P , controlled by (113). Insert this into the hidden updates with $r = -y + O(t)$ and $h = h(0) + O_{\|\cdot\|_n}(t^2)$, then integrate. The normalized outer-product identities $\|uv^\top\|_F/\sqrt{n} = \|u\|_n\|v\|_2$ and $\|uh^\top\|_F/n = \|u\|_n\|h\|_n$ justify the matrix-norm remainder.

Instead of expanding each forward feature, test its increment against P . Put $E_{1,n} = \|\ddot{W}^{(1)}(0)\|_F^2/n$ and $E_{\ell,n} = \|\ddot{W}^{(\ell)}(0)\|_F^2$ for $\ell \geq 2$. The integral formula for an activation increment and (113) give

$$\langle P_a^{(\ell)}, \Delta h_a^{(\ell)} \rangle_n = \langle B_a^{(\ell)}, \Delta z_a^{(\ell)} \rangle_n + o_*(t^2), \quad \langle u, v \rangle_n = u^\top v/n.$$

The error is bounded by the RMS gate difference times P , multiplied by $\|\Delta z\|_n = O(t^2)$. In $\Delta z = \Delta W h(0) + W(0) \Delta h + \Delta W \Delta h$, the last term is $O(t^4)$ in RMS. Summing the first term's pairings with $y_a B_a$ and using (109) gives $t^2 E_{\ell,n}/(2k^2) + o_*(t^2)$; the second pairs the lower feature increment with $P_a^{(\ell-1)}$. Telescoping to the first layer yields

$$\sum_a y_a \langle P_a^{(\ell)}, \Delta h_a^{(\ell)} \rangle_n = \frac{t^2}{2k^2} \sum_{j \leq \ell} E_{j,n} + o_*(t^2). \quad (115)$$

All energies have positive limits. Cauchy–Schwarz, with bounded $\sum_a y_a^2 \|P_a^{(\ell)}\|_n^2$, therefore proves the feature-motion lower bound in every layer on one deterministic small interval.

The tangent kernel is the readout Gram plus the hidden terms

$$K_{ab}(t) = \langle h_a^{(L)}, h_b^{(L)} \rangle_n + (v_a^\top v_b) \langle \delta_a^{(1)}, \delta_b^{(1)} \rangle_n + \sum_{\ell=2}^L \langle \delta_a^{(\ell)}, \delta_b^{(\ell)} \rangle_n \langle h_a^{(\ell-1)}, h_b^{(\ell-1)} \rangle_n.$$

Equation (112) gives $K(t) - K(0) = O(t^2)$. Its readout part, contracted with y , is twice the top pairing in (115), plus $O(t^4)$. By (114) and the squared outer products in (109), the hidden contribution has the same coefficient. Consequently

$$y^\top [K(t) - K(0)] y = \frac{2t^2}{k^2} \sum_{\ell} E_{\ell,n} + o_*(t^2).$$

Finally $\dot{f}_n = kK(t)(y - f_n)$, whereas the frozen flow uses $K(0)$. Variation of constants gives their difference as $k \int_0^t e^{-kK(0)(t-s)} [K(s) - K(0)](y - f_n(s)) ds$. Here $f_n(s) = O(s)$ and the exponential is $I + O(t-s)$ because $K(0)$ is bounded. Pair with y and integrate the last display. The additional terms are $O(t^4)$, giving $2/(3k) = m/3$ in (111). $\square$

Lemma 7.4 (The nonlinear feature component moves). *Let $V = \text{span}\{x_1, \dots, x_m\}$, with normalized surface measure σ_V on its unit sphere. For sufficiently small fixed $t > 0$, with probability tending to one,*

$$\frac{1}{n} \sum_i \inf_{a \in V, b \in \mathbb{R}} \int_{S(V)} |h_i^{(1)}(t, \sqrt{d}v) - h_i^{(1)}(0, \sqrt{d}v) - a^\top v - b|^2 d\sigma_V(v) \geq c_* t^4. \quad (116)$$

Proof. Since $\dim V \geq 2$, for nonzero $g, r \in V$ the function $\phi'_1(g^\top v)(r^\top v)$ is not affine on $S(V)$. Otherwise its values on $r^\perp$ and antipodes force the affine function to be $\lambda r^\top v$. Off the equator, division and then continuity make ϕ'_1 constant on $[-\|g\|, \|g\|]$, contradicting analyticity and nonaffineness. Project the initial first-layer row onto V and call it g_i ; call its acceleration row $r_i \in V$. By Lemma 7.2, their joint empirical law converges with second moments; g is nondegenerate Gaussian and $\mathbb{E}\|r\|^2 > 0$. The squared distance of $\phi'_1(g^\top v)(r^\top v)$ from affine functions in $L^2(\sigma_V)$ is continuous in (g, r) , bounded by $\|\phi'_1\|_\infty^2 \|r\|^2$, and positive when both are nonzero. Its empirical mean therefore tends to a positive constant.

The first-layer expansion in (114) and bounded ϕ'_1 reduce the actual increment to the fixed row direction $t^2 r_i/2$. On $\|r_i\| \leq D$, its Taylor remainder divided by t^2 is at most $\|\phi''_1\|_\infty t^2 D^2/8$; on the complement it is at most $\|\phi'_1\|_\infty \|r_i\|$. The deterministic second-moment limit, choosing D and then time, gives $o_*(t^2)$ in neuron-index-times-sphere L^2 . Orthogonal projection off affine functions is contractive, so the positive limiting projected acceleration proves the claim. $\square$

Proof of Theorem 1.2. The four-point combination (108), initial velocity convergence and (112) give a positive multiple of t for the dense predictor on a width-independent small interval. It annihilates every affine predictor and has coefficient absolute sum one, so it bounds their uniform distance below. The positive energy limit in (111), divided by $\|y\|_1$, similarly gives a training-input gap of order t^3 from the frozen kernel. The two preceding lemmas supply all internal feature conclusions.

For each fixed positive time, the vanishing absolute compression errors in Propositions 4.2, 6.5 and 6.6 transfer both prediction gaps by the triangle inequality; eventual estimates at every confidence imply convergence in probability. Choose common smaller $c_*, t_* > 0$. Legendre and Harmonic already cover the witnesses. Taylor appends them before initialization, without labels. Its bound replaces p by at most $p + 4$, multiplying the quadratic panel factor by at most 9 and the linear data factor by at most 3, since $m + p \geq 2$. No endpoint gap or identification of compressed coordinates with dense neurons is used or asserted. $\square$

8 Assembly of the headline theorem

Proof of Theorem 1.1. The preceding constructions provide the stated retained counts and initialization-only provenance. Here we verify the probabilistic comparison, whose reference is actual dense variability rather than an upper bound. Fix a confidence budget $0 < \delta_0 < 1$. By Proposition 5.2, for sufficiently large width,

$$\Pr \left\{ \|f_n - \tilde{f}_n\|_{\mathcal{X}} \geq c_{\phi, L, \delta_0} \frac{Y \sqrt{\gamma}}{\sqrt{n} [\log(en)]^{5/2}} \right\} \geq 1 - \delta_0.$$

Its witness is a training input, so the same lower bound applies on either promised domain. The source and fitting events give all required limits. The Legendre order of Proposition 4.2 gives error at most $CY/(\sqrt{n} [\log(en)]^3)$; its ratio to the displayed lower bound is at most a fixed-problem, fixed-confidence multiple of $[\log(en)]^{-1/2}$. For the two selected models, Propositions 6.5 and 6.6

give error at most Y/n , and the corresponding ratio is at most

$$\frac{[\log(en)]^{5/2}}{c_{\phi,L,\delta_0}\sqrt{\gamma n}} \longrightarrow 0.$$

Intersecting each approximation event with the lower event costs at most $2\delta_0$ by the union bound, without any independence requirement. For every fixed positive ratio tolerance the limiting superior of its failure probability is therefore at most $2\delta_0$. Letting $\delta_0 \downarrow 0$ proves (7), including the stated failure convention. For (11), use the selected-model event at the requested confidence. The label assumption stays fixed throughout; no label inequality is used to reduce a displayed sample/gap power. $\square$